MANUAL
ON
DEVICES

ROAD SIGNS, SIGNALS AND ROAD MARKINGS

Second edition November ٢٠٠١

THE HASHEMITE KINGDOM OF JORDAN

MINISTRY OF PUBLIC WORKS AND HOUSING

٢٠٠١-١١-٢٩

CONTENTS

١. GENERAL PROVISIONS

١,١. *Glossary*

For the purpose of this Manual, the following expressions shall have the meanings hereby assigned to them:

Built-up area: Area with entries and exits specially sign-posted as such.

Road: The entire surface of any way or street open to public traffic.

Carriageway: The part of a road normally used by vehicular traffic; a road may comprise several carriageways clearly separated from one another by, for example, a dividing strip or a difference of level.

Lane: Any one of the longitudinal strips into which the carriageway is divisible, whether or not defined by longitudinal road markings, which is wide enough for one moving line of motor vehicles other than motor cycles.

Intersection: Any level crossroad, junction or fork, including the open areas formed by such crossroads, junctions or forks.

Level-crossing: Any level intersection between a road and a railway track with its own track formation.

Expressway: A divided road with two or more lanes for traffic in each direction with partial or total grade separated intersections.

Freeway: A road specially designed and built for motor traffic, which does not serve properties bordering on it, and which:

- Is provided, except at special points or temporarily, with separate carriageways for the two directions of traffic, separated from each other either by a dividing strip not intended for traffic or, exceptionally, by other means;
- Does not cross at level with any road, railway track, or footpath; and,
- Is specially sign-posted as a motorway.

Other roads: All public roads and streets other than Freeways and Expressways.

Standing: Vehicle stationary for the time needed to pick up or set down persons or to load or unload goods; and

Parking: Vehicle stationary for any reason other than the need to avoid interference with another road-user or collision with an obstruction or to comply with traffic regulations, and if the period during which the vehicle is stationary is not limited to the time needed to pick up or set down persons or goods.

Cycle: Any vehicle that has at least two wheels and is propelled solely by the muscular energy of the persons on that vehicle, in particular by means of pedals or hand-cranks;

Moped: Any two-wheeled or three-wheeled vehicle which is fitted with an internal combustion engine having a cylinder capacity not exceeding ٥٠ cc and a maximum design speed not exceeding ٥٠ km per hour.

Motor cycle: Any two-wheeled vehicle, with or without a side-car, which is equipped with a propelling engine whose unladen mass does not exceed ٤٠٠ kg.

Power-driven vehicle: Any self-propelled road vehicle, other than a moped, and other than a rail-borne vehicle.

Motor vehicle: Any power-driven vehicle which is normally used for carrying persons or goods by road or for drawing on the road, vehicles used for the carriage of persons or goods except agricultural tractors.

Trailer: Any vehicle designed to be drawn by a power-driven vehicle and includes semi-trailers.

Semi-trailer: Any trailer designed to be coupled to a motor vehicle in such a way that part of it rests on the motor vehicle and that a substantial part of its mass and of the mass of its load is borne by the motor vehicle.

Driver: Any person who drives a motor vehicle or other vehicle, including a cycle, on a road.

Laden weight: The actual mass of the vehicle as loaded, with the crew and passengers on board.

Give Way to other vehicles: Means that drivers must not continue or resume their advance or manoeuvre if by so doing they might compel the drivers of other vehicles to change the direction or speed of their vehicles abruptly.

١,٢. *Road Signs in General*

١,٢,١. Requirements of Road Signs and other Traffic Control Devices.

This manual sets forth the basic principles that govern the design and usage of road signs and some other traffic control devices.

Traffic control devices give essential information to road users and are thus very important for providing drivers with guidance and warning for safe operation. Used in a proper way, traffic control devices contribute considerably to road safety.

The road signs must, however, be used with proper judgement and the number must be restricted. Otherwise signs tend to lose credibility and effectiveness. It is essential that signs are to be used according to this Manual and only when the Manual indicates the need for placement. Warning signs shall for instance never be used where a hazard is self-evident.

١,٢,٢. Responsibility for Traffic Control Devices.

The responsibility for decisions on the use of Road signs and other Traffic Control Devices on roads and streets rests upon the Ministry of Public Works and Housing for rural roads and certain through road in certain municipalities. The responsibility for decisions on the use of Road signs and other Traffic Control Devices on other streets and roads within the Municipalities rests upon respective Municipality. Respective authority is also responsible for the installation and maintenance of the Traffic Control Devices.

No other authorities, organisations or private persons are allowed to erect any Traffic Control Devices on the public road and streets without a written authorisation from the responsible authority.

١,٢,٣. Application of signs.

Uniformity of use and placement of signs is essential. Identical conditions shall always be indicated with the same type and placement of sign.

The use of Traffic Control Devices should be based on the following general principles.

The devices should:

* fulfil a need
* command attention
* convey a clear and simple meaning
* command respect of road users
* give adequate time for proper response

١,٢,٤. Reflectorization.

All signs shall be reflectorized or illuminated.

Coefficient of Retroreflection.

The minimum initial coefficient of retroreflection R' ($cd \cdot lx^{-١} \cdot m^{-٢}$) of retroreflective signs when measured in accordance with the procedure specified in CIE٥٤, using CIE standard illuminant A, shall conform to tables ١,٢,٤.A and ١,٢,٤.B, as appropriate.

The coefficient of retroreflection (R') of all printed colours, except white, shall be not less than ٧٠ % of the values in tables ١,٢,٤.A and ١,٢,٤.B for Class ١ (C١) and Class ٢ (C٢) signs respectively.

TABLE ١,٢,٤.A COEFFICIENT OF RETROREFLECTION R': C١.
Unit: $cd \cdot lx^{-١} \cdot m^{-٢}$

Geometry of measurement		Colour							
α	β١	White	Yellow	Red	Green	Blue	Brown	Orange	Grey
١٢'	+٥º	٧٠	٥٠	١٤,٥	٩	٤	١	٢٥	٤٢
	+٣٠º	٣٠	٢٢	٦	٣,٥	١,٧	٠,٣	١٠	١٨
	+٤٠º	١٠	٧	٢	١,٥	٠,٥	#	٦	٦
٢٠'	+٥º	٥٠	٣٥	١٠	٧	٢	٠,٦	٢٠	٣٠
	+٣٠º	٢٤	١٦	٤	٣	١	٠,٢	٨	١٤,٤
	+٤٠º	٩	٦	١,٨	١,٢	#	#	٢,٢	٥,٤
٢º	+٥º	٥	٣	١	٠,٥	#	#	١,٢	٣
	+٣٠º	٢,٥	١,٥	٠,٥	٠,٣	#	#	٠,٥	١,٥
	+٤٠º	١,٥	١,٠	٠,٥	٠,٢	#	#	#	٠,٩

α indicates Observation angle
β١ indicates Entrance angle
# indicates “Value greater than zero but not significant or applicable”

TABLE ١,٢,٤.B COEFFICIENT OF RETROREFLECTION R': C٢.
Unit: $cd \cdot lx^{-١} \cdot m^{-٢}$

Geometry of measurement		Colour							
α	β١	White	Yellow	Red	Green	Blue	Brown	Orange	Grey
١٢'	+٥º	٢٥٠	١٧٠	٤٥	٤٥	٢٠	١٢	١٠٠	١٢٥
	+٣٠º	١٥٠	١٠٠	٢٥	٢٥	١١	٨,٥	٦٠	٧٥
	+٤٠º	١١٠	٧٠	١٥	١٢	٨	٥	٢٩	٥٥
٢٠'	+٥º	١٨٠	١٢٠	٢٥	٢١	١٤	٨	٦٥	٩٠
	+٣٠º	١٠٠	٧٠	١٤	١٢	٨	٥	٤٠	٥٠
	+٤٠º	٩٥	٦٠	١٣	١١	٧	٣	٢٠	٤٧
٢º	+٥º	٥	٣	١	٠,٥	٠,٢	٠,٢	١,٥	٢,٥
	+٣٠º	٢,٥	١,٥	٠,٤	٠,٣	#	#	١	١,٢
	+٤٠º	١,٥	١,٠	٠,٣	٠,٢	#	#	#	٠,٧

α indicates Observation angle
β١ indicates Entrance angle
# indicates “Value greater than zero but not significant or applicable”

<u>Chromaticity and luminance factors.</u>

Retroreflective materials for road signs and other traffic control devices shall be of colour specified in table ١,٢,٤.C and ١,٢,٤.D. The tables indicate the chromaticity limits when tested against the CIE standard light source D٦٥, measure geometry ٤٥/٠.

TABLE ١,٢,٤.C CHROMATICITY AND LUMINANCE FACTORS CLASS C١ MATERIAL

Colour	Limit ١		Limit ٢		Limit ٣		Limit ٤		Luminance factor β
	x	y	x	y	x	y	x	y	
White	٠,٣٥٥	٠,٣٥٥	٠,٣٠٥	٠,٣٠٥	٠,٢٨٥	٠,٣٢٥	٠,٣٣٥	٠,٣٧٥	≥ ٠,٣٥
Yellow	٠,٥٤٥	٠,٤٥٤	٠,٤٨٧	٠,٤٢٣	٠,٤٢٧	٠,٤٨٣	٠,٤٦٥	٠,٥٣٤	≥ ٠,٢٧
Red	٠,٧٣٥	٠,٢٦٥	٠,٦٧٤	٠,٢٣٦	٠,٥٦٩	٠,٣٤١	٠,٦٥٥	٠,٣٤٥	≥ ٠,٠٣
Green	٠,٠٠٧	٠,٧٠٣	٠,٢٤٨	٠,٤٠٩	٠,١٧٧	٠,٣٦٢	٠,٠٢٦	٠,٣٩٩	≥ ٠,٠٣
Blue	٠,٠٧٨	٠,١٧١	٠,١٥٠	٠,٢٢٠	٠,٢١٠	٠,١٦٠	٠,١٣٧	٠,٠٣٨	≥ ٠,٠١

Brown	٠,٤٥٥	٠,٣٩٧	٠,٥٢٣	٠,٤٢٩	٠,٤٧٩	٠,٣٧٣	٠,٥٥٨	٠,٣٩٤	≥ ٠,٠٣
Orange	٠,٦١٠	٠,٣٩٠	٠,٥٣٥	٠,٣٧٥	٠,٥٠٦	٠,٤٠٤	٠,٥٧٠	٠,٤٢٩	≥ ٠,١٤
Grey	٠,٣٥٠	٠,٣٦٠	٠,٣٠٠	٠,٣١٠	٠,٢٨٥	٠,٣٢٥	٠,٣٣٥	٠,٣٧٥	≥ ٠,١٢

TABLE ١,٢,٤.C CHROMATICITY AND LUMINANCE FACTORS CLASS C٢ MATERIAL

Colour	Limit ١		Limit ٢		Limit ٣		Limit ٤		Luminance factor β
	x	y	x	y	x	y	x	y	
White	٠,٣٠٥	٠,٣١٥	٠,٣٣٥	٠,٣٤٥	٠,٣٢٥	٠,٣٥٥	٠,٢٩٥	٠,٣٢٥	≥ ٠,٣٥
Yellow	٠,٤٩٤	٠,٥٠٥	٠,٤٧٠	٠,٤٨٠	٠,٤٩٣	٠,٤٥٧	٠,٥٢٢	٠,٤٧٧	≥ ٠,٢٧
Red	٠,٧٣٥	٠,٢٦٥	٠,٧٠٠	٠,٢٥٠	٠,٦١٠	٠,٣٤٠	٠,٦٦٠	٠,٣٤٠	≥ ٠,٠٥
Green	٠,١١٠	٠,٤١٥	٠,١٥٠	٠,٤١٥	٠,١٥٠	٠,٤٥٥	٠,١١٠	٠,٤٥٥	≥ ٠,٠٤
Blue	٠,١٣٠	٠,٠٨٦	٠,١٦٠	٠,٠٨٦	٠,١٦٠	٠,١٢٠	٠,١٣٠	٠,١٢٠	≥ ٠,٠١
Brown	٠,٤٥٥	٠,٣٩٧	٠,٥٢٣	٠,٤٢٩	٠,٤٧٩	٠,٣٧٣	٠,٥٥٨	٠,٣٩٤	≥ ٠,٠٣
Orange	٠,٦١٠	٠,٣٩٠	٠,٥٣٥	٠,٣٧٥	٠,٥٠٦	٠,٤٠٤	٠,٥٧٠	٠,٤٢٩	≥ ٠,١٤
Grey	٠,٣٠٥	٠,٣١٥	٠,٣٣٥	٠,٣٤٥	٠,٣٢٥	٠,٣٥٥	٠,٢٩٥	٠,٣٢٥	≥ ٠,١٢

CHOICE OF REFLECTIVE MATERIALS ON PERMANENT ROAD SIGNS					
Type of Road Sign	Placement	Environment			
		No road lighting or low standard road lighting and no disturbing lights in the road vicinity		Road lighting of good standard and/or disturbing lights in the road vicinity	
		First-class Freeway, Expressway	Other roads	First-class Freeway, Expressway	Other roads
Stop, Give Way	Ground	C٢	C٢	C٢	C٢
Mandatory sign Pass This Side placed in the middle of the road	Ground	C٢	C٢	C٢	C٢
Pedestrian Crossing	Ground	-	C٢	-	C٢
Informative signs	Ground	C٢	C١	C٢	C١
	Overhead	C٢*	C٢	C٢*	C٢
All other signs	Ground	C٢	C١	C٢	C٢
	Overhead	C٢*	C١	C٢*	C٢

*) These signs may require separate illumination in certain environment. As an alternative to separate illumination so called “Microprismatic” (C٣) reflective material may be considered.

Signs with different types of reflective materials should not be used on the same road section and especially not on the same post.

Traffic control devices must be kept clean and in good condition. The function of the devices should be tested periodically under both day and night conditions. Signs with damaged (scratches, cracks, holes etc.) or worn out reflective material should be replaced with new ones. If the measured reflective values (R') are less than those in the table below, the function of the reflective material is not satisfactory and the signs should be replaced. The reflectivity units are cd · $lx^{-١}$ · $m^{-٢}$. The values in the table are with ١٢' observation angle (α) and +٥° entrance angle ($\beta_{١}$). For measuring the reflectivity on road signs a reflectometer should be used.

Sign placement	**Colour**				
	White	**Yellow**	**Red**	**Blue**	**Green**
Ground	٢٠,٠	١٤,٠	٤,٠	١,٢	٢,٨
Overhead	٣٠,٠	٢١,٠	٦,٠	١,٨	٤,٢

١,٣. *Position.*

a) A Warning Sign is placed ahead of the condition to which it calls attention.
b) A Regulatory Sign is normally placed where its mandate or prohibition applies or begins.
c) Informative Signs are placed where needed to keep the drivers informed as to their route and destination.

١,٤. *Sign Categories*

Road signs are divided into the following principal categories:

١,٤,١. Warning Signs

These signs call attention to conditions that are potentially hazardous to traffic operations. The Warning Signs are subdivided into:

٢,١ Warning Signs, other than those placed at intersections
٢,٢ Warning Signs at intersections
٢,٣ Other Warning Devices

١,٤,٢. Regulatory Signs

These signs inform road users of special obligations, restrictions or prohibitions. The Regulatory Signs are subdivided into:

٣,١ Priority Signs
٣,٢ Prohibitory or Restriction Signs
٣,٣ Mandatory Signs

١,٤,٣. Informative Signs

These signs guide the road users and provide relevant information. The Informative Signs are subdivided into:

٤,١ Advance Direction Signs
٤,٢ Direction Signs
٤,٣ Confirmatory Signs
٤,٤ Road Number signs
٤,٥ Pedestrian Crossing
٤,٦ Useful Information Signs
٤,٧ Service Facilities Signs
٤,٨ Signs for Recreational and Cultural Areas
٤,٩ Direction Symbols
٤,١٠ Signs for Emergency Escape Ramps
٤,١١ Signs for truck weighing stations

١,٤,٤. Standing and Parking Signs

These signs are prohibiting or restricting standing or parking.

- ٥,١ Parking Prohibited
- ٥,٢ Standing and Parking Prohibited
- ٥,٣ Parking

١,٤,٥. Temporary Signs for Road Work Areas (in separate Manual)

١,٥. *Sign Dimensions*

Road sign dimensions shall be:

a) Large size
b) Normal size
c) Small size

١,٥,١. Sizes on standard road signs.

SIGN	LARGE mm	NORMAL mm	SMALL mm
WARNING (SIDE)	١٣٥٠	٩٠٠	٦٠٠
REGULATORY (DIAMETER)	٩٠٠	٦٠٠	٤٠٠
MANDATORY (DIAMETER)	٩٠٠	٦٠٠	٤٠٠
INFORMATIVE; SERVICE (WIDTH x HEIGHT)	Width x Height ٩٠٠ x ٩٠٠	Width x Height ٦٠٠ x ٦٠٠	-

Large size signs shall be used:

- On expressways or other highways with similar standard
- Other roads with two or more lanes in the same direction (divided highways) and a speed limit of more than ٨٠ km/h.
- At any other location where a greater legibility or emphasis is required.

Normal size signs shall be used:

- In all cases other than stated above (large size) or below (small size).

Small size signs may be used:

- At special locations where normal sized signs do not fit and the speed limit is low
- Inside urban areas on narrow streets/roads

According to the “**AGREEMENT ON INTERNATIONAL ROADS IN THE ARAB MASHREQ**” the following sign dimensions shall be used on roads included in the agreement.

SIGN DIMENSION

SIGN	MAXIMUM SPEED (km/h)		
	٦٠ ـ ٧٥	> ٧٥ ـ ٩٠	> ٩٠
	DIMENSIONS (mm)		
Warning sign (side length)	٦٠٠	٩٠٠	١٣٥٠

STOP sign (diameter)	٦٠٠	٩٠٠	١٢٠٠
Give Way (side length)	٦٠٠	٩٠٠	١٣٥٠
Priority Road (side length)	٦٠٠	٦٠٠	٦٠٠
Priority Over Oncoming Traffic (side length)	٦٠٠	٦٠٠	٦٠٠
Other Regulatory signs (diameter)	٦٠٠	٩٠٠	١٢٠٠

١,٦. *Lettering.*

١,٦,١. Letters on Informative Signs.

Style. All informative signs shall have text in both Arabic and English. The Arabic text shall be in AF-Najed style.

The Latin letters shall be in small letters with the initial letter Capital unless otherwise specified. Latin letters shall be Arial Bold.

Colour. White text and symbols on light blue background on Direction signs for public roads.

White text and symbols on brown background on Direction signs for cultural and recreational areas.

Letter sizes.

Arabic text mm	English text mm	Application
١١٥	٧٥	Inside parking lots or other areas with low speed traffic
١٩٠	١٢٥	Side mounted signs on low traffic Secondary Roads, on Village Roads, and on narrow streets
٣٠٠	٢٠٠	Side mounted signs on Primary and Secondary Roads
٣٠٠	٢٠٠	Side mounted signs on Freeways, Expressways and similar roads. Overhead mounted signs inside urban areas. Signs for Emergency Escape Ramps.
٤٥٠	٣٠٠	Overhead mounted signs on Freeways, Expressways and similar roads

Table ١,٦,١ - ١ Letter sizes on Informative Signs

According to the "**AGREEMENT ON INTERNATIONAL ROADS IN THE ARAB MASHREQ**" the following sign dimensions shall be used on roads included in the agreement.

MINIMUM CHARACTER HEIGHT IN WRITING ON INFORMATIVE SIGNS (ENGLISH LOWER CASE)

Maximum Speed km/h	Pre-advance informative signs	Advance informative signs	Informative signs
	Character height (mm)	Character height (mm)	Character height (mm)
٦٠ – ٧٥	١٥٠	١٠٠	١٠٠
> ٧٥ – ٩٠	٢٠٠	١٥٠	١٠٠
> ٩٠	٣٠٠	٣٠٠	٣٠٠

Note: - The principal difference between the pre-advance informative signs and advance informative signs referred to in the table above lies in the distance from the signs to the intersection before which they are placed.
- The height of the Arabic letter “alif” shall be at least ١,٥ times the height of the lower-case English letter.
- The space between the lines should be equal to the English letter height.

١,٦,٢. Letters on Additional Panels.

Additional panels for warning signs and regulatory signs.

The colour of the panels shall be white with black legends and red border.

The Latin letters shall be in small letters with the initial letter Capital unless otherwise specified.

Arabic text mm	English text mm	Application
١٥٠	١٥٠	Below large size signs
١٠٠ ٧٥	١٠٠ ٧٥	Below normal size signs Speed limit > ٨٠ km/h Speed limit ≤ ٨٠ km/h
٦٠	٦٠	At the entries of, or within parking areas or other areas with low speed traffic
٦٠	٦٠	Below small size signs

Additional panels for informative signs.

The colour of the panels shall be the same as the sign it refers to.

The Latin letters shall be in small letters with the initial letter Capital unless otherwise specified. Latin letters shall be Arial Bold.

The letter size on additional panels shall be the same as on the sign the panel refers to. In case the additional panel refers to an informative sign without letters, the letter size on the additional panel to be chosen to fit the sign size.

١,٧. *Sign Location.*

Signs shall be located in such a way that they are readily visible and easily identifiable from a distance.

They are normally placed on the right hand side of the road. Direction signs might, however, be placed above the road or on the left hand side. When visibility is limited because of road alignment, side obstacles etc., signs may have to be duplicated on both sides of the road.

On divided highways, signs are normally placed on both sides of the carriageway.

Additional rules for location of signs are described in respective sign-class chapter.

Large Informative signs should be slightly turned away from the longitudinal axis of the road to avoid disturbing glaring from vehicle headlights during darkness.

Overhead mounted signs should be tilted towards the road to improve reflectivity during darkness.

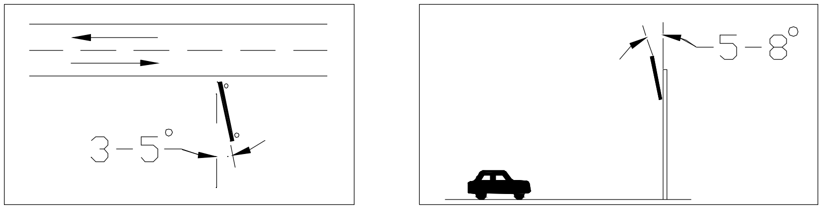


Figure ⅔a Tilting of side mounted signs and overhead mounted signs.

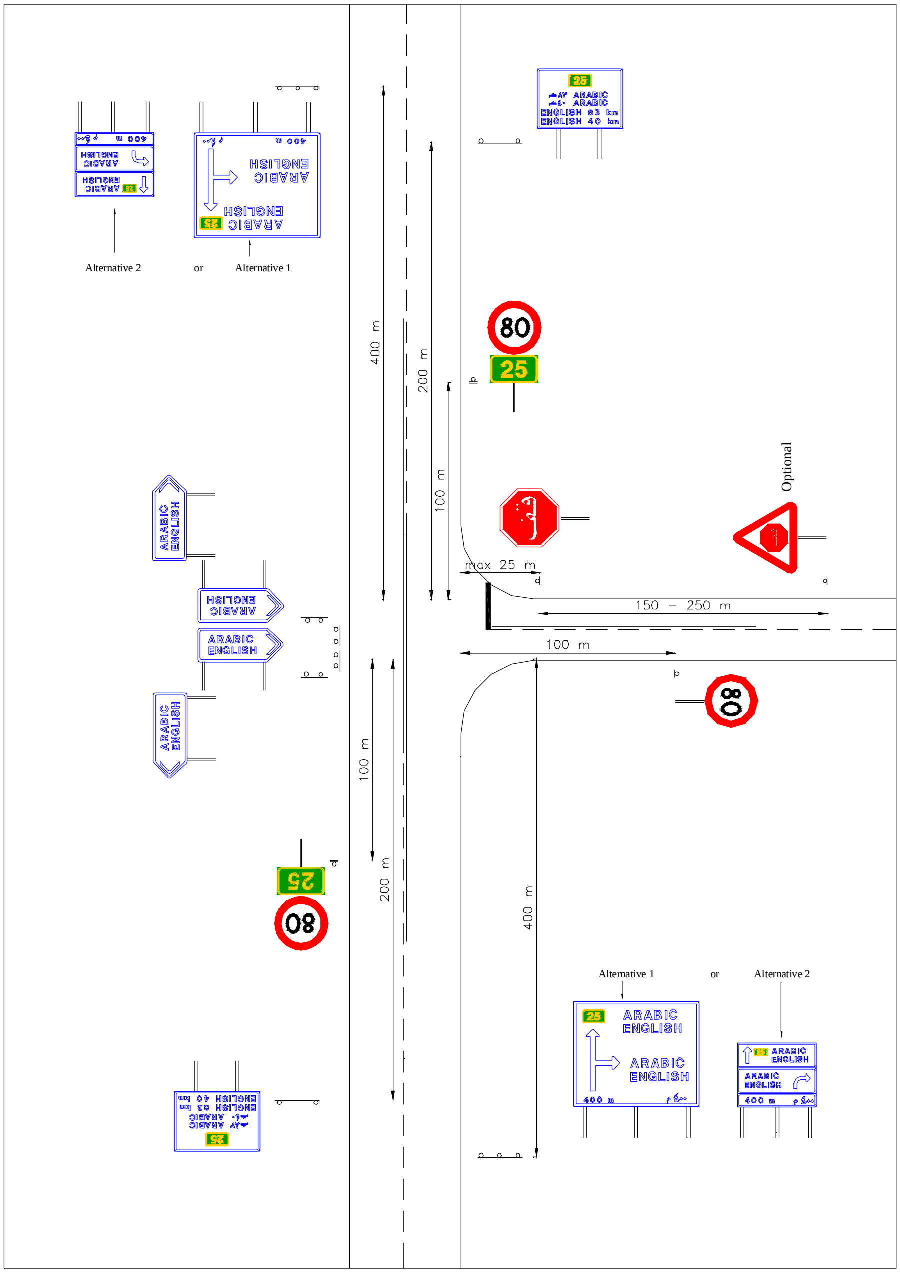


Fig. ١,٧b Typical example of sign locations at intersections in rural areas.

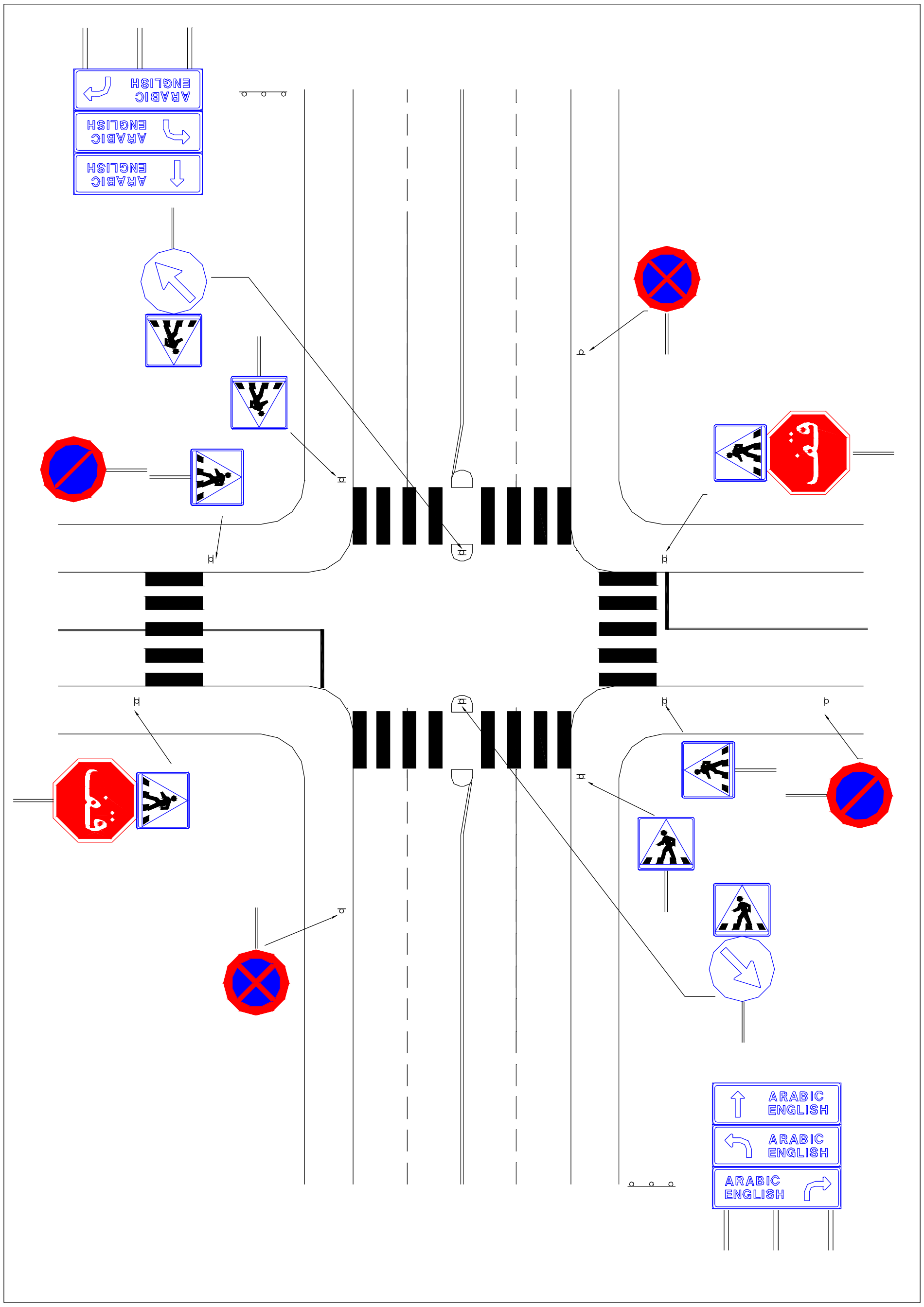


Fig ١,٧c Typical example of sign location at intersections within urban areas.

١,٨. Mounting height and lateral clearance.

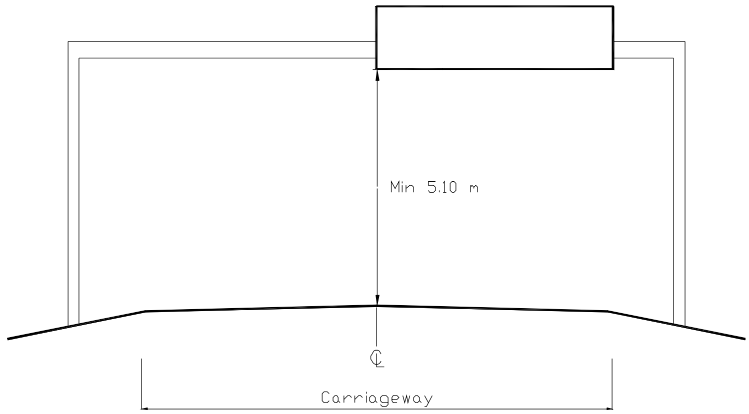


Fig. ١,٨a Vertical clearance for overhead signs.

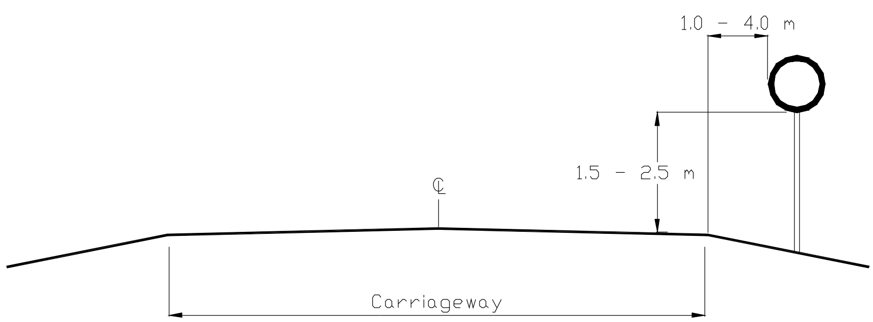


Fig. ١,٨b Mounting height and lateral clearance for standard signs outside urban areas.

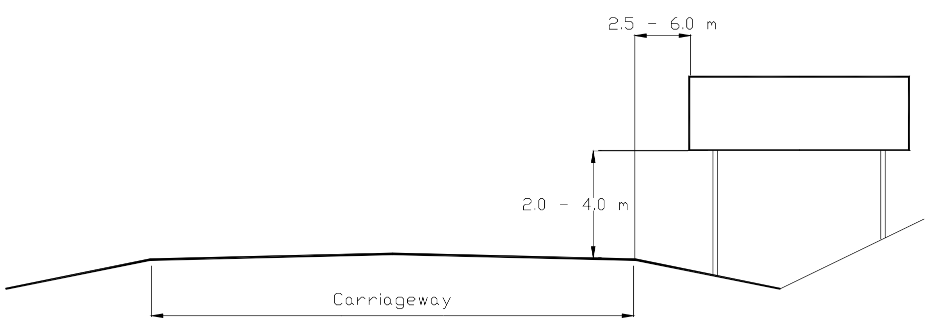
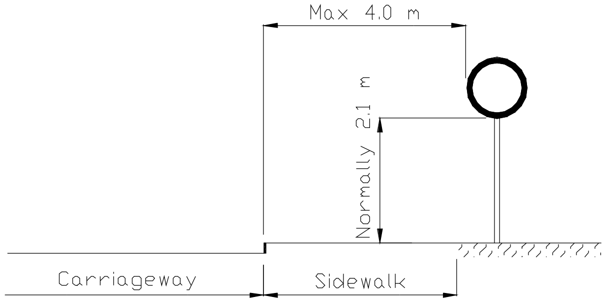


Fig. ١,٨c Mounting height and lateral clearance for large direction signs outside urban areas. (On roads with shoulders the lateral clearance should still be measured from the travelled way.)



Signs shall be placed off the sidewalk (for all types of signs)

Fig. ١,٨d Road with sidewalk, signpost outside the sidewalk

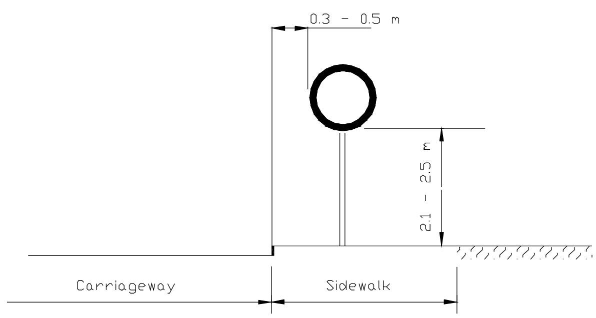


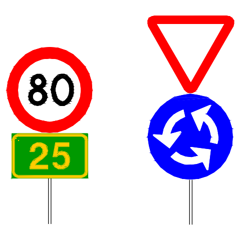
Fig. ١,٨f Road with sidewalk, signpost on the sidewalk

If the signs cannot be placed by the sidewalk they may be placed on the sidewalk as shown. Alternatively; signs may be placed on a console on a house wall less than ٤ meters from the carriageway.

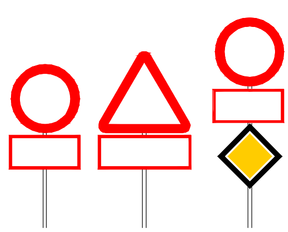
If two signs are placed on the same post, the height should be measured to the lower edge of the lower sign.

١,٩. *Relative Placement.*

١. If signs of the same category are placed on the same post, the one that is most important to road safety or traffic conduct shall be placed in the lower position.
٢. A warning sign normally is not combined with a regulatory sign because of their different significance.
٣. Below are some typical examples of combination of different types of signs.



٤. More than three signs are normally not placed on the same post.
٥. An additional panel may be placed directly below the sign it refers to. See example below.



٦. **No Traffic Control Device or its support shall bear any advertising or commercial message, or any other message that is not essential to traffic control.**

٢. WARNING SIGNS

٢,٠. *In General*

a: Shape

Warning signs shall have the shape of an equilateral triangle having one side horizontal and the opposite vertex above it. The ground is white and the border is red. Unless the description specifies otherwise, the symbols shall be black.

b: Location

Areas and Road type	Location
<u>In Rural areas</u>	
Expressways and similar roads	٤٠٠ meters ahead of the hazard
On other roads	١٥٠-٢٥٠ meters ahead of the hazard
<u>In Urban areas</u>	
Speed limit maximum ٦٠ km/h	٢٥ - ٥٠ meters ahead of the hazard
Speed limit more than ٦٠ km/h	١٠٠ meters ahead of the hazard

Special circumstances may justify other locations, e.g. longer distances between the Warning Signs and the hazards. If any other placement than the normal is used, an additional panel shall indicate the distance between the warning sign and the hazard, for example "٨٠٠ m".

c: Special location

Hazards, on an intersecting road, may be indicated by a warning sign on the road. In those cases the warning sign shall have an additional panel with a bent arrow showing the direction of the hazard. (See figure below as an example).

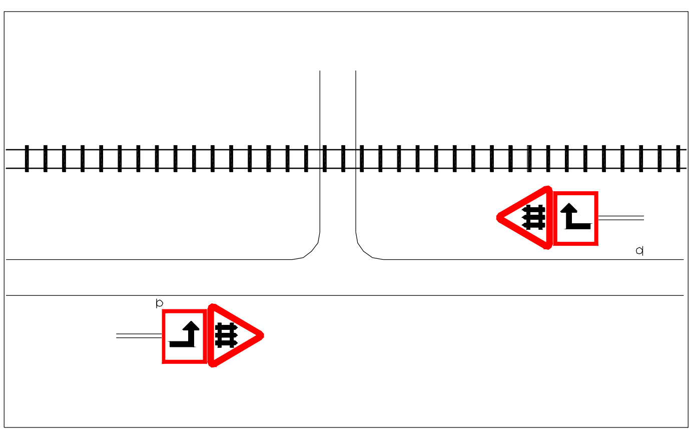


Fig. ٢,٠a Warning of a railway level crossing located close to the road.

٢,١. *Warning signs, other than those placed at intersections*

W١. DANGEROUS BENDS

Warning of a dangerous bend or a succession of dangerous bends is given with one of these symbols, whichever is appropriate.

If warning applies to more than two bends, the length of the section with bends should be indicated on an additional panel, for example ١,٢ km.

These signs are used only to warn of a bend or bends that are considered especially dangerous compared with the standard of the major part of the road and therefore a speed reduction is necessary.

W ⅔

Right Bend

W ⅔

Left Bend

W ⅔

Double Bend or succession of more than two bends, the first to the right

W ⅔

Double Bend or succession of more than two bends, the first to the left

W٢. DANGEROUS DESCENT OR ASCENT

W ⅔ Steep Descent

W ⅔ Steep Ascent

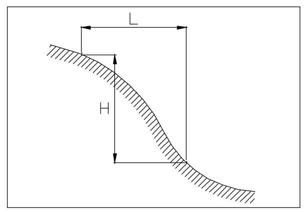
These signs give warning of a road section with either a steep descent or a steep ascent.

These signs are used to warn of descents or ascents that are considered especially dangerous. That is normally when the gradient is as follows:

٥٪ and the distance is more than ١،٠٠٠ meters long
٦٪ and the distance is more than ٦٠٠ meters long
٧٪ and the distance is more than ٣٠٠ meters long
٨٪ and the distance is more than ٢٥٠ meters long
٩٪ and the distance is more than ١٥٠ meters long

The gradient indicated on the signs is the steepest gradient for the road section of which the warning is given.

The gradient may be calculated as follows:



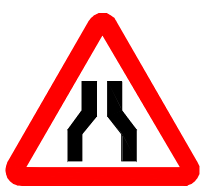
Gradient in % = H x ١٠٠ / L

If the warning applies to a descent or ascent of some considerable length, the length shall be indicated on an additional panel, for example ١،٢ km.

W٣. CARRIAGEWAY NARROWS

These signs indicate that the carriageway ahead is narrowing.

These signs shall be used only if the road suddenly narrows by one lane or more and when paved shoulders end.



W ⅔

From both sides

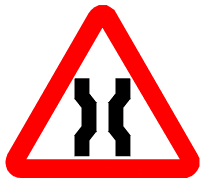
W ⅔

From the right

W ⅔

From the left

W٤. NARROW BRIDGE



This sign indicate that the carriageway ahead is narrowing due to a narrow bridge.

The actual width of the bridge may be shown on an additional panel.

W٥. ROAD LEADS ON TO QUAY OR RIVER BANK

This sign give warning that the road leads to a quay or a riverbank and there is a risk for drivers to drive into the water. The sign may be used to warn of a ferry berth.

W٦. UNEVEN ROAD

Sign W٦ Uneven Road gives warning of dangerous road damages and unevenness such as cracks, potholes, etc.

If the road section in poor condition is more than ١٠٠ m in length, the length may be indicated on an additional panel.

W٧. HUMP

Signs W٧ Hump shall be used where humps have been deliberately installed to reduce vehicle speed.

W٨. DIP

Signs W٨ Dip shall be used where dips have been deliberately installed to reduce vehicle speed.

The sign may also be used for warning of an Irish Bridge.

W٩. ROAD WORKS

This sign indicates that road work is in progress on a section of the road.

Since road works always increase accident risk to public, placement and the condition of the signs are very important. They shall be placed in the best position for good visibility, even at night. They shall never be obstructed by other items such as houses, trees, other signs etc. They shall always be kept clean and in good condition. If the retroreflective sheeting is damaged, the sign shall be replaced.

Signage of roadwork areas including other traffic control devices shall be implemented according to the details provided in the Manual on Traffic Control of Road Work Areas.

All warning signs and regulatory signs, except Mandatory signs, used at road work areas shall have yellow background red border and black symbols or text. Informative signs should be yellow with black border and text. Informative signs may be have their ordinary colour if considered appropriate.

W١٠. SLIPPERY ROAD

This sign gives warning of a slippery road condition.

This sign shall be used for warning of slippery conditions on the road. In some cases the sign may be permanently erected and provided with an additional panel, for example if a road surface becomes slippery during rain or sand. The additional panel must then bear the text "In Rain" respectively “In Sand”.

W١١. FALLING ROCKS

W١١,١ From the right

W١١,٢ From the left

These signs give warning of risk of slides or falling rocks on a section of the road ahead. The signs also indicate risk of the presence of rocks and stones on the carriageway.

The symbol used shall indicate the actual direction of slides or falling rocks.

W١٢. LOOSE GRAVEL

This sign gives warning of a road section with loose gravel on the carriageway.

The sign may be used when applying surface dressing and there is loose gravel remaining on the road. The sign may also be used on other occasions when loose gravel occurs unexpectedly.

The sign shall never be used on gravel roads even if there is more loose gravel on a section than normally.

When the warning concerns a longer section of the road, the sign may have an additional panel indicating the length of that section, for example "١,٥ km". In such a case it has to be repeated at each kilometre.

W١٣. PEDESTRIAN CROSSING

This sign gives warning of a pedestrian crossing ahead. It is used in locations where the visibility is poor or if speed limit exceeds ٦٠ km/h.

The sign shall not be used for warning of a signalised pedestrian crossing.

The warning may be combined with a low speed limit such as ٣٠ km/h in order to increase road safety for pedestrians.

W١٤. CHILDREN

This sign gives warning of a section of road frequented by children such as school and playground places.

The sign should only be used when exits from school and playground are in direct connection with a road or a street.

The warning may be combined with a low speed limit such as ٣٠ km/h in order to increase road safety for children.

W١٥. CYCLISTS ENTERING OR CROSSING

This sign gives warning of a point at which cyclists frequently enter or cross the road.

W١٦. ANIMAL CROSSING

W١٦,١ Camels

W١٦,٢ Sheep

These signs give warning of a road section with particular danger due to animal crossing.

The symbol on the sign indicates kind of animals, camels or sheep, that are likely to occur on the road.

W١٧. DANGEROUS SHOULDERS

This sign gives warning of dangerous shoulders or when there is a difference in level between the roadway and its shoulder.

W١٨. LOW FLYING AIRCRAFT

This sign gives warning of low flying aircraft.

It may be used on roads close to an airfield where aircraft may be flying over the road at a low altitude when taking off or landing.

W١٩. STRONG CROSS-WIND

W١٩,١

These signs warn of a road section where strong cross-winds often occur. Such places could be on high bridges for instance.

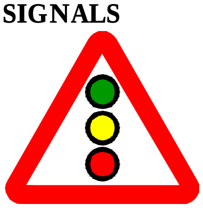
The symbol on the sign indicates the common wind direction.

Strong cross-wind from the right

W١٩,٢

Strong cross-wind from the left

W٢٠. TRAFFIC LIGHT SIGNALS



This sign gives warning of a traffic light signal.

It warns of a three-colour traffic light signal where drivers do not expect them.

This sign should not be used inside urban areas where traffic light signals are common.

W٢١. TWO-WAY TRAFFIC

This sign indicates a road section carrying two-way traffic.

The sign shall be used when a section of a road with one-way traffic, temporarily or permanently, changes into two-way traffic. It could be repeated every ٥٠٠ m or as necessary.

W٢٢. DIVIDED ROAD AHEAD

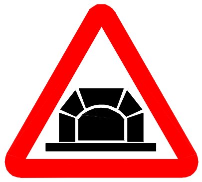
This sign gives warning of a section of the road where a physical barrier separates opposing traffic.

It shall not be used to warn of traffic islands etc. at intersections.

W٢٣. DIVIDED ROAD ENDS AHEAD

This sign gives warning of a road narrowing from a divided road to a two-way road.

W٢٤. TUNNEL



This sign is used to warn of a tunnel ahead.

The sign may have an additional panel with text "Turn on Light" unless such a sign is erected separately in the direct vicinity of the tunnel entrance.

W٢٥. RAIL LEVEL-CROSSING WITH GATES

This sign warns of a railway level-crossing with gates or staggered half-gates on both sides of a railway line.

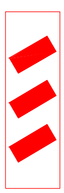
Beneath this sign, sign W٢٧,١ or W٢٧,٤ should be added.

W٢٦. RAIL LEVEL-CROSSING WITHOUT GATES

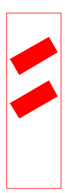
This sign warns of a railway level-crossing without gates or half gates.

Beneath this sign, sign W٢٧,١ or W٢٧,٤ should be added.

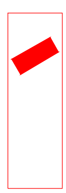
W٢٧. ADDITIONAL SIGNS AT APPROACHES TO RAIL LEVEL-CROSSING



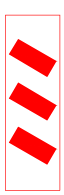
W٢٧,١



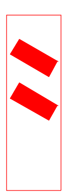
W٢٧,٢



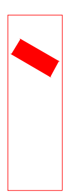
W٢٧,٣



W٢٧,٤



W٢٧,٥



W٢٧,٦

These signs are used ahead of a level-crossing to increase awareness of drivers of the crossing and its distance.

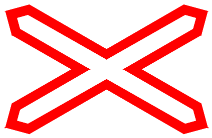
Sign W٢٧,١ and/or W٢٧,٤ shall be placed as additional panels to the warning signs W٢٥ or W٢٦.

Signs W٢٧,٢ and W٢٧,٥ shall be placed at ١/٣ of the distance from the warning signs to the crossing and signs W٢٧,٣ and W٢٧,٦ at ٢/٣ of that distance.

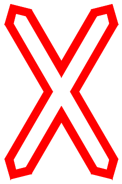
The bars on the signs shall slope down towards the road.

Signs on left side of the road are necessary only when enhanced warning is needed.

W۲۸. RAIL LEVEL-CROSSING SIGN



W۲۸,۱



W۲۸,۲

W۲۸,۳

These signs are placed in the immediate vicinity of the level crossing. They are placed at each side of rail level-crossings without gates to indicate the exact position of the railway crossing.

These signs are normally not to be used in the following cases:

- at a railway crossing where the trains pass slowly and road traffic is regulated by a guard
- at a railway crossing with an earth track with very small traffic volumes
- at a railway crossing with a cycle path or a footpath

The sign W۲۸,۳ shall be used only if the railway line comprises at least two tracks.

The signs are placed at the right hand side of the road and ٥,٠ meters from the nearest railway line. If the width of the road is more than ٥ meters, the sign W۲۸ may also be erected at the left hand side.

If there is a stop sign or a light signal sign W۲۸ may be placed on the same post as the light signal or the stop sign.

Mounting height of the Rail Level-crossing Sign shall not be less than ١,٢٠ meters.

If sufficient space is not available, sign W۲۸,۲ may be placed instead of sign W۲۸,۱ with bars pointing upwards and downwards.

W۲۹. OTHER DANGERS

This sign may be used for warning of hazards for which there are no specific symbol, for instance; Fog, Smoke, Flooding, Road Ends, Drifting Sand, etc. The type of hazard may be indicated on an additional panel.

۲,۲. *Warning signs at intersections*

W۳٠. INTERSECTION

This sign gives warning of an intersection ahead where the general priority rule is in force; that is priority for vehicles coming from the right. This sign shall be used to warn of an intersection with poor visibility.

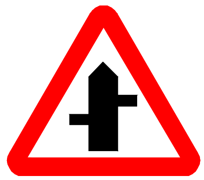
Figures tabulated below can be used as a guide for the sight distances required.

Speed limit km/h	**Sight distance to the junction meters**
١٠٠	٣٠٠
٨٠	٢٠٠
٦٠	١٥٠
< ٦٠	١٢٥

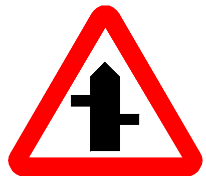
The possibility to inform of a junction by using direction signs should also be consider.

W٣١ INTERSECTION WITH A ROAD THE USER OF WHICH MUST GIVE WAY

W٣١,١



W٣١,٢



W٣١,٣

W٣١,٤

W٣١,٥

W٣١,٦

W٣١,٧

These signs may be used to warn of intersections where drivers coming from the intersecting road must yield or to stop before entering or crossing the main road. If this is obvious, or if the main road has the Priority Road Sign, no warning should be needed, unless the warning is due to poor sight distances.

The symbol of the sign shall indicate the actual conditions.

These signs are used only when either STOP signs or YIELD signs are placed on the oncoming roads.

W٣٢. ROUNDABOUT

This sign is used to warn about a roundabout ahead. Such warning may be necessary when the visibility is poor or when a roundabout creates a hazard for some other reason. When assessing the necessity of the sign, the figures tabulated under sign W٣٠ can be applied.

Also consider the possibility to give information of a roundabout by using direction signs.

W٣٣. STOP AHEAD

This sign gives warning of a STOP sign placed on an intersection ahead. It is used in the following cases:

- when the sight distance to the STOP sign is less than ١٥٠ meters at a speed limit of less than ٨٠ km/h,
- when the sight distance to the STOP sign is less than ٢٠٠ meters at a speed ٨٠ km/h or more,
- in any other case when it is considered necessary to emphasise the STOP sign.

W٣٤. GIVE WAY AHEAD

This sign gives warning of a Give Way sign placed on an intersection ahead. It is used in the following cases:

- when the sight distance to the Give Way sign is less than ١٥٠ meters at a speed limit of less than ٨٠ km/h,
- when the sight distance to the Give Way sign is less than ٢٠٠ meters at a speed ٨٠ km/h or more,
- in any other case when it is considered necessary to emphasise the Give Way sign.

٢,٣. *Other Warning Devices.*

In addition to the use of ordinary warning signs, Other Warning Devices may be useful at some locations. They may be used to complement warning signs. They may also be erected by their own in some situations.

Other Warning Devises should not be used unless there is a real need for emphasised warning. An excessive use of these devices will cause them to lose credibility and therefore effectiveness.

Other Warning Devices are Warning Text Signs, Chevron Markers and Side Obstacle Markers.

W٣٥. WARNING TEXT SIGNS

Text signs may be used to give warning of specific hazards not included in the warning sign symbols.

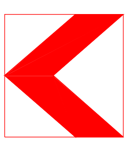
Text signs are red with white text and border. The letter sizes on warning text signs shall be based on the same criteria as for Informative signs.

Text on signs can be difficult for drivers to apprehend. The amount of text should therefore be limited as much as possible.

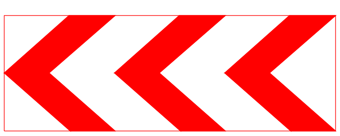
W٣٦. CHEVRON MARKERS

The Chevron Markers may be used at sharp bends to improve the visibility of the road alignment and provide better guidance to drivers.

Single Chevron Markers are used along the outer edges of long and hazardous bends. These Markers to be angled towards traffic along the bend. That mean special care must be given to the erection of each marker to ensure it will have the correct angle towards traffic. These Markers may be doubled at each position and directed towards both directions of traffic.



W٣٦,١ One Chevron



W٣٦,٢ Three Chevrons

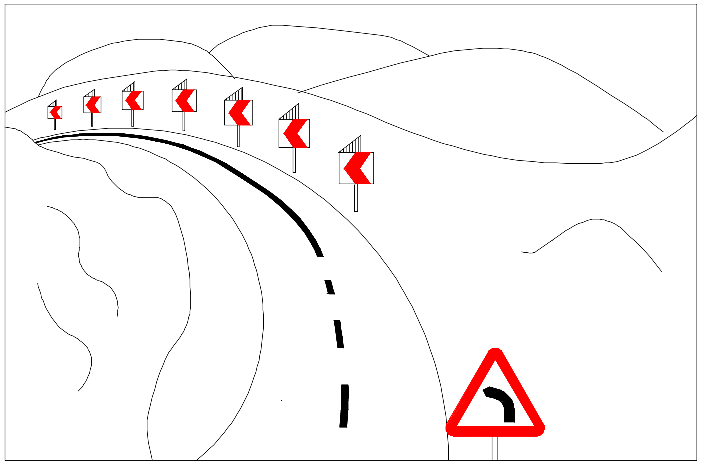


Figure ٢،٣a Typical example of the use of Chevron Markers with one chevron.

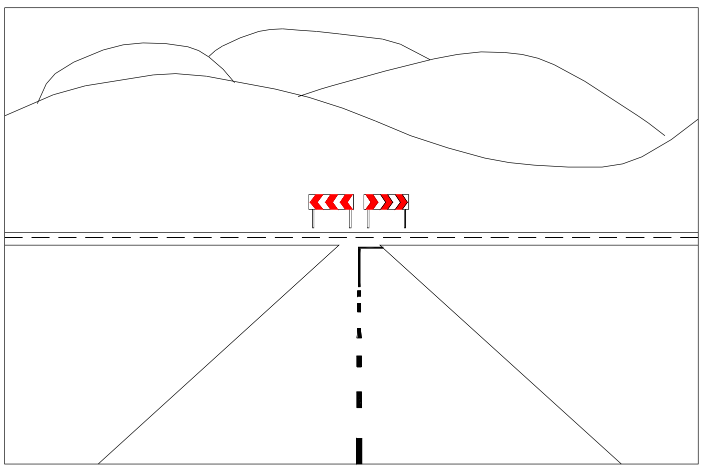


Figure ٢،٣b Typical example of the use of Chevron Markers with three chevrons at a T-intersection

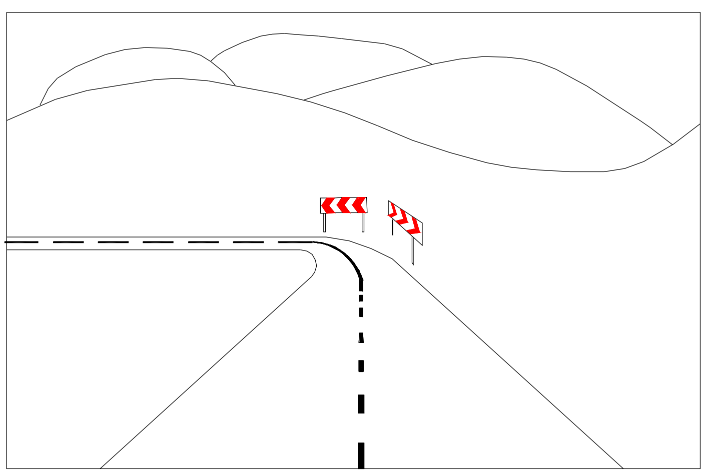
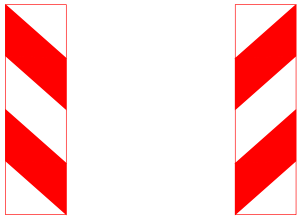


Figure ٢،٣c Typical example of the use of Chevron Markers with three chevrons at a sharp single bend located so that only one is seen from each direction of traffic.

W٣٧. SIDE OBSTACLE MARKERS



Side Obstacle Markers may be used to mark points where the carriageway narrows in a way that is not clearly visible to the drivers.

The Side Obstacle Markers shall always be erected so that the bars are sloping down towards the carriageway. Thus markers on the right hand side shall have bars sloping down from the right hand side of the marker, and markers on the left hand side shall have bars sloping down from the left hand side of the marker. Figure ٢,٣d shows a typical example of the use of Side Obstacle Markers when the carriageway narrows on the right hand side.

Side obstacle markers may be used on top of guardrails, concrete barriers and bridge parapets to warn drivers. Such markers should be of small size; width ٢٥٠ mm and height ٤٠٠ mm.

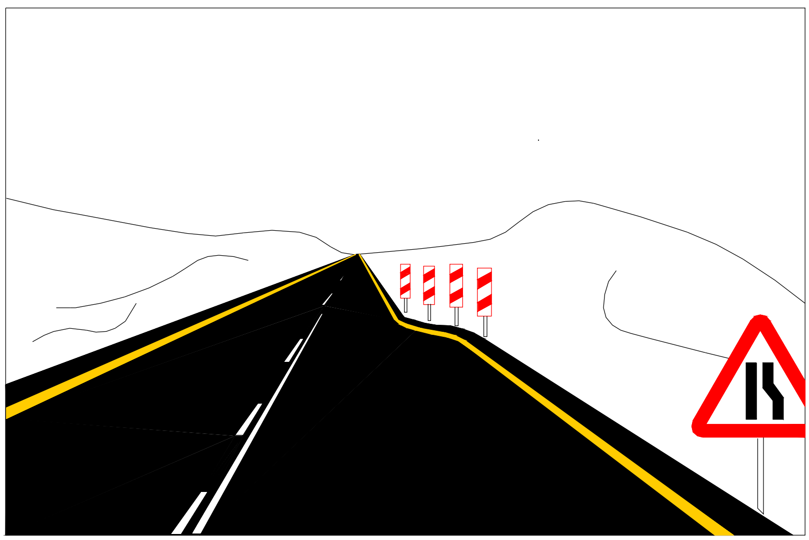


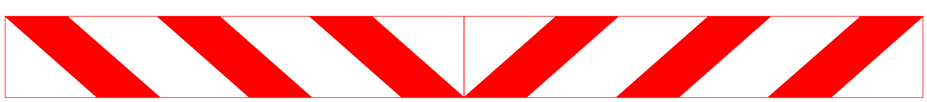
Figure ٢,٣d. Typical example of the use of Side Obstacle Markers.

W٣٨. HEIGHT OBSTACLE MARKERS

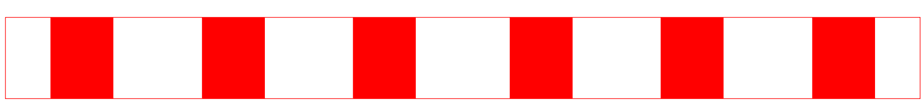
Height Obstacle Markers may be used to mark height obstacles such as bridges or tunnels.

Marker W٣٨,١ with ٤٥° bands may be used above the right hand lane on a two-way road and Marker W٣٨,٢ with vertical bands on the left hand side (see paragraph ٣,٢ Prohibitory and restrictive signs, figure ٣,٢c).

The height of the markers may be ٣٠٠ mm or ٤٠٠ mm depending on available space.



W٣٨,١. HEIGHT OBSTACLE MARKERS WITH ٤٥° ANGEL BANDS.



W٣٨,٢. HEIGHT OBSTACLE MARKERS WITH VERTICAL BANDS.

٣. REGULATORY SIGNS

˜ ⅔. *GENERAL*

Regulatory signs are used to inform road users of certain restrictions. Unless otherwise indicated on an additional panel, the restriction starts to apply at the point the sign is erected.

Regulatory signs shall be placed in a way to make them easily visible and legible to road users. They must be kept in good condition and be clean. Since most Regulatory signs convey information which is illegal to violate, the importance of keeping signs visible and in good condition is obvious.

Except as specified below, the shape of the regulatory signs is round with black symbols on white background. The border is red and so is the oblique bar when it occurs.

The STOP sign is octagonal with red background and white border and message. The word STOP shall be in Arabic and English.

The Yield sign is an equilateral triangle with one side horizontal and the opposite vertex pointing downwards. The colour is white with red border.

The "End of Overtaking" signs are round with white background and black symbols. They have no borders. A diagonal band, consisting of five parallel black lines, sloping downwards from the right to the left.

Mandatory signs are round with a blue background and white symbols. They have no borders.

The "Priority Road" sign consists of a square with one diagonal vertical. The sign shall be yellow with a white rim and black border.

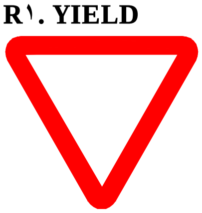
The "End of Priority Road" sign has the same design but with a black band perpendicular to the lower left and upper right sides of the square.

The "Priority Over Oncoming Traffic" sign is rectangular with blue background. The arrow pointing upwards is white and the other is red.

Regulatory signs shall not be used indiscriminately because using such signs if really not warranted may week compliance by drivers.

٣,١. *Priority signs*

R١. YIELD



This sign states that at an intersection where the sign is placed, drivers must yield to vehicles on the other approaches to the intersection.

Warrants: The sign may be needed in the following cases:

– On a minor road at an intersection where the general right-of-way rules would not work but stopping is not necessary if no other vehicles are approaching,
– On an entrance ramp to a main road without adequate acceleration lane,
– On separate right turn lanes without adequate acceleration lanes,
– In any other case where engineering study indicates a need for a yield sign.

The sign is not needed on paths or on earth-tracks leading to a road, since road users on such paths or tracks always must give way to vehicles on the main road.

The sign may also be used for advance warning of Yield at an intersection ahead. In such cases the Yield sign shall have an additional panel indicating the distance from the advance warning sign to the intersection.

Advance warning sign shall be used in the following cases:
– when the sight distance to the Yield sign is less than ١٥٠ meters at a speed limit of less than ٨٠ km/h,
– when the sight distance to the Yield sign is less than ٢٠٠ meters at a speed limit of ٨٠ km/h or more,
– in any other case when it is considered necessary to emphasise the Yield sign.

Location: The sign shall be placed within ٢٥ m from the intersection. Where there is a marked crosswalk on the pavement, the sign shall be erected approximately ١ m in advance of the crosswalk line nearest to approaching traffic.

Design: The Yield sign has the same design as the ordinary warning sign but without any symbol. Thus the sign is an equilateral triangle with one side horizontal and its opposite vertex pointing downwards.

R٢. STOP

This sign states that drivers must stop before entering an intersecting road and give way to all vehicles on that road.

The sign may also be used to indicate obligation to stop before crossing a railway level-crossing without gates.

Warrants: STOP sign may be warranted at intersections where one or more of the following conditions exist:
– Intersection of a less important road with a main road where application of the general right-of-way rule is unduly hazardous
– Street entering a through highway or street
– Unsignalised intersection on a street having signals at other nearby intersections
– Other intersections where a combination of high speed, restricted view, and serious accident record indicate a need for control by the STOP sign

STOP sign shall never be used in such cases:
– On through roadways of main roads
– On intersections where traffic control signals are operating
– As portable or part time STOP signs, except for emergency purposes
– For speed control
– On paths or on earth-tracks leading to a road.

Advance warning of a STOP sign shall be given as stated under item W٣٣.

If STOP sign is erected at intersections outside urban areas and the "main" road is not declared as a Priority road, the warning sign *W٣١ Intersection with a road the user of which must give way* should be placed on the "main" road ahead of the intersection.

Location: The sign shall be placed as near as possible to the stop-line, and shall never be more than ٢٥ meters from the nearest edge of the intersecting road.

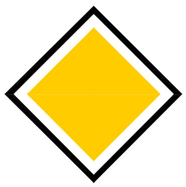
In the following cases the sign may be erected at both sides of the road:

- when the road has two or more lanes in the same direction
- when the design of the intersection admits two or more vehicles to approach side by side
- when there are poor sight conditions at the junction and
- in other cases when the STOP signs need to be emphasised

Design: The sign is octagonal with red background, white border and the text STOP in Arabic and English.

Size: Normal size is ٩٠٠ mm and large size is ١٢٠٠ mm. In special cases small size signs (height ٦٠٠ mm) may be used.

R٣. PRIORITY ROAD



This sign informs the road users they are driving on a Priority road and that drivers on intersecting roads must give way to all vehicles on the main road.

Primary roads, main streets inside urban areas, and other roads or streets with through traffic may be declared as Priority roads. Other streets inside urban areas may also be classed as Priority roads if, for instance, the general obligation to give way to traffic coming from the right does not work.

There may be two Priority roads intersecting each other. In such a case one of the Priority roads shall have STOP or Yield signs at the intersection. Advance warning of the STOP or Yield signs shall be given in those cases. (See example below)

Location: The sign shall be set up at the beginning of the road and repeated after each intersection. It may also be set up before the intersection.

Design: The Priority sign consists of a square with one diagonal vertical. The border of the sign shall be black with a white rim on a yellow background. The black border shall be ٢٥ mm wide and the white rim ٥٠ mm. The normal size is ٦٠٠ mm and the small size is ٤٠٠ mm.

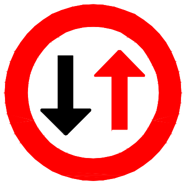
R٤. END OF PRIORITY ROAD

This sign indicates where a Priority road ends.

Location: The sign shall be placed at the ending point of the Priority Road and, when considered necessary, on both sides of the road.

Design: The sign face is the same as Priority Road but with a black band (١٠٠ mm wide) perpendicular to the lower left and upper right sides of the square.

R٥. PRIORITY FOR ONCOMING TRAFFIC



This sign indicates Priority for oncoming traffic on a narrow section of the road where two-way traffic is difficult or impossible. The sign means that entering the narrow section is prohibited if the narrow section is occupied by approaching traffic and entering will disturb approaching traffic flow.

The driver must have a clear view over the whole length of the section, both at night and by day.

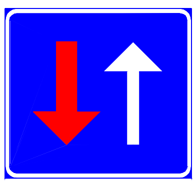
On the opposite direction sign R   Priority over oncoming traffic may be erected.

Note: The red arrow shall always be pointing upwards.

The sign may be provided with an additional panel showing a truck, if the prohibition applies only to trucks.

Size: Normally ٦٠٠ mm diameter.

R٦. PRIORITY OVER ONCOMING TRAFFIC



This sign indicates Priority over oncoming traffic on a narrow section of the road.

This sign must be used only if sign R٥ Priority for oncoming traffic is erected towards the opposing traffic.

Note: The red arrow shall always be pointing downwards.

Design: The sign is rectangular with blue background. The arrow pointing upwards is white and the other is red.

Size: Large size: ٦٠٠ x ٦٠٠ mm.
Normal size: ٤٠٠ x ٤٠٠ mm.
(Small size shall not be used.)

٣،٢. *Prohibitory or restriction signs*

Prohibitory and restrictive signs shall apply as from the place they are displayed until the point where a contrary sign is displayed, otherwise until the next intersection. If the prohibition or restriction should continue to be applied after the intersection the sign shall be repeated.

R٧. CLOSED TO ALL VEHICLES IN BOTH DIRECTIONS

This sign gives notification that all vehicular traffic is prohibited in both directions.

This sign may be used for example on streets where only pedestrians are allowed. In business areas there might be a need for delivery trucks to enter such streets. In such cases the sign may have an additional panel bearing the text "Except for delivery traffic". This text may be in Arabic only. If such an exception is applicable during business hours only, the time could also be added on the panel, for example; "٧ a.m. - ٧ p.m.". The sign R٧ may also be used on roads temporary closed for some reason such as landslide etc. In such cases the sign must be complemented with physical barriers and, at night, also with traffic control signals showing a red light. (See example below.)

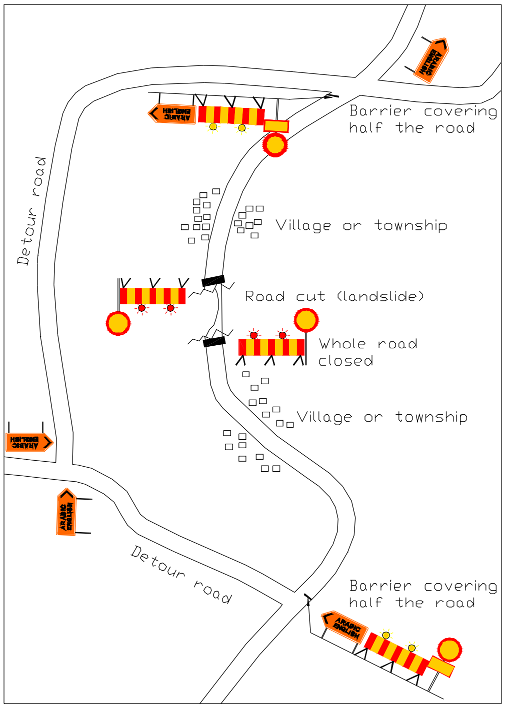
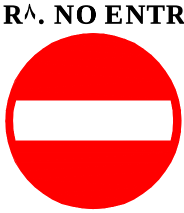


Figure ٣,٢a Example of signing a road closure due to some unforeseen event.

Note: Sign R٧ must be erected at the intersection where the detour starts. On the additional panel distance to the closure point shall be indicated. At night hazard lamps at the detour point shall show a flashing amber light, and lamps at the closure point, a steady red light.

R٨. NO ENTRY

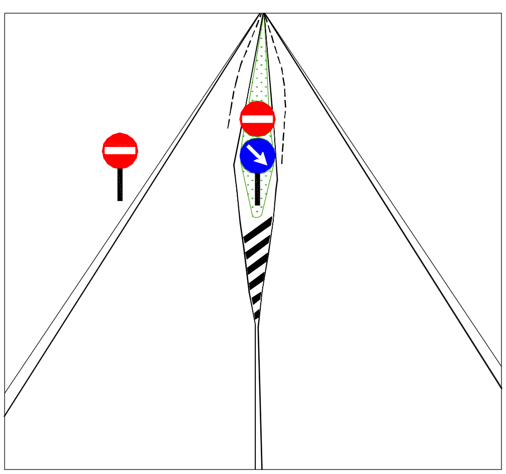


This sign gives notification that entry by all vehicles is prohibited.

The sign normally indicate that a street or a road has one-way traffic from the other direction.

The sign is often used to control traffic within urban areas where two-way traffic is not suitable or not wanted. The sign is also used where a divided road starts and the Mandatory sign, M٢,١ "Pass on the right side" is not enough to prevent drivers from entering the wrong direction.

Since the sign has a significant importance to road safety it should be clearly placed in the most appropriate position. It shall always be mounted at the right-hand side of the road. At some locations a supplementary sign on the left-hand side may be needed for additional emphasis.



It is very important to prevent drivers from entering the wrong direction. For that reason both the "No Entry" sign and the "Pass on right side" sign may be needed as well as the extra "No Entry" sign on the left-hand side of the opposing direction.

Figure ٣,٢b Beginning of a divided road section.

R٩. NO ENTRY FOR POWER-DRIVEN VEHICLES

This sign gives notification that entry is prohibited to all kinds of power-driven vehicles.

This sign may be used on roads and streets where cycles and animal drawn vehicles can be allowed but no other vehicles. Such a restriction may be suitable inside residential areas to prevent disturbing through traffic outside school buildings, hospitals etc.

On an additional panel exceptions can be given such as "Except traffic to the Hospital", "Except Residents", etc. If prohibition is valid only at night, an additional panel indicating that time is used. The text on such panel may be for example "١٠ p.m. - ٥ a.m.".

The text on the additional panels may be in Arabic only or in both Arabic and English.

R١٠. NO ENTRY FOR MOTOR CYCLES

This sign gives notification that entry is prohibited to motor cycles.

The sign may be used on roads and streets, mainly inside urban areas, where motor cycle traffic is considered unsuitable, for example due to disturbances from the noise of motor cycles.

R١١. NO ENTRY FOR CYCLES

This sign gives notification that entry is prohibited to cycles.

The sign may be used to prevent cyclists from entering roads and streets where it could be especially dangerous to cycle or where the cyclist would cause other traffic hazards or inconvenience to the motorists.

R١٢. NO ENTRY FOR MOPEDS

This sign gives notification that entry is prohibited to mopeds (cycle with a small petrol engine).

The sign may be used to prevent moped drivers from entering roads and streets where it could be especially dangerous for them, or where the presence of mopeds would cause other traffic hazards or inconvenience to other road users.

R١٣. NO ENTRY FOR GOODS VEHICLES

This sign gives notification that entry is prohibited to goods vehicles.

The prohibition may be used where heavy traffic is unsuitable, for instance in narrow city areas or where the noise or the air pollution from trucks would be especially disturbing.

By using an additional panel indicating a certain weight figure, for example ٥ tons, the prohibition is limited to trucks with the permissible maximum mass exceeding that figure.

Advance information of the prohibition may be given by putting an additional panel below the sign. That panel should indicate the distance from the advance information to the place the prohibition starts, for example ١,٣ km. The reason for giving such advance information is to give a truck driver the chance to choose an alternative route rather than be forced to turn back when reaching the prohibition.

R١٤. NO ENTRY FOR ANY POWER DRIVEN VEHICLE DRAWING A TRAILER OTHER THAN A SEMI-TRAILER OR A SINGLE AXLE TRAILER

This sign gives notification that entry is prohibited to power driven vehicles drawing a trailer other than a semi-trailer or a single axle trailer.

This sign may be used where heavy traffic is unsuitable, for instance in narrow city areas or where noise or air pollution from trucks would be especially disturbing.

Advance information of the prohibition may be given by putting an additional panel below the sign. That panel indicates the distance from the advance information to the place the prohibition starts, for example ١,٣ km. The reason for giving such advance information is to give drivers a chance to choose an alternative route rather than be forced to turn back when reaching the prohibition.

R١٥. NO ENTRY FOR ANY POWER DRIVEN VEHICLE DRAWING A TRAILER

This sign gives notification that entry is prohibited to any power driven vehicles drawing a trailer.

This sign may be used where heavy traffic is unsuitable, for instance in narrow city areas or where the noise or the air pollution from trucks would be especially disturbing.

By using an additional panel indicating a certain weight figure, for example ١٠ tons, the prohibition is limited to trailers with the permissible maximum mass exceeding that figure.

Advance information of the prohibition may be given by putting an additional panel below the sign. That panel indicates the distance from the advance information sign to the place the prohibition starts, for example ١,٣ km.

The reason for giving such advance information is to give drivers a chance to choose an alternative route rather than be forced to turn back when reaching the prohibition.

R١٦. NO ENTRY FOR PEDESTRIANS

This sign gives notification that entry is prohibited to pedestrians.

The sign is used only at places where it would be especially dangerous for pedestrians to enter, for instance on high speed multi-lane highways without sidewalks, other roads ending in such highways, or in other places where pedestrians are prohibited by law.

R١٧. NO ENTRY FOR ANIMAL-DRAWN VEHICLES

This sign gives notification that entry is prohibited to animal-drawn vehicles.

The sign may be used to prevent animal-drawn vehicles from entering roads and streets where such vehicles would be dangerous or inconvenient, for instance on high-speed multi-lane highways.

R١٨. NO ENTRY FOR HANDCARTS

This sign gives notification that entry is prohibited to handcarts.

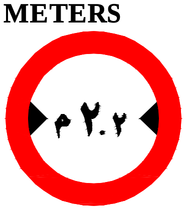
The sign may be used to prevent handcarts from entering roads and streets where presence of such vehicles would be dangerous or inconvenient, for instance on high speed multi-lane highways.

R١٩. NO ENTRY FOR POWER DRIVEN AGRICULTURAL VEHICLES

This sign gives notification that entry is prohibited to power driven agricultural vehicles.

The sign may be used to prevent power driven agricultural vehicles from entering roads and streets where such vehicles would be dangerous or inconvenient, for instance on high-speed multi-lane highways.

R٢٠. NO ENTRY FOR VEHICLES HAVING AN OVERALL WIDTH EXCEEDING ..... METERS



This sign gives notification that entry is prohibited to vehicles having an overall width exceeding the figures indicated on the sign.

This sign may be needed if a road has a narrow section, such as a narrow bridge or a narrow underpass, etc.

When this sign is erected at the prohibition point, provided that point is not at an intersection, it must also be erected at the latest intersection where it is possible to choose an alternative road. This advance information sign must have an additional panel indicating the distance from the sign to the place of the prohibition.

R٢١. NO ENTRY FOR VEHICLES HAVING AN OVERALL HEIGHT EXCEEDING ...... METERS

This sign gives notification that entry is prohibited to vehicles having an overall height exceeding the figures indicated on the sign.

This sign may be needed if the vertical clearance at some point of a road is less than permitted maximum vehicle height according to the Traffic Regulations, in other words, if the vertical clearance is less than ٤,٦ meters. Such places may be for example a bridge underpass.

The structure with clearance restriction shall be marked with obstacle markers. Such markers shall bear white and red bars, vertically oriented on the overhead markings and diagonally oriented on side markers. (See example below.) The markers must be provided with reflective sheeting to be clearly visible also in dark.

When this sign is erected at the prohibition point, provided that point is not at an intersection, it must also be erected at the previous intersection where it is possible to choose an alternative road. This advance information sign must have an additional panel indicating the distance from that sign to the place of the prohibition.

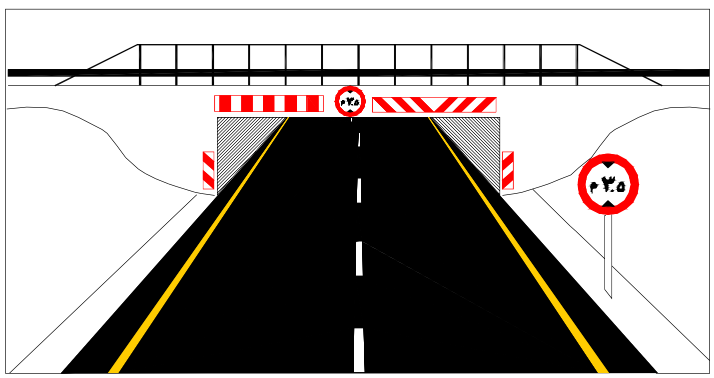


Figure ٣,٢c Signing of a bridge underpass with limited height clearance.

Note: In order to draw attention to the warning, side obstacle markers may be used even if the lateral clearance is not obstructed.

R٢٢. NO ENTRY FOR VEHICLES EXCEEDING ..... TONS LADEN WEIGHT

This sign gives notification that entry is prohibited to vehicles having a laden weight exceeding the figures indicated on the sign.

This sign may be used on roads where the permissible laden weight must be lower than the normal weight stated in the Traffic Regulations. The limitation may be caused by bridges with limited bearing capacity or temporary weakening of the road due to heavy rain, landslide, etc.

When this sign is erected at the prohibition point, provided that point is not at an intersection, it must also be erected at the previous intersection where it is possible to choose an alternative road. This advance information sign must have an additional panel indicating the distance from that sign to the place of the prohibition.

R٢٣. NO ENTRY FOR VEHICLES HAVING A WEIGHT EXCEEDING .... TONS ON ONE AXLE

This sign gives notification that entry is prohibited to vehicles having a weight on one axle exceeding the figures indicated on the sign.

This sign may be used on roads where the permissible axle load must be lower than the normal axle load stated in the Traffic Regulations. The limitation may be caused by bridges with limited bearing capacity or temporary weakening of the road due to heavy rain, landslide, etc.

When this sign is erected at the prohibition point, provided that point is not at an intersection, it must also be erected at the previous intersection where it is possible to choose an alternative road. This advance information sign must have an additional panel indicating the distance from that sign to the place of the prohibition.

R٢٤. NO ENTRY FOR VEHICLES OR COMBI-NATION OF VEHICLES EXCEEDING ..... METERS IN LENGTH

This sign gives notification that entry is prohibited to vehicles, or a combination of vehicles, with a length exceeding the figures indicated on the sign.

This sign may be used to prevent long vehicles from entering streets or roads where such vehicles would cause traffic disturbances or other problems.

R٢٥. NO LEFT TURN

This sign gives notification that left turns are prohibited.

The sign is erected at intersections where left turns are dangerous or unsuitable for some other reason. It should be placed just before the intersecting road. If considered necessary the sign is placed on both sides of the road.

On approaches with more than one lane, the sign should be placed on the left hand side of the carriageway and, if necessary, on the right hand side as well.

If certain vehicle categories are excluded from the prohibition this could be indicated on an additional panel, for instance: Except Taxi, Except Bus, etc.

R٢٦. NO RIGHT TURN

This sign gives notification that right turns are prohibited.

The sign is erected at intersections where right turns are dangerous or unsuitable for some other reason. It should be placed just before the intersecting road. If considered necessary the sign is placed on both sides of the road.

If certain vehicle categories are excluded from the prohibition this could be indicated on an additional panel, for instance: Except Taxi, Except Bus, etc.

R٢٧. NO U-TURN

This sign gives notification that U-turns are prohibited.

The sign is erected at intersections where U-turns are dangerous or unsuitable for some other reason. It should be placed just before the intersecting road or when applicable just before an opening in a median. If considered necessary the sign could be placed on both sides of the carriageway.

On approaches with more than one lane, the sign is placed on the left hand side of the road and, if necessary, on the right hand side as well.

If certain vehicle categories are excluded from the prohibition this could be indicated on an additional panel, for instance: Except Taxi, Except Bus, etc.

R٢٨. OVERTAKING PROHIBITED

This sign gives notification that overtaking of power-driven vehicles other than two-wheeled mopeds and two-wheeled motor cycles without side-car is prohibited.

The sign is used to prevent overtaking in such places where it is especially dangerous to overtake, for instance where there is a vertical curve or a horizontal curve and that curve is not visible to the drivers, or ahead of intersections with limited sight distance.

As a complement to this sign, the road marking centre line shall be solid.

R٢٩. OVERTAKING BY GOODS VEHICLES PROHIBITED

This sign gives notification that overtaking is prohibited to goods vehicles having a permissible maximum mass exceeding ٣,٥ tons.

The sign is used to prevent goods vehicles overtaking in such places where overtaking would be especially dangerous, for instance on long steep ascents where trucks need an abnormally long distance to overtake.

R٣٠. END OF PROHIBITION OF OVERTAKING

This sign gives notification that the prohibition of overtaking ceases to apply.

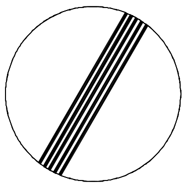
The sign shall always be used when sign R٢٨ “Overtaking Prohibited” is erected in order to mark the point where the prohibition ends.

R٣١. END OF PROHIBITION OF OVERTAKING BY GOODS VEHICLES

This sign gives notification that the prohibition of overtaking by goods vehicles ceases to apply.

The sign shall always be used when sign R٢٩ “Overtaking by Goods Vehicles Prohibited” is erected in order to mark the point where the prohibition ends.

R٣٢. END OF PROHIBITION OR RESTRICTION



This sign gives notification that all prohibitions notified by prohibitory signs for moving vehicles cease to apply. The sign may be used if a prohibition or restriction ceases to apply at another point than an intersection.

This sign is circular and have a white ground. It have no border and a diagonal band, sloping downward from right to left, consisting of five black parallel lines.

The sign may be placed on the reverse side of the prohibitory or restrictive sign intended for traffic coming in the opposite direction.

R٣٣. SPEED LIMIT

R٣٣,١

R٣٣,٢

These signs gives notification of a speed limit.

These signs are erected at:

- The point where speed limits change.
- Past major intersections.
- Country borders
- City limits
- Through speed limit zones, at intervals (١٠ km) as a reminder

On divided roads, signs that state a change of speed limit shall be erected on both sides, they may also be erected on both sides of other roads if considered necessary.

Speed limit shown on the sign shall be in multiples of ١٠ km/h.

When speed is reduced from, for example, ١٠٠ km/h to ٦٠ km/h the reduction shall be in steps with ٢٠ km/h intervals. Thus the signing shall be; ١٠٠ → ٨٠ → ٦٠. The distance between each sign should be approximately ١٠٠ m.

In order to determine the proper speed limit, the following factors are to be considered:

١. Road surface characteristics, shoulder conditions, grade, alignment, sight distance and other related geometric design elements
٢. ٨٥-percentile speed and pace speed
٣. Roadside development and culture
٤. Safe speed for curves hazardous locations
٥. Parking practices and pedestrian activities
٦. Accident experience for the recent ١٢-month period.

If one speed limit applies to trucks with a gross weight exceeding ٥ tons and another speed limit to private cars, the sign R٣٣,٢ should be used.

Speed limit starts to apply at the point where this sign is erected and ceases to apply at:

١. The “End of Speed Limit” sign
٢. The next main intersection
٣. The next sign that has different speed limit

Below are some examples of location of speed limit signs.

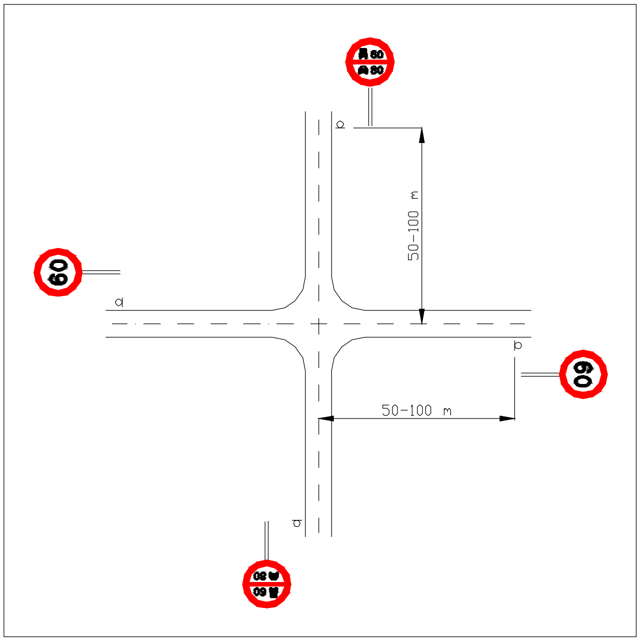


Figure ٣،٢d
Typical example of location of Speed Limit Signs at a rural intersection.

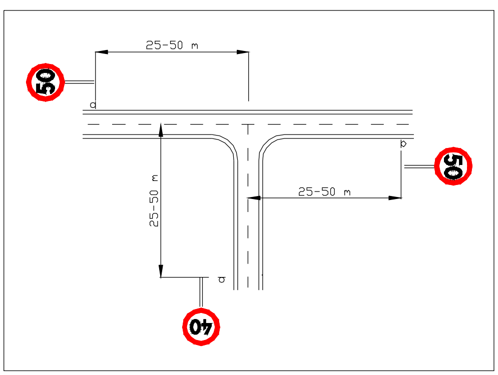


Figure ٣،٢e
Typical example of location of Speed Limit Signs at an urban intersection.

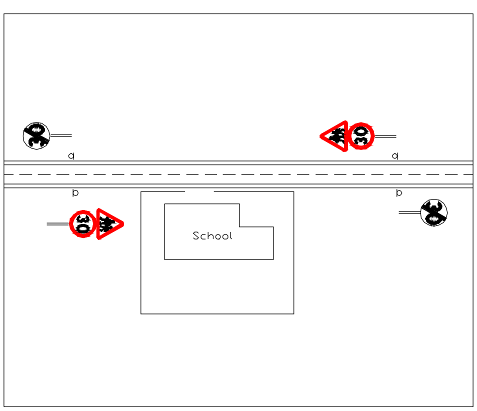


Figure ٣،٢f
Typical example of location of Speed Limit Signs outside a school place.

Note: Speed limit signs indicating the lower speed at school place and the warning signs may be erected on both sides of the street/road when considered appropriate.

R٣٤. END OF SPEED LIMIT

This sign gives notification that a speed limit on a certain section of a road ceases to apply and the former speed limit signed applies.

The sign shall be used only where a lower speed limit is imposed a short distance, for example outside a school.

R٣٥. USE OF AUDIBLE WARNING DEVICES PROHIBITED

This sign gives notification that use of audible warning devices is prohibited.

The sign may be erected where audible warning devices would be disturbing. Such places may be for instance, near hospitals, schools, nursery schools, within residential areas, etc.

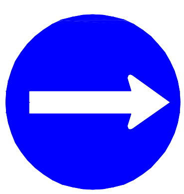
R٣٦. PASSING WITHOUT STOPPING PROHIBITED

This sign gives notification that drivers must stop for custom clearance.

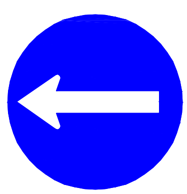
The sign may also be used at police checkpoints and military checkpoints. In those cases the word "Custom" must be replaced by the word “Police” or "Control".

٣,٣. *Mandatory signs*

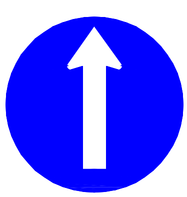
M١. DIRECTION TO BE FOLLOWED



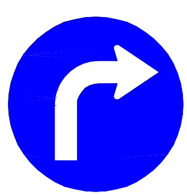
M١,١
To the Right



M١,٢
To the Left



M١,٣
Straight On



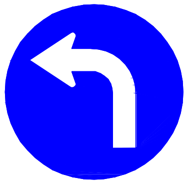
M١,٤
Turn Right

These signs gives notification that drivers are obliged to follow only the direction or directions indicated on the sign.

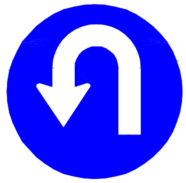
If, for example buses are excluded from the obligation, there shall be an additional panel with the text " Bus excluded". The additional panel shall have the same colour as the sign, blue with white border and letters.

Location: These signs shall normally be placed ahead of intersections. At some places these signs will be more visible if they are located at the intersection, for example if any of the signs M١,١ or M١,٢ are used at a T-intersection (see example below).

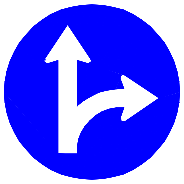
The sign M١,٦ Mandatory U-turn must be used only if there is a separate lane for U-turns (see example below).



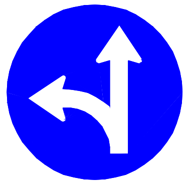
M١,٥
Turn Left



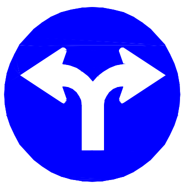
M١,٦
U-turn



M١,٧
Straight On or Turn Right



M١,٨
Straight On or Turn Left



M١,٩
Turn Right or Turn Left

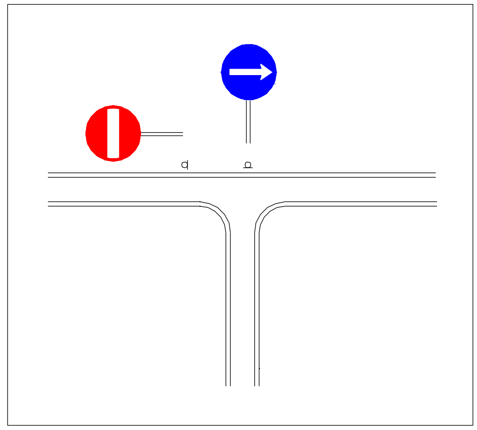


Figure ٣,٣a Typical example of the use of sign M١,١ at an intersection with a one-way street.

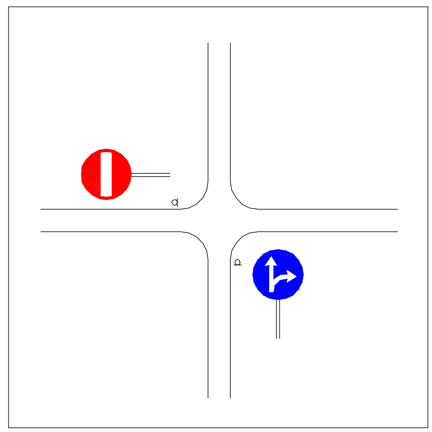


Figure ٣,٣b Typical example of the use of sign M١,٧ at a ٤-way intersection with a one-way street.

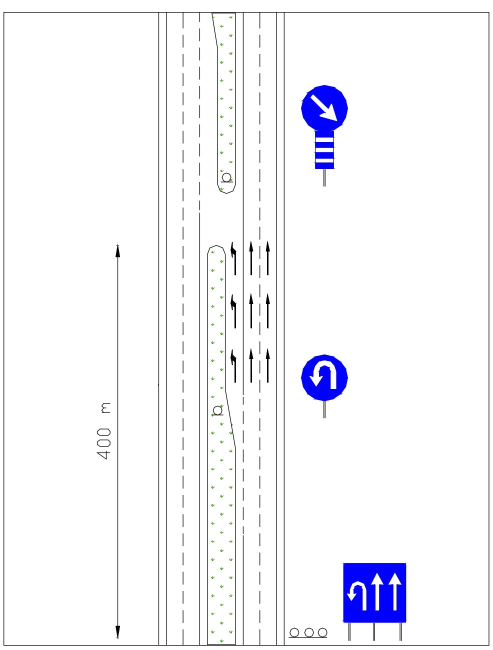


Figure ٣,٣c Typical example of the use of the sign M١,٦ at a median opening for U-turns.

Note: The Mandatory sign M١,٦ must not be used unless there is a separate lane for U-turning traffic. To give advance information of the U-turn lane, an informative sign for preselection of lanes shall be used. That sign is placed approximately ٤٠٠ m ahead of the opening in the median at rural areas. That distance may be ٥٠-٢٠٠ m within urban areas

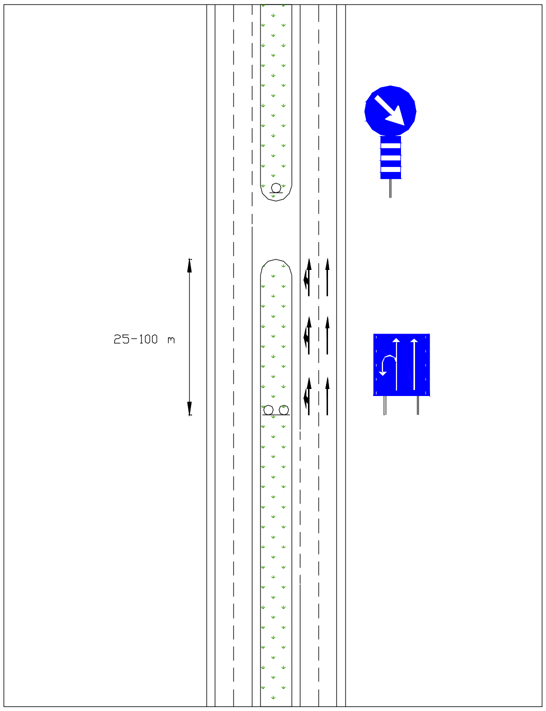
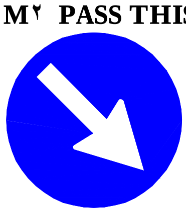


Figure ٣,٣d Typical example of signing of U-turn opening in a median.

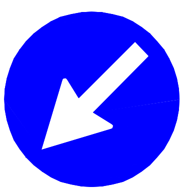
Note: U-turns without separate lane as this example shows may be allowed within urban areas only. No mandatory sign for U-turn (M١,٦) shall be used since the left lane may be used by drivers going straight as well.

U-turns from a lane not separated from straight traffic should only be used in exceptional cases since the traffic safety is heavily jeopardised by such a traffic solution.

M٢ PASS THIS SIDE



M٢,١ Pass to the Right Side



M٢,٢ Pass to the Left Side

M٢,٣ Pass to Either Side

These signs gives notification that drivers are obliged to pass the sign on the side(s) indicated by the arrow(s) on the sign.

Sign M٢,٣ should not be used to indicate the gore area on a divided road with exit ramps.

If, for example buses are excluded from the obligation, there shall be an additional panel with the text " Bus excluded". The additional panel shall have the same colour as the sign, blue with white border and letters.

In order to improve the visibility of the sign, a Signpost Marker may be attached to the signpost below the sign. Signs erected outside urban areas without street lightning are to be complemented with such Signpost Marker. Signpost Marker shall be rectangular with blue and white horizontal bars (see example below).

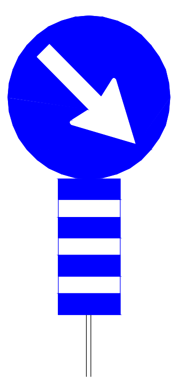
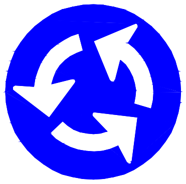


Figure ٣,٣e Sign "Pass this side" with signpost marker.

Sign M.٢,٣ Pass either side shall **not** be used to mark the gore area between a main road and an exit ramp. On such places a "Gore Area Marker" is used.

M٣. COMPULSORY ROUNDABOUT



This sign shows the prescribed direction of movement through a roundabout.

This signs is placed at every entrance of the roundabout and should always be combined with sign R١ Give Way.

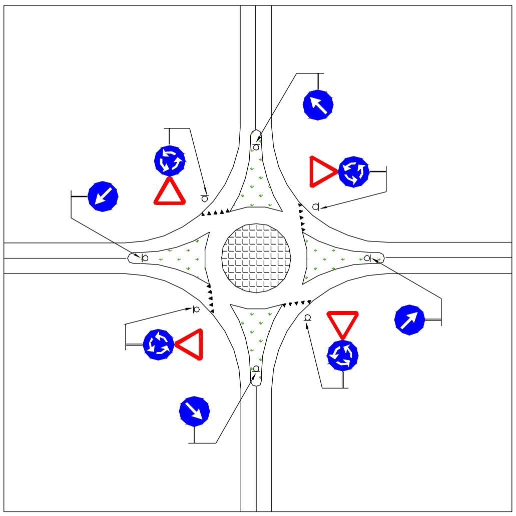


Figure ٣,٣f Typical example of signing of a roundabout.

Note: Signs R١ Give Way and M٣ Compulsory Roundabout shall be placed not more than ٢٥ meters from the Yield Line.

M٤. COMPULSORY CYCLE TRACK

This sign gives notification that cyclists are obliged use the cycle track and that other vehicle are not entitled to use this track.

This sign may be erected at the beginning of the special track constructed solely for cyclists. The sign is repeated after each intersection with roads or streets.

M٥. COMPULSORY FOOT-PATH

This sign gives notification that pedestrians must use the path at the entrance of which it is placed. It also notifies other road users that they are not allowed to use the path.

The sign shall only be used at foot-paths separated from any adjacent street or road. Thus it must not be used on sidewalks.

M٦. COMPULSORY TRACK FOR RIDERS ON HORSEBACK

This sign gives notification that riders on horseback must use the track at the entrance of which it is placed. The sign also notifies other road users, except for pedestrians, that they are not allowed to use the track.

This sign may be used when, for example, a separate track for riders is constructed in a park.

M٧. COMPULSORY MINIMUM SPEED

This sign gives notification that vehicles using the road at the entrance to which it is placed shall travel at not less than the speed specified, the figure shown on the sign.

The sign can apply to one or more lanes on a road with two or more lanes in the same direction. The sign shall in such case be located above the actual lane or lanes or included in I.٤. L Changing Signs.

M٨. END OF COMPULSORY MINIMUM SPEED

This sign gives notification that that the compulsory minimum speed imposed by sign M٧ is no longer in effect.

The sign is normally posted on the right hand side of the road even if sign M٧ is placed above the road.

٤. INFORMATIVE SIGNS

~ ⅔ *General.*

Directional signing is very important as to road safety. A driver who does not know the way to his destination is very dependent on the directional signing. Good directional signing shall, in a clear and easy manner, provide the driver with the information he or she needs for the choice of way.

A driver who hesitates and is uncertain which direction to follow, will create hazards by breaking, stopping, reversing, and maybe even, in the worst of cases, turn his vehicle in the wrong direction on a one-way road.

In order to achieve a good directional signing some main principles are of vital importance to follow:

- All directional signing must be well **planned**. Routing plans should be established Kingdom wide for the primary and secondary road network. Those plans should be provided by the Ministry of Public Works and Housing in Amman.
- There must be **continuity** in the choice of destinations on each sign. A destination once signed must appear on the following signs all the way to that destination.
- The number of destinations on a sign must be **limited**. There should never be more than four destinations on the same sign or sign combination. That means all directional signing shall be based on the assumption that drivers have a road map and have a general idea about the route to choose. The number of destinations for each direction must be maximum two.
- If, for some reason, the number of destinations cannot be limited to maximum four, the information may be given on two signs after each other. In such case the Arabic text should be on one sign and English on the other.
- All destinations given on direction signs should be **confirmed** by a Place Identification sign unless the location of the destination is obvious without signs.
- Similar places or situations should be **consistently** signed. The design of the sign should also be the same for similar locations.
- All signs should be manufactured according to the **standard drawing** or to **special drawings**.

Colour: All direction signs shall be blue and the legends and borders shall be white. Other road signs or symbols included in a direction sign shall be of the original colour(see paragraph ٤,٩ for these signs and symbols).

Location: Advance information should normally be given ahead of interchanges and intersections on Primary Roads. The distance between an Advance Direction Sign and the interchange/ intersection varies from ٢ km on Freeways and Expressways and similar roads to ٤٠٠ m on other roads. Inside urban areas the advance information may be given from ٥٠ m to ٢٠٠ m ahead of the intersection.

At the interchange/intersection Direction Signs should be erected at the point where drivers have to turn or in the very close vicinity of that point.

٤,١. *Advance Direction Signs*

I١. DIAGRAMMATIC SIGN

Diagrammatic signs are used when volume of the turning traffic is high or whenever advance information is considered important to traffic safety.

Diagrammatic sign shows a graphic view of the interchange or intersection ahead. When appropriate, a road number sign may be displayed on the signs combined with destinations. Other symbols or road signs may also be displayed on the signs.

At the bottom of the signboard the distance from the sign to the interchange or intersection shall be displayed. The distance shall be given in Latin numerals and letters at the left hand side of the sign and in Arabic at the right hand side.

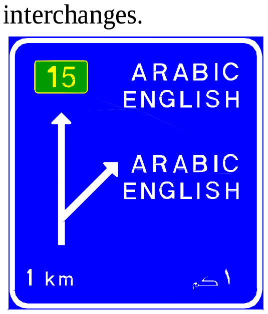
Instead of giving the distance at the bottom of the signboard, the distance may be indicated on an additional panel to the Diagrammatic sign.

Diagrammatic signs normally are erected at the right hand side of the road.

Each sign must never contain more than four destinations. However, Direction symbols like for example Airport or Trucks may be added.

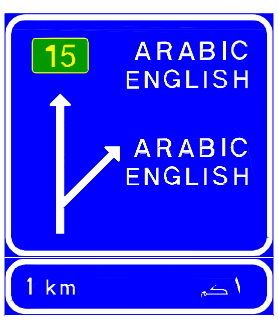
Diagrammatic signs shall always be followed by a Direction sign at the interchange or intersection. At interchanges that direction sign is normally an Exit Direction sign while at intersections a Stack type sign or a Chevron type sign is used.

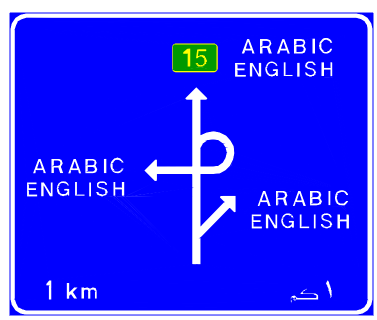
Î ⅔ Diagrammatic sign for interchanges.



On Freeways, Expressways and highways with similar standard, advance signing shall always be given. On such roads Diagrammatic sign I١,١ shall be used provided the exit is designed with a deceleration lane, continuing in an exit ramp.

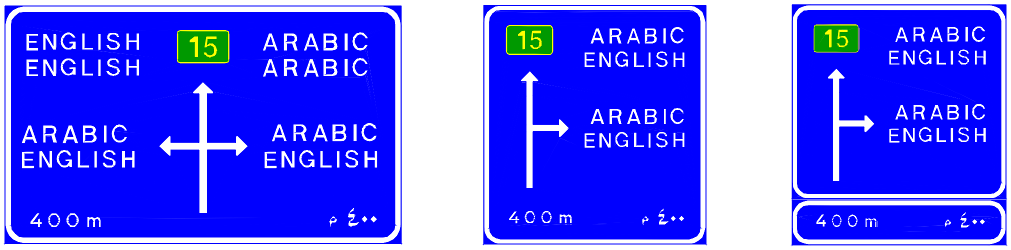
On Freeways, Expressways and similar roads with high-speed traffic, the Diagrammatic sign should be doubled and placed ١ km and ٥٠٠ m ahead of the exit.





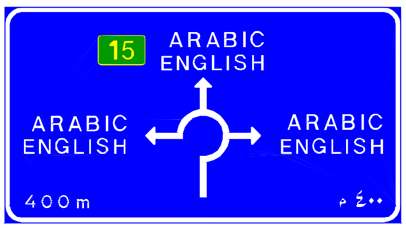
I ⅔ Diagrammatic signs for intersections

These signs are normally placed approximately ٤٠٠ m ahead of the intersection.

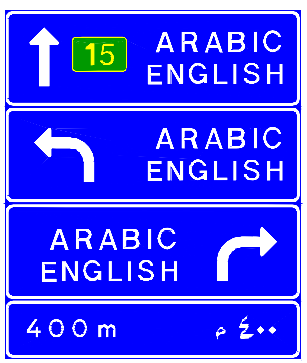


I ⅔ Diagrammatic sign for roundabouts.

This sign is normally placed approximately ٤٠٠ m ahead of the roundabout on rural roads and ٥٠ – ٢٠٠ m in urban areas.



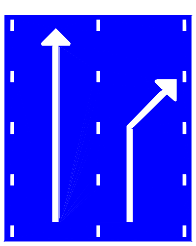
I٢. STACK TYPE SIGN



Stack type signs may be used as an advance direction sign when traffic volume is rather low but advance information nevertheless is needed.

When Stack type sign is used as an advance direction sign it shall have an additional panel showing the distance from the sign to the intersection. That distance is normally approximately ٤٠٠ meters outside urban areas. Inside urban areas the distance may be ١٠٠ - ٢٠٠ meters. If placed closer to the intersection (٢٥ - ١٠٠ m) the sign will be a Direction sign and without additional panel showing the distance (see I٧).

I٣. LANE PRESELECTION SIGNS

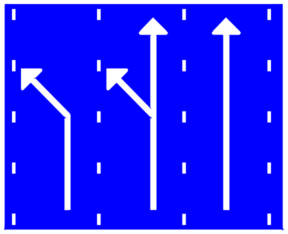


I٣,١ Two lanes, one straight on and one for right turn

These signs are intended to inform drivers of the use of lanes at intersections. The number and direction of lanes are shown in schematic form.

These signs shall show all existing lanes at the intersection approach. If considered appropriate, road number signs may be included above the arrows.

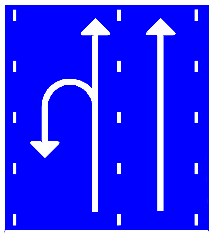
These signs may be used both inside urban areas and in rural areas.



I٣,٢ Three lanes, one straight on and one for left turn. The centre lane is for both

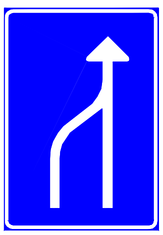
Lane Preselection signs shall never be used at intersections with more than three lanes. If there are more than three lanes, an overhead Lane Assignment sign is used.

Location: Lane Preselection signs are erected at the point where drivers have to choose lanes for their preceding through an intersection. They are normally erected on the right hand side of the road. However, if appropriate, they may be erected on left hand side or on both sides.



I٣,٣ Two lanes, the right for straight on and the left for straight on and U-turn

I٤. LANE CHANGING SIGNS

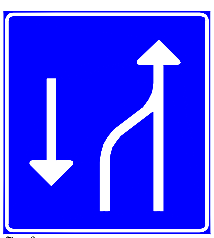


I٤,١ Reduction of one lane on a one-way road

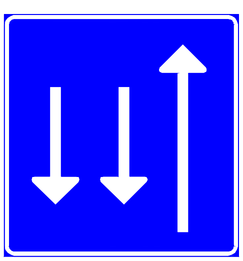
These signs are used to give information of changes of lane configuration ahead. They are not used at interchanges or intersections. Other designs than those illustrated may be used as well to fit a specific situation.

In a two-way traffic road, lane or lanes allocated to oncoming traffic shall be shown on the sign as well.

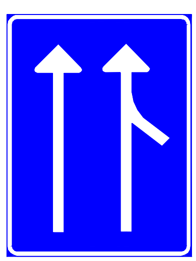
By including other road signs on the arrow shaft, different regulations for different lanes can be shown, for example different speed limit for different lanes. Prohibition for trucks to use a lane may also be shown in that way. Signs, included on the arrows, shall have their ordinary design and colour, but may be smaller than a separately erected sign.



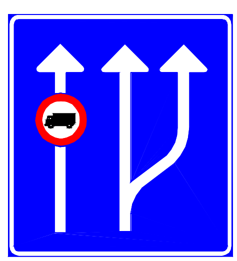
Ĩ ⅔ Reduction of one lane on a two-way road



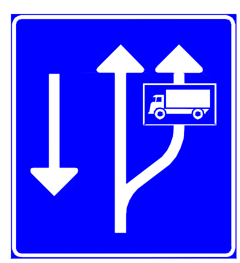
I٤,٣ Information that opposing traffic has two lanes



I٤,٤ Information of a merging lane from right



I٤,٥ Addition of one lane at the right hand side. Trucks are prohibited from using the extreme left lane



I٤,٦ Climbing Lane on a two-way road. Trucks must use the right lane

٤,٢. *Direction Signs*

I٥. LANE ASSIGNMENT SIGN

Lane Assignment signs are used to give information of the destination of the different lanes in a multi-lane intersection or interchange.

Lane Assignment signs shall always be mounted above the road. Number of arrows on the sign shall be the same as the number of lanes.
Arrows on the signboard shall, if possible, be located over the centre of the lane that the arrow refers to or shall at least be within lane limits, marked by road markings.

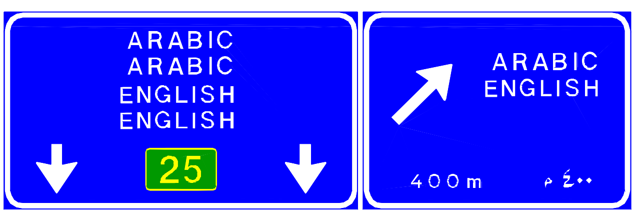
The arrows on the signboard shall be pointing downwards except for exiting lanes where the arrows shall be pointing upwards and be leaning towards the direction of the exit.

Decisions about erection of overhead signs are based on the following criteria:

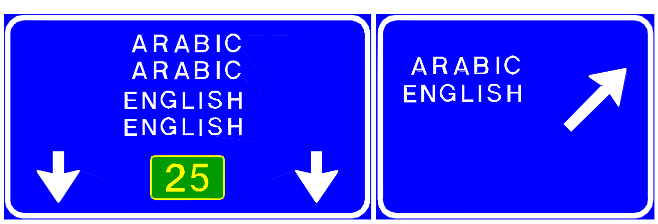
* More than three approaching lanes
* High traffic volume
* Complex interchange design
* Restricted sight distances
* High speed traffic
* Large percentage of trucks
* Insufficient space for ground mounted signs

Existence of one or more of these conditions does not automatically justify the use of overhead signs. An engineering study must be done at each separate location to judge the need and to see if some other measure may reduce the need of overhead signs.

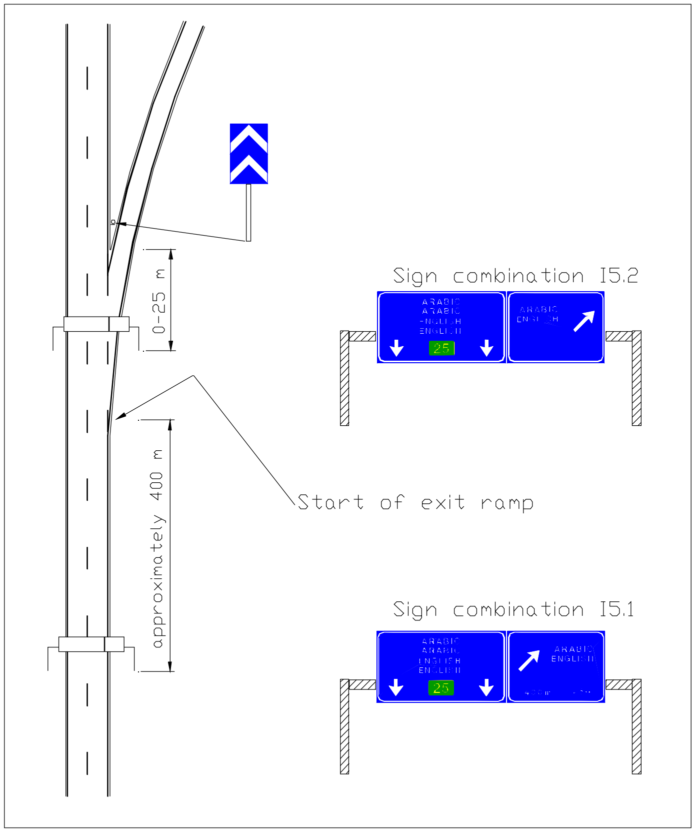
Below are some examples of the design of Lane Assignment signs.



I٥,١ Two through lanes, the right lane is also for exiting traffic. Arrow on the sign for exit direction shall be placed to the left on the sign so it is located above the lane. These signs are placed approximately m in advance of the exit.



I٥,٢ Two through lanes and one separate lane for exiting traffic. Arrow on the sign for exit shall be placed to the right on the sign to show it is a separate lane. These signs are placed where the exiting lane has full width.

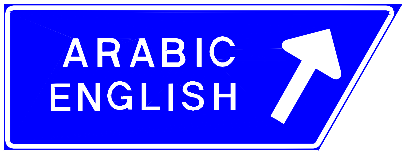


Figure˜ ⅔⅔a Typical example on use of Lane Assignment signs Ĩ ⅔ and Ĩ ⅔

I٥,٣ Three through lanes, the extreme right is also for exiting traffic.

I٥,٤ Intersection inside urban area. Two through lanes, the extreme left also for left turning traffic. A separate lane for right turning traffic. The sign for left turning traffic shall have the arrow to the right so it is above the common lane for straight on traffic. The sign for the right turning traffic shall have the arrow to the right so it is located over the separate lane for right turning traffic.

I٦. EXIT DIRECTION SIGN



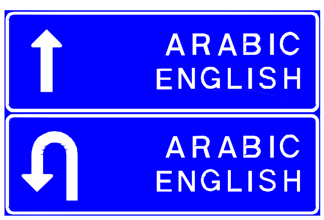
Exit Direction sign shall be used at exits from expressways and other roads with deceleration lanes. The sign shall repeat the destination(s) given on the Advance Direction sign for the diverging direction.

Exit Direction sign shall be placed at the beginning of the taper. It is normally placed at the right hand side of the road but may also be placed on a gantry or cantilever over the starting point of the deceleration lane.

The sign shall have an upwards slanting arrow at the right edge of the sign.

There should not be more than two destinations on Exit Direction signs.

I٧. STACK TYPE SIGNS



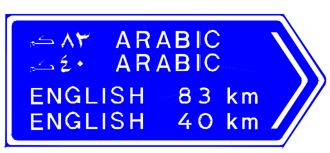
Stack Type signs may be used at intersections where placement of the sign close to the intersecting roads is impossible or unsuitable for some reason. This sign replaces the Chevron Type sign and is not erected further from the intersection than ١٠٠ meters.

The sign may show only two directions whenever appropriate.

Stack Type sign used as a direction sign may be most useful inside urban areas. According to the previous section the sign may be used as an Advance Direction sign when provided with an additional panel indicating the distance to the intersection.

Stack type signs may also be used to show destinations that can be reached after a u-turn. Advance information of this sign may be given by sign I٧ showing the same arrows and destinations but with an additional panel showing the distance to the u-turn.

I٨. CHEVRON TYPE SIGN



Chevron Type sign may be used at level intersections to show the destinations of the intersecting road.

On the sign the distances to the destinations may be displayed. Road number or any other appropriate symbol may also be included on the sign.

This sign shall be placed in close vicinity of the road it concerns. That normally means at most ٢٥ meters from the nearest edge of the intersecting road.

There should not be more than two destinations on a Chevron Type sign.

Below are some typical examples of directional signing of exits on different types of interchanges. The examples are showing ground mounted signs which are the most common types of signs for the directional signing on those road types.

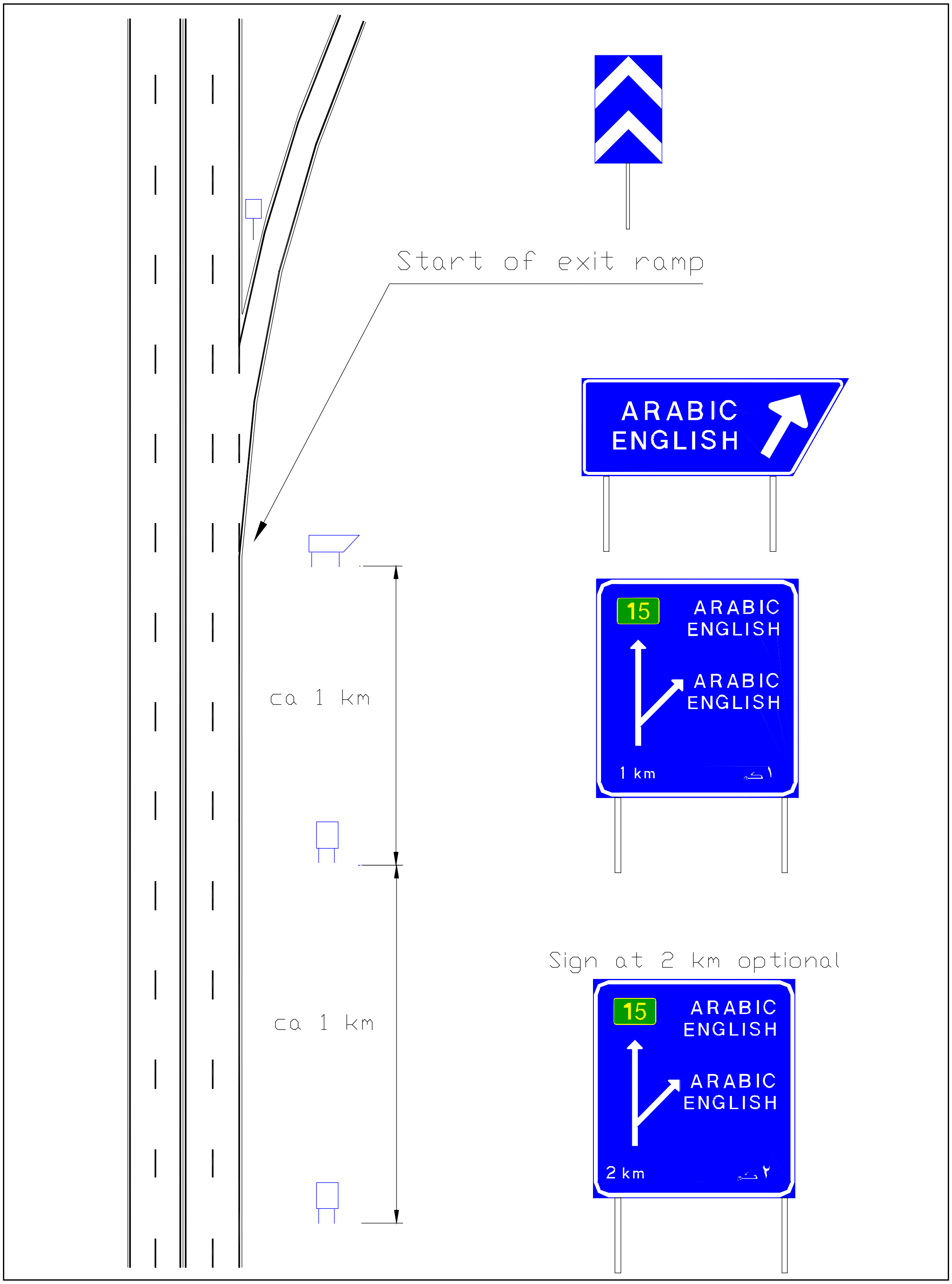
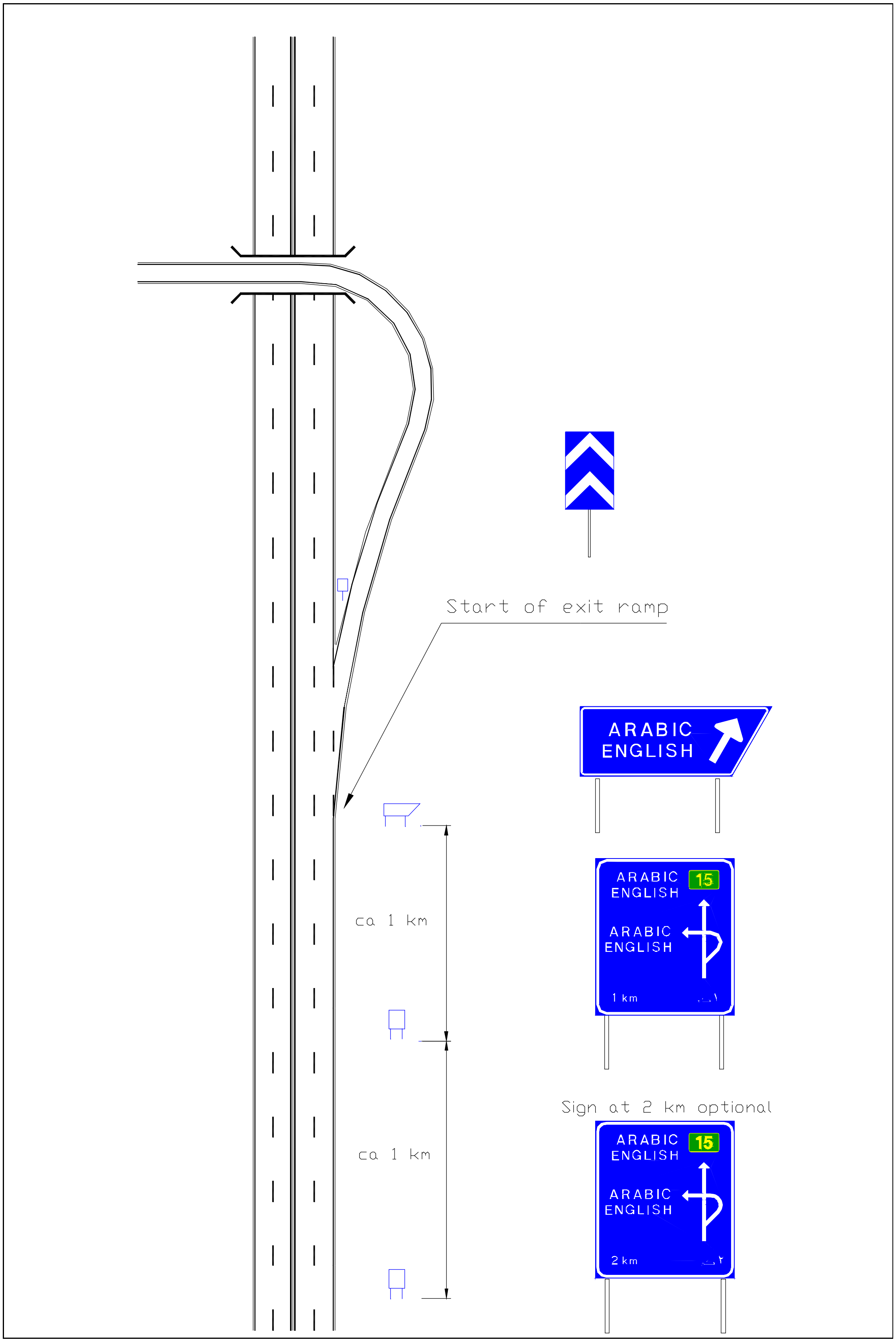


Figure ⅔ ⅔ b Directional signing of a common exit on divided road.



Figure˜ ⅔⅔c Directional signing of an exit for left turning traffic.

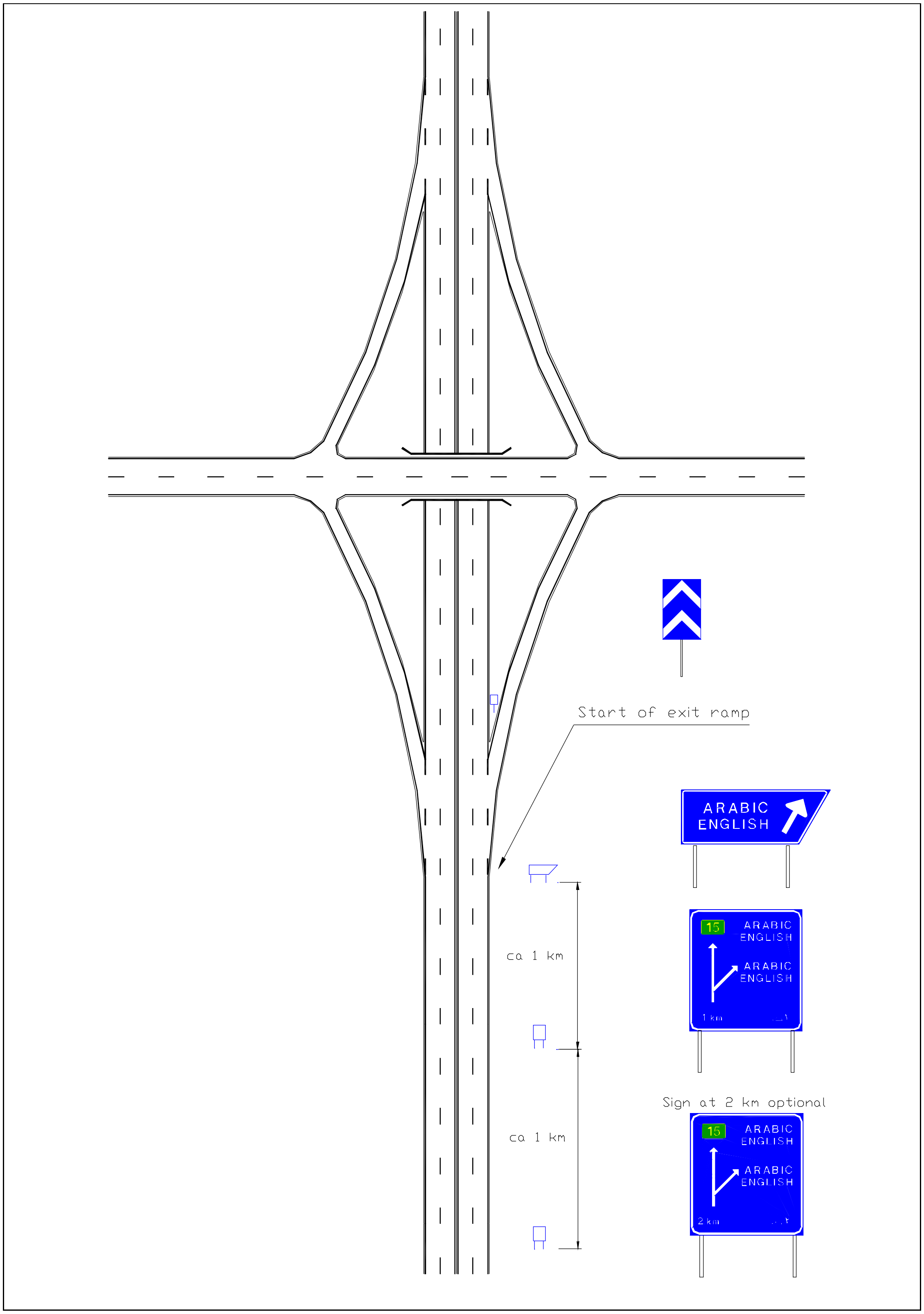
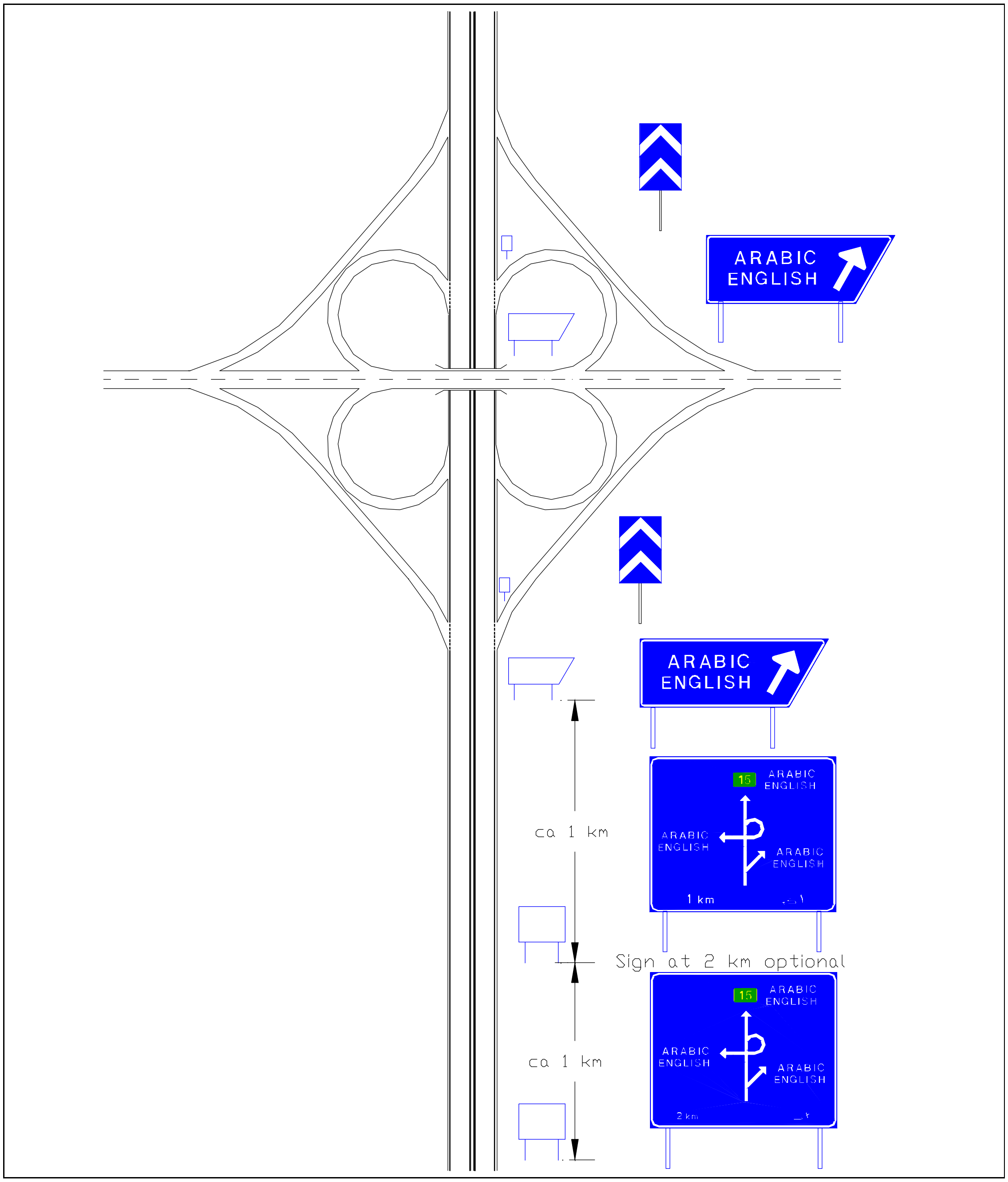


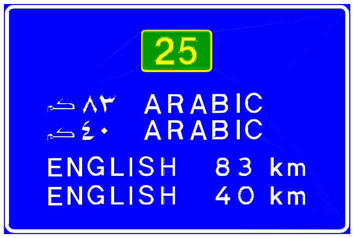
Figure ˜ ⅔ ⅔ d Directional signing of exit on a diamond interchange



Figure˜ ⅔⅔e Directional signing of exit
on a cloverleaf interchange

٤,٣. *Confirmatory Signs*

I٩. CONFIRMATORY SIGN



This sign shall be used on roads to confirm the destinations previously given on a direction sign. They are not normally used inside urban areas unless there is a Freeway, Expressway or similar. These signs are important since they reduce hesitation among drivers whether they have chosen the correct road or not.

These signs are normally erected approximately ٢٠٠ meters after the intersection, or if there is an acceleration lane, ٢٠٠ meters from the end of that lane. The Confirmatory sign may be repeated along the road at approximately every ١٠ - ٢٠ km.

These signs may contain main destination of the road and the nearest city and distances to the destinations. On primary roads the road number may be displayed on the sign.

The sign should contain maximum four destinations. If more destinations are needed another sign with those destinations may be erected approximately ٢٠٠ m after the first sign.

I١٠. PLACE IDENTIFICATION SIGN

This sign is used to confirm destinations given by direction signs. If the destination is a city, the sign is erected at the city limits. It may also be used to confirm direction signing to a city area, such sign is erected at the limit of that area.

The sign may also be used to inform road users of places of special interest such as rivers, wadis, bridges, etc.

The sign may be used to show administrative border of cities, municipalities, etc.

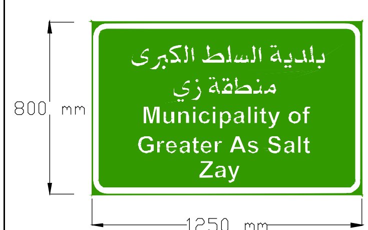
To inform drivers of greater municipalities and municipalities within greater municipalities the signs shown below may be used. The signs may be used to inform driver they are leaving a municipality as well.

Design. The colour of these signs shall be green with white text and border. The measures of the signs shall be; width ٢٥٠٠ mm and the height ١٢٥٠ mm for the larger signs and width ١٢٥٠ mm and height ٨٠٠ mm for the smaller.

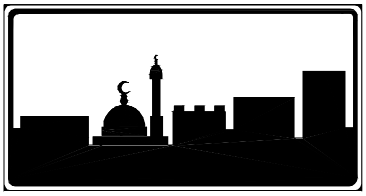
I١٠,١ GREATER MUNICIPALITY

I١٠,٢ LEAVING GREATER MUNICIPALITY

I١٠,٣ SMALLER MUNICIPALITY



I١١. BUILT-UP AREA



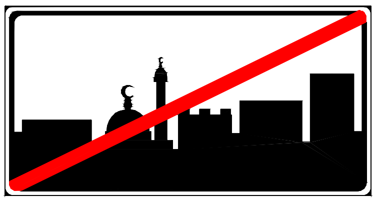
This sign is used to indicate the beginning of a built-up area.

If no Speed Limit sign is erected, the Built-up Area sign also means a speed limit as required by regulations. The speed limit indicated by the Built-up Area sign is in force to the point where the End of Built-up Area sign is erected or to a point where a Speed Limit sign indicate another speed limit.

Design. The Built-up Area sign shall be rectangular with black border and symbol on a white background. Additional panel to this sign shall be white with black text.

The name of the built-up area may be shown on the sign above the symbol. In such case the name shall be in both Arabic and English. An additional panel showing “Built-Up Area” in Arabic may be added to the sign.

I١٢. END OF BUILT-UP AREA



This sign is used to indicate the end of a built-up area.

Since the Built-up Area Sign means a speed limit as required by regulations, the End of Built-up Area sign indicates that the speed limit is the same as before the built-up area unless some speed limit sign indicates otherwise.

Design. The End of Built-up Area sign shall have the same design as the Built-up Area sign but with a red oblique bar running from the upper right edge to the lower left edge.

٤,٤. *Road Number Signs*

I١٣. ROAD NUMBER SIGNS FOR THE INTERNATIONAL ROADS IN THE ARABIC MASHREQ

Road Number Signs for the international roads in the Arabic Mashreq shall be erected along all roads included in the “Agreement on international roads in the Arabic Mashreq”. According to the Agreement the signs must be repeated at intervals of approximately ١٠ km on freeways and expressways and every ٢٠ km on other roads. The road number should be indicated before and after every point of entry or exit to or from the international road, whether at interchanges or at-level intersections.

Design: The sign is rectangular with blue background and white numerals and border. The numerals shall be English numerals.

If the Road Number Sign is included in some other Directional Sign, the size of the Road Number Sign shall be adjusted to fit the other Directional Sign.

I١٤. ROAD NUMBER SIGN, PRIMARY ROADS

Road Number Signs for Primary roads should be erected along all Primary roads in the Kingdom except those roads included in the “Agreement on international roads in the Arabic Mashreq”.

Road Number Signs may be posted solely or as a symbol on other Direction signs. A preferable way of showing road number is to combine the Road Number Sign with a Speed Limit Sign after each major intersection or to include the Road Number Sign in the Confirmatory Signs.

Road Number Sign is normally posted solely or included in some other sign, at approximately every ١٠ km along Primary roads.

Design: The sign is rectangular with green background and yellow numerals and border. The numerals shall be English numerals.

If the Road Number Sign is included in some other Directional Sign, the size of the Road Number Sign shall be adjusted to fit the other Directional Sign.

I١٥. ROAD NUMBER SIGN, SECONDARY ROADS

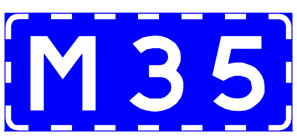
Road Number signs for Secondary roads may be erected along the Secondary road network wherever it is considered appropriate. The sign is displayed in the same way as described above for Primary roads. The distance between Road Number Signs on Secondary roads is approximately ٢٠ km.

Design. The sign is a circle with green background and yellow numerals and border. The numerals shall be English numerals.

If Road Number Sign is included in some other Directional Sign, its size shall be adjusted to fit that Directional Sign.

Road number signs with broken border may be used to indicate that the road with the shown number will intersect or divert ahead. Such road number (with broken border) should normally appear together with destination(s) that belongs to that road number.

I١٣-١. ROAD NUMBER SIGNS ON ROAD LEADING TO AN INTERNATIONAL ROAD IN THE ARABIC MASHREQ



I١٤-١. ROAD NUMBER SIGN ON ROAD LEADING TO A PRIMARY ROAD

I١٥-١. ROAD NUMBER SIGN ON ROAD LEADING TO A SECONDARY ROADS

٤,٥. *Pedestrian Crossing*

I١٦ PEDESTRIAN CROSSING

This sign indicates the location of a pedestrian crossing.

The sign is not used if the speed limit is more than ٦٠ km/h. Furthermore this sign is only used in combination with road marking for Pedestrian Crossing.

At pedestrian crossing not located at intersection, the sign should be of large size. Otherwise normal size sign is used.

This sign is not normally used at pedestrian crossings with traffic light signals.

If traffic-calming devices are installed at a pedestrian crossing, speed limit signs indicating a speed less than ٦٠ km/h must be erected ahead of the traffic-calming devices.

Before any installation of a pedestrian crossing a careful engineering study must be run. The purpose with a pedestrian crossing is to provide a safe place for crossing a road or street, but if criteria such as location, traffic volumes and speed, number of crossing pedestrians etc. are not considered, the pedestrian crossing may create hazards to the pedestrians instead of safety. The following graph may be used when judging the need for pedestrian crossings. However, such things as speed limit, sight distance, percentage of children among pedestrians, etc. must also be considered.

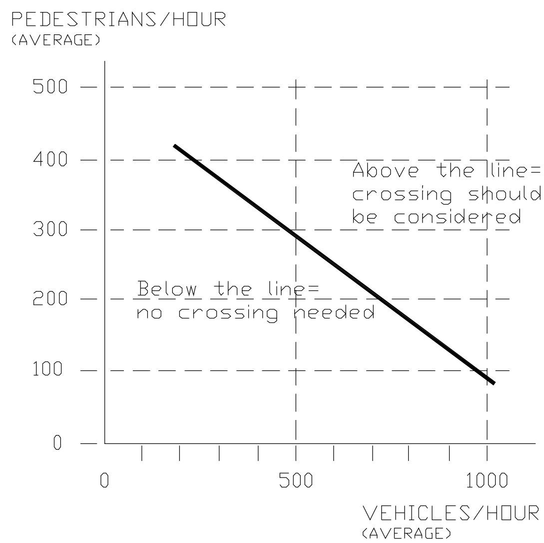


Figure ٤,٥a Criteria for estimating the need of pedestrian crossings.

Either vehicle flow or pedestrian flow during peak hour shall be used as dimensioning value. The flow shall be measured along a section, ٣٠ - ٥٠ meters in each direction from the location of the pedestrian crossing.

Below are some typical examples of the use of sign I١٦ Pedestrian Crossing.

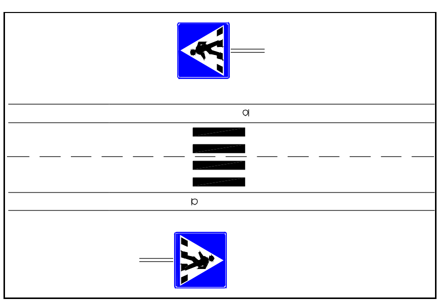


Figure ٤,٥b Road section with one lane in each direction. Each sign is double sided (facing both directions).

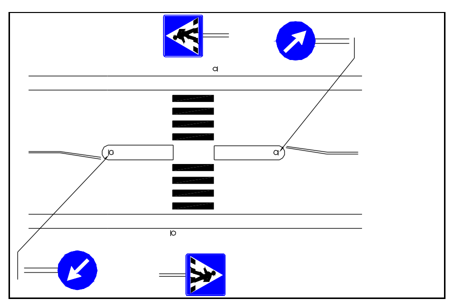
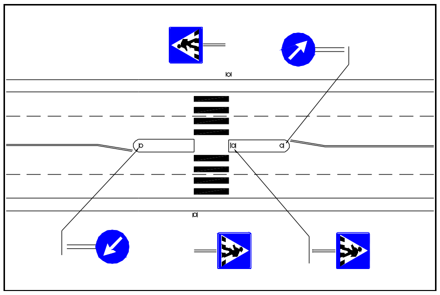


Figure ٤,٥c Road section with one lane in each direction and a centre island. The Pedestrian Crossing signs are double sided (facing both directions)



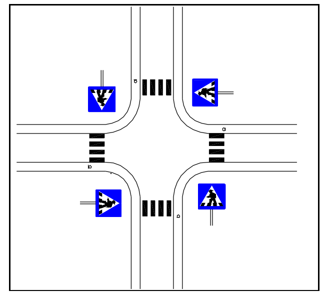


Figure ٤,٥d Road section with two lanes in each direction and a centre island.
The Pedestrian Crossing signs are double sided (facing both directions).

Figure ٤,٥e Four-legged intersection with pedestrian crossings on all roads. The sign is needed only on the right hand side of each approach road.
All Pedestrian Crossing signs are double sided (facing both directions).

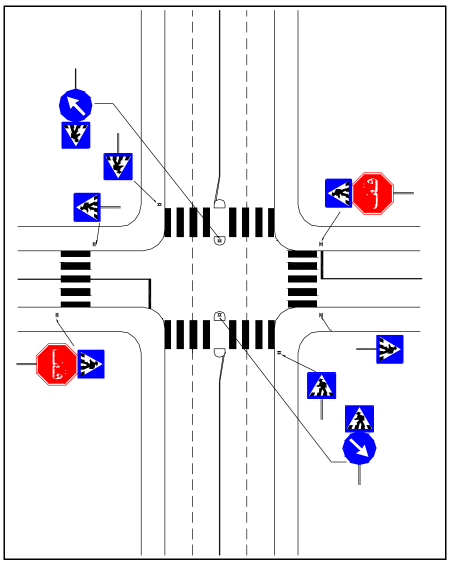


Figure ٤,٥f Intersection with multi-lane approaches in two directions and two lane approaches in the other directions. Note that at multi-lane approaches, Pedestrian Crossing signs are needed on the right hand side and on the traffic island but not on the left hand side.
All Pedestrian Crossing signs are double sided (facing both directions).

٤,٦. *Useful Information Signs*

I١٧. ONE-WAY ROAD

This sign indicates that the road it is erected at, is a one-way road.

It is used only if sign R.٨. No Entry is erected at the entrance to the road/street.

This sign informs drivers that they must follow the direction of the arrow when entering a one-way road, either at the beginning of the road or from any intersecting road.

This sign is placed approximately parallel to the one-way road and in such positions that is visible to drivers concerned. It is erected at a height approximately ٠,٨ - ١,٢ meters above the road in order to be more visible to drivers.

Below are some examples on placement of sign I١٧.

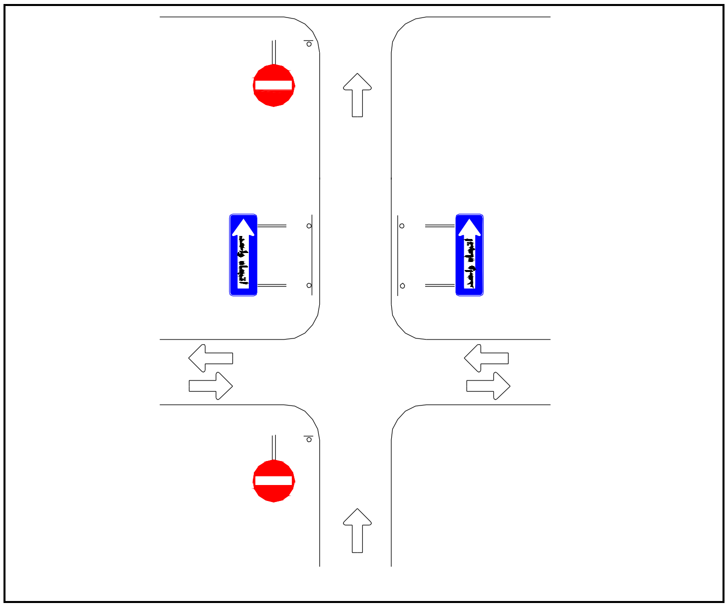


Figure ٤,٦,١a A two-way traffic road intersecting a one-way road.

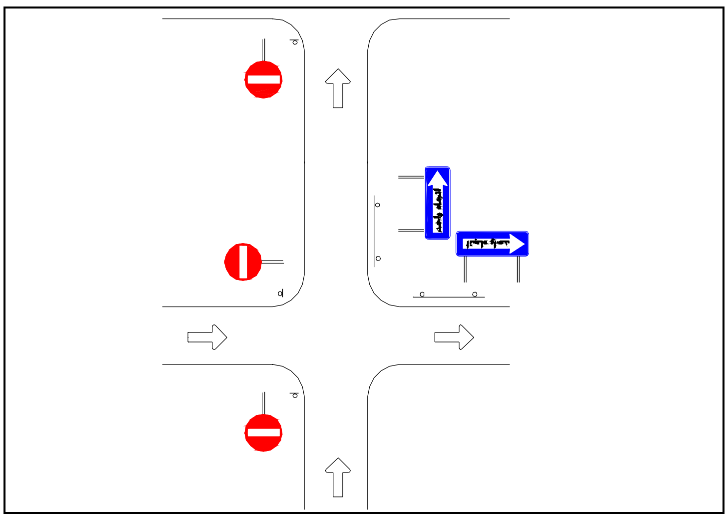


Figure ٤,٦,١b Intersection of two one-way roads.

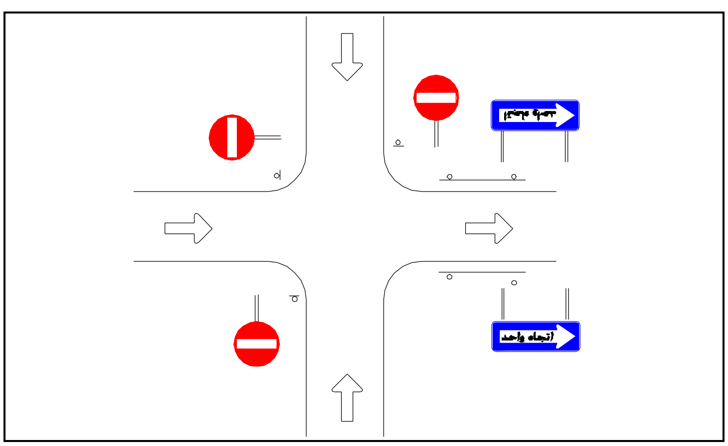
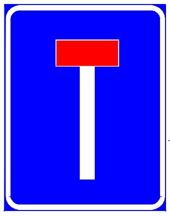


Figure ٤,٦,١c Intersection of a one-way road and two other one-way roads approaching from each direction.

I١٨. NO THROUGH ROAD



This sign gives information of a no through road or dead end street.

I١٩. HOSPITAL

This sign gives information of a hospital.

It is used to show the location of ٢٤ hours emergency hospitals.

Signing to hospitals without emergency facilities and to First Aid stations is done with sign I٣١ First Aid.

The sign may be erected separately and, when needed, provided with an additional panel with an arrow showing the direction to the facility.

The sign may also be included as a symbol on other directional signs.

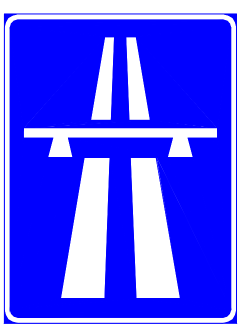
I٢٠. ADVISORY SPEED

This sign is used to show the speed at which it is advisable to drive if circumstances permit and if the driver is not required to comply with a lower speed limit specific to his category of vehicle.

The advisory speed indicated on the sign shall be chosen after a careful engineering study of the actual place. It is essential that the recommended speed is safe for all categories of vehicles during normal weather conditions.

The reason for the use of Advisory Speed sign must be obvious to drivers, for example a very sharp bend with restricted sight distance. Thus there is no need for sign indicating end of advisory speed.

I٢١. FREEWAY



This sign is used to indicate the start of a freeway.

It is erected at both sides of the carriageway.

Highways may be classed as freeway if they have no level intersections and the opposing traffic directions are separated by a physical barrier. The physical barrier may be a dividing strip, a guard-rail, a concrete barrier, etc.

Access to a freeway is possible only at specially arranged entrances, and exit is possible only at exit ramps.

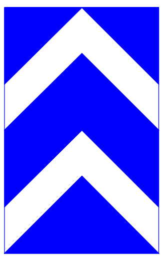
If those criteria mentioned above are not fulfilled, the road is not classified as a freeway.

I٢٢. END OF FREEWAY

This sign is used to indicate the end of an expressway.

It is erected at both sides of the carriageway.

I٢٣. GORE AREA MARKER



The Gore Area Marker is used to indicate the gore nose at an exit from freeway, expressway and other roads with similar standards.

The marker shall be blue with white bars pointing upwards from the vertical edges towards the center of the sign. Normal size shall be ١،٢٨٠ x ٢٠٤٠ mm and small size ٦٤٠ x ١،٠٢٠ mm.

The figure below shows an example of the use of the Gore Area Marker.

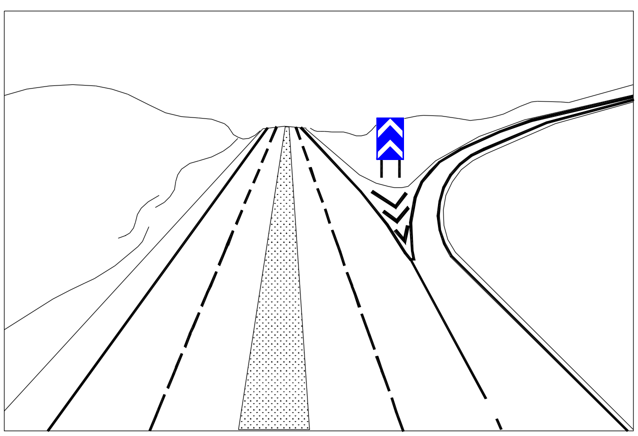


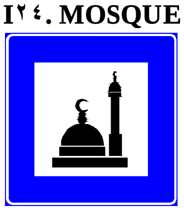
Figure ˜ ⅔ a Gore Area Marker at a freeway (similar) exit.

٤,٧. *Service Facility Signs*

The signs I٢٦ – I٣١ should be set up only on roads on which such facilities are rare.

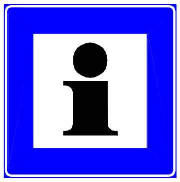
Service Facilities signs may be separately mounted or included as a symbol on other directional signs and, when needed, provided with an additional panel with an arrow showing the direction to the facility or the distance. Such panel shall have blue background, white border and white arrow.

I٢٤. MOSQUE



This sign indicates a Mosque.

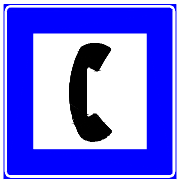
I٢٥. INFORMATION OFFICE



This sign indicates information office or other place where tourist information or other information directed to road users can be achieved.

Information signs on buildings, tourist sites, etc can be of other colours.

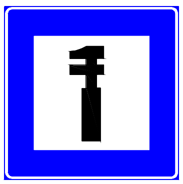
I٢٦. TELEPHONE



This sign may be used to indicate a telephone, available to road users.

If the telephone is an emergency telephone, an additional panel with the symbol SOS may be placed beneath the sign.

I٢٧. WORKSHOP



This sign indicates a car repair shop.

I٢٨. PETROL STATION

This sign indicates a petrol station.

The sign shall be used only to show petrol stations providing ٢٤-hour service.

I٢٩. REFRESHMENT

This sign indicates a place where refreshments are available.

It may be used to show the way to a cafe, a tea-room, snack bar, etc. where only light food such as sandwiches, coffee, tea etc. is served.

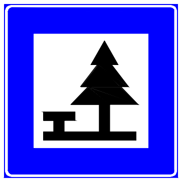
I٣٠. RESTAURANT

This sign indicates a restaurant.

I٣١. HOTEL

This sign indicates a hotel.

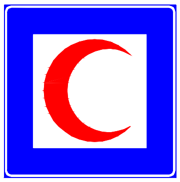
I٣٢. PICNIC SITE



This sign indicates a picnic site.

To be approved for signing, the picnic site should be equipped with tables, chairs, and dustbins. The site must be separated from the road in such a way that children can play safely and so that traffic does not disturb people using the picnic site. The entrance and the exit must be arranged in a safe way.

I٣٣. FIRST-AID STATION



This sign indicates a First Aid Station.

It is used to show the location of a small hospital or a physician where first aid can be obtained. The sign is not used for signing to hospitals with emergency treatment, sign I١٩ Hospital is used for signing to such hospitals.

I٣٤. YOUTH HOSTEL

This sign indicates a Youth Hostel.

A Youth Hostel is a type of accommodation for younger persons who want to find simple and cheap accommodation during travel. This kind of accommodation normally demands something from the guests, for instance bringing their own bedclothes and towels. The guests are normally also requested to make up the rooms before leaving the hostel.

I٣٥. CARAVAN SITE

This sign indicates a Caravan Site.

To use this sign, a camping site must be equipped to host caravans, providing for example; special parking spaces, facilities to connect the caravan to electrical net, possibility to empty sanitary equipment from the caravan, filling of water tanks, etc.

I٣٦. CAMPING SITE

This sign indicates a Camping Site.

To use this sign, a camping site must be equipped with service facilities for the campers, such as showers, toilets, etc.

I٣٧. INDUSTRIAL AREA

This sign indicates an industrial area.

If the area is a free zone an additional panel with the text “Free Zone” may be added to the sign.

٤,٨. *Recreational and Cultural Areas (RCA) signs*

٤,٨,١. General.

Signs for RCA may be used for signing of ancient monuments, historical places, buildings of special interest, etc. These signs may also be used for signing of areas, specially prepared for recreation such as parks, rest areas, forests with foot-paths etc.

Signing to RCA shall normally start close to the objects or areas. If RCA is located within a city, municipality, village etc that is shown on ordinary direction signs (blue/white), the signing to RCA shall normally not start until at the border/limits of such city, municipality, village etc.

Signing to RCA must not be done indiscriminately since too much signs will lessen the significance of the special type of objects that shall be shown on this special type of signs. In order to limit the number of RCA that is shown on the RCA signs, a nationwide plan containing all objects should be done. Such plan should be made in co-operation with Ministry of Tourism.

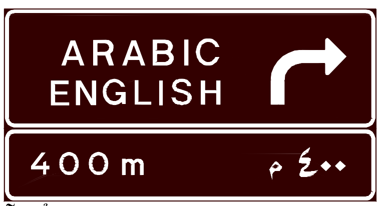
Approved symbols illustrating RCA or cities or other places with RCA may be included in all types of RCA signs.

The colour of RCA signs shall be brown with white text, border, and symbols.

RCA signs and symbols may be included in other direction signs.

The design shall be as described below and according to standard drawings for each sign.

I٣٨. ADVANCE DIRECTION RCA SIGN



I٣٨,١ STACK TYPE SIGN, RIGHT DIRECTION

I٣٨,٢ STACK TYPE SIGN, LEFT DIRECTION

The Stack type sign may be used as an advance direction sign for showing the name and direction of RCA areas or facilities.

When the Stack type sign is used as an advance direction sign it shall have an additional panel showing the distance from the sign to the intersection. That distance is normally approximately ٤٠٠ meters outside urban areas, and ١٠٠ - ٢٠٠ meters inside urban areas.

When an advance direction sign is erected a Chevron Type sign at the intersection shall follow it.

I٣٩. CHEVRON TYPE RCA SIGN

The Chevron Type (RCA) is erected at intersections in order to show the direction to RCA. If the distance from the sign to the destination is more than ١ km, the distance shall be displayed on the sign as well.

I٤٠. PLACE IDENTIFICATION SIGN

This sign is erected at the place of RCA. If there has been directional signing to the place, a Place Identification sign is erected as a confirmation to drivers, unless the destination is self-evident.

The sign may be used to mark places that have not been shown by an advance RCA sign.

٤,٩. *Direction Symbols.*

In all Directional Signing use of symbols, instead of text or as a complement to text, is considered. Symbols are normally easier to comprehend than text. They must however be simple in design and easy to understand, otherwise they may confuse drivers.

Number of symbols used on the sign shall therefore be limited. No other symbols than those described in this Manual are used without approval of MPWH. In addition to the symbols described below, some prohibitory signs may be included as symbols on Direction signs. In such cases these signs shall have their ordinary design and colour, size may be adjusted to fit the size of the Direction sign.

Prohibitory signs included on Direction signs shall only provide information. Thus the sign that has been included on the Direction Sign must also be erected separately at their normal location e.g. at the roadside.

The following Prohibitory signs may be used as symbols on Direction signs:
- R٧. Closed to all vehicles in both directions
- R٨. No entry
- R٩. No entry for power-driven vehicles
- R١٣. No entry for goods vehicles
- R١٤. No entry for any power driven vehicle drawing a trailer other than a semi-trailer or a separately axle trailer
- R١٥. No entry for any power driven vehicle drawing a trailer
- R٢٠. No entry for vehicles having an overall width exceeding ..... meters
- R٢١. No entry for vehicles having an overall height exceeding ..... meters
- R٢٢. No entry for vehicles exceeding ..... tons laden weight
- R٢٣. No entry for vehicles having a weight exceeding .... tons on one axle
- R٢٤. No entry for vehicles or combination of vehicles exceeding ..... meters in length
- R٢٥. No left turn
- R٢٦. No right turn
- R٢٧. No U-turn

I٤١. AIRPORT

This symbol is used for signing of an Airport.

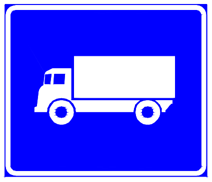
The sign may be posted on its own with an additional panel having a direction arrow, or may also be included on a Direction Sign, indicating the direction to the Airport.

When appropriate, the name of the airport may be indicated together with the symbol.

Design: Rectangular with white border and symbol on a blue background.

If the symbol is included on a Direction Sign, no separate border around the symbol is needed.
Whenever this sign is combined with an arrow the symbol should be turned in the same direction as that arrow to avoid confusion to drivers.

I٤٢. TRUCKS



This symbol is used for directing trucks.

This sign is normally used to detour trucks, for example from:

- a low bridge,
- a road section with low bearing capacity,
- a hospital,
- a residential area,
- etc.

This sign may be posted on its own with an additional panel having a direction arrow. The symbol on the sign may also be included on a Direction sign, indicating the direction to be followed by trucks.

Design: Rectangular with white border and symbol on a blue background.

If the symbol is included on a Direction sign, no separate border around the symbol is needed.

Whenever this sign is combined with an arrow the symbol should be turned in the same direction as that arrow to avoid confusion to drivers.

٤,١٠. *Emergency Escape Ramps*

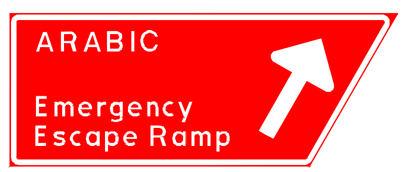
Emergency escape ramps are ramps constructed in steep downhill providing an opportunity for heavy vehicles to be stopped in a relatively safe way if the brakes fail. Such ramps are constructed so that vehicles are stopped, either by an arrester bed, consisting of loose gravel or sand, or by turning the ramp uphill.

Signing of Emergency Escape Ramps must be clear and visible, especially the exit point, so the driver is prepared to turn into the exit when it comes. On the other hand there should not be too much signs since they concern only a very few drivers and others should not be distracted by these signs. The following signs are used as advance information signs and exit direction signs:

I٤٣,١. ADVANCE INFORMATION SIGN FOR EMERGENCY ESCAPE RAMPS, ١ km

I٤٣,٢. ADVANCE INFORMATION SIGN FOR EMERGENCY ESCAPE RAMPS, ٥٠٠ m

I٤٤. EXIT DIRECTION SIGN FOR EMERGENCY ESCAPE RAMPS



At the summit a map type sign may be placed to inform drivers of the dangerous section ahead and where escape ramps are located. Such sign should preferably be placed at a parking space where drivers can study the map without obstructing the road.

The text size on the signs above shall be in accordance with chapter ١,٦,١ Letters on Informative Signs.

Below is a typical example of signing of Emergency Escape Ramp:

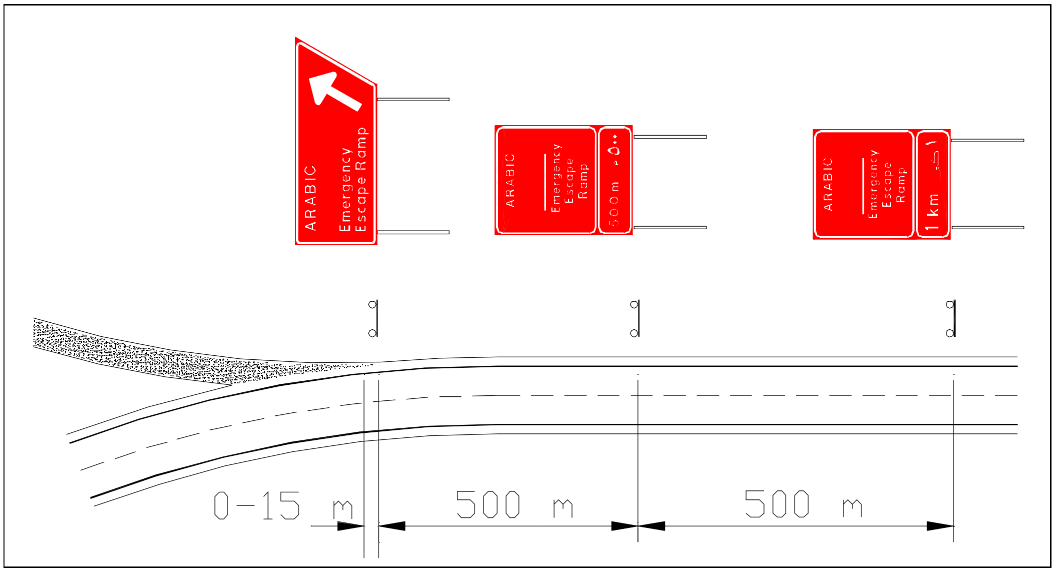


Figure ⅔ a Signing of Emergency Escape Ramp

٤,١١. *Truck Weighing Stations*

٤,١١,١. General

Use of road signs at Truck Weighing Stations where weighing is compulsory for all trucks must be clearly visible and easy to comprehend for truck drivers. It must also be obvious for other road users that the signing does not concern them. The signing must be legally correct so that drivers who does not follow the instructions can be sued and punished. Thus there must be both regulatory signs and informative signs.

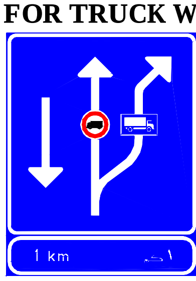
٤,١١,٢. Road Signs

I٤٣. FIRST ADVANCE INFORMATION FOR TRUCK WEIGHING STATION

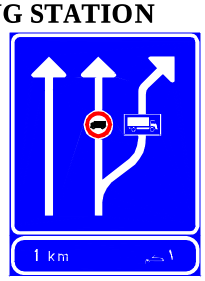
This sign gives information of a truck weighing station where weighing of all trucks is compulsory.

The sign shall be placed approximately ٢ km in advance of the exit to the station. It shall be located on the right hand side of the road but may be located on the left hand side as well on a divided road.

I٤٤. SECOND ADVANCE INFORMATION FOR TRUCK WEIGHING STATION



I٤٤,١ ADVANCE INFORMATION ON TWO-WAY ROADS

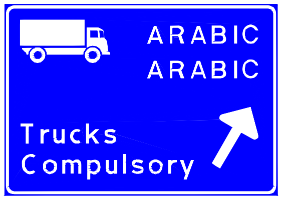


I٤٤,٢ ADVANCE INFORMATION ON DIVIDED ROADS

This sign is a lane changing sign giving information that trucks are directed onto the exit to the truck weighing station and that they are prohibited to continue along the main road.

Sign I٤٤,١ shall be used on two-way roads and sign I٤٤,٢ on divided roads.

I٤٥. EXIT SIGN FOR TRUCK WEIGHING STATION



This sign gives information of the exit to truck weighing station and that truck drivers shall follow the direction of the arrow.

The sign shall be located at the beginning of the exit road or, when appropriate, at the beginning of the taper of an exit ramp.

Below are examples on signing of Truck Weighing Stations:

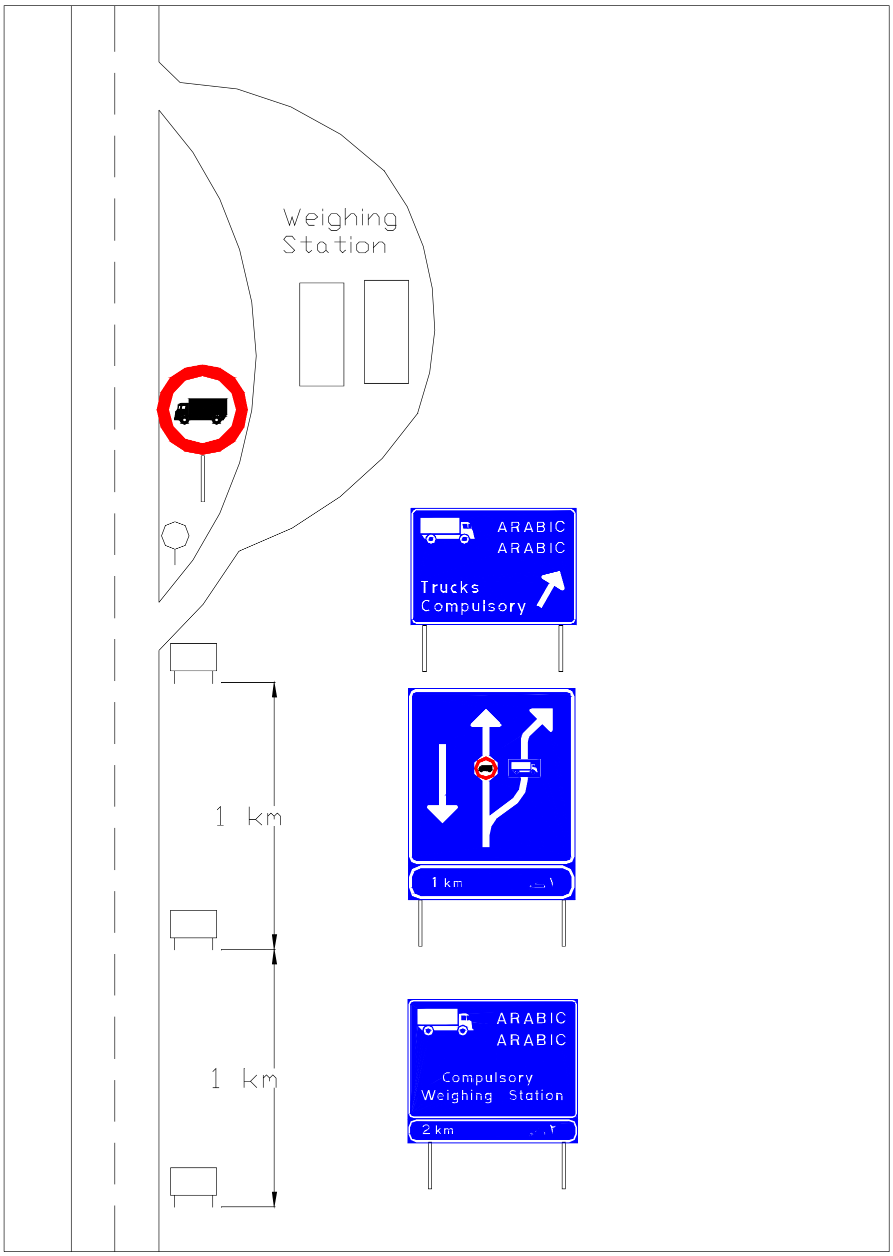


Figure ˜ ⅔ a Signing of a truck weighing station on a two-way road.

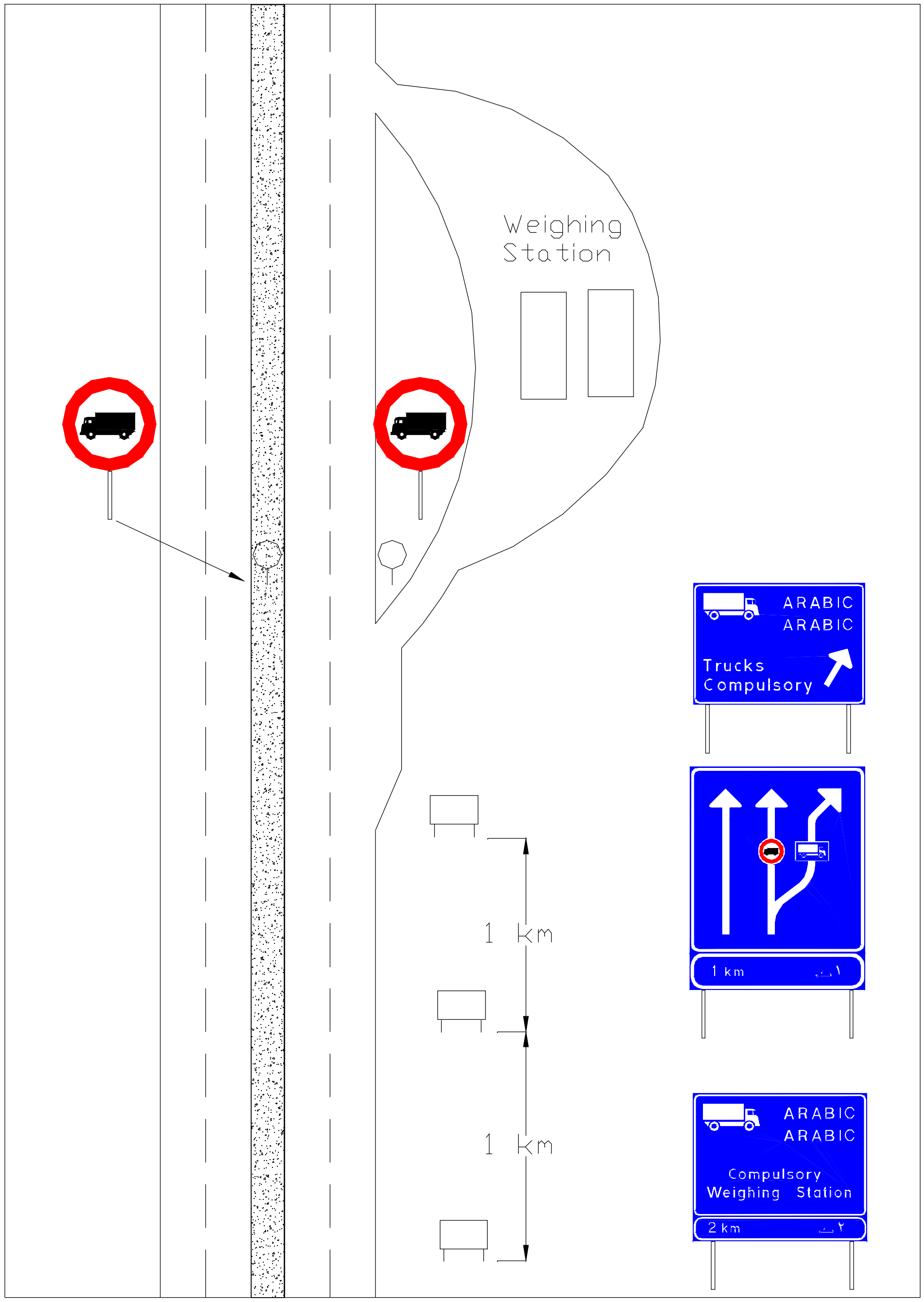


Figure ٦/٣ b Signing of a truck weighing station on a divided road.

Note: In both examples above the prohibitory sign R١٣ No Entry For Goods Vehicles are placed immediately beyond the exit. On divided roads sign R١٣ shall be placed on both sides of the carriageway.

٥. STANDING AND PARKING SIGNS

٥,١. *General*

Standing and Parking signs are used to control and regulate areas where drivers may park or stop their vehicles. On some streets and roads parked vehicles may create hazards or other problems. Parking in such places shall be prohibited or regulated. In other places even stopping may create unacceptable hazards or problems. In such places standing and parking shall be prohibited.

On the other hand, when drivers need to park their cars in areas where parking may create problems, Parking Places must be provided, otherwise they may park or stand where parking and standing is prohibited. Those Parking Places must be identified by parking signs.

A careful engineering study must always be made before any of these signs are used. Excessive use of these signs lessens drivers adherence to them, thereby defeating their object.

Prohibition indicated by a No Parking sign and a No Standing and Parking sign starts to apply at the point where the sign is erected and ceases to apply at the next intersection. If the distance is ٢٠٠ m or more the sign may be repeated at suitable intervals according to an engineering decision.

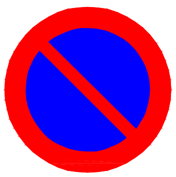
If the prohibition ceases to apply before the next intersection, the sign shall have an additional panel indicating the length of the prohibition, provided that that length is a maximum of ٥٠ meters. Otherwise a new sign with an additional panel, displaying a downwards pointing arrow, shall be erected at the point where the prohibition ceases to apply.

The permission to park indicated by the Parking sign also applies up to the next intersection unless another condition is obvious, for example road markings indicating parking spaces.

The general rule for all Standing and Parking signs is that a prohibition or a regulation, indicated by a road sign, ceases to apply if another sign is erected between the first sign and the next intersection.

The prohibition or regulation applies to the side of the road or street on which the sign is erected on. Thus, if a prohibition or a regulation is to apply to the left hand side of a one-way street, the sign must be erected on the left hand side.

P١. PARKING PROHIBITED



This sign gives notification that parking is prohibited.

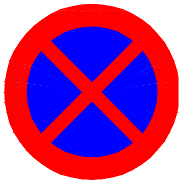
Stopping is allowed only to pick up or set down persons or to load or unload goods.

The prohibition may be needed during a certain time of the day only. Then the sign should have an additional panel indicating the time the prohibition applies, for example "٧ a.m. - ٧ p.m.".

If the parking prohibition only applies to a certain category of vehicles, there shall be an additional panel bearing a symbol of that vehicle category, for example a truck.

Design: Circular with a red border and a red bar, sloping from the upper left to the lower right. The background shall be blue.

P٢. STANDING AND PARKING PROHIBITED



This sign gives notification that standing and parking is prohibited.

The prohibition may be needed during a certain time of the day only. In that case the sign shall have an additional panel indicating the prohibited time, for example "٧ a.m. - ٧ p.m.".

Exception from the prohibition of standing and parking shall not be given to any vehicle category.

Design: Circular with a red border and a red diagonal cross. The background shall be blue.

P٣. PARKING

This sign gives notification that parking is allowed.

The sign may be used to allow parking along a street or road. The sign may also be used at entrances of parking areas.

If parking meters are used for parking spaces, sign P٣ is not needed. Instead a sticker with the letter P and the parking fee may be fixed on the post of the meter. If the time for parking is limited or if the fee only applies to certain times of the day, this may also be displayed on the sticker.
If parking is restricted to a limited time period, that period is indicated on an additional panel, for example "٢ hours". If parking is allowed during a certain time, that time is indicated on an additional panel, for example "٨ p.m. - ٧ a.m.".

A parking space may be reserved for disabled persons. In such cases additional panel AP١١ shall be used.

Sign combination indicating a parking space reserved for disabled persons.

٦. ADDITIONAL PANELS

٦,١. *General*

Additional panels may be used to show the distance from a sign to the beginning of a hazardous section of a road or of the zone to which regulation applies. Additional panels may also be used to show the length of a hazardous section of a road or of the zone to which regulation applies.

Additional panels may be used together with regulatory signs to specify the days of the week or month or the times of the day during which a regulation applies. Additional panels may also be used to indicate exceptions from regulations granted for certain classes of road users.

Additional panels may be used together with informative signs to show the distance from a sign to an interchange or intersection or to a service facility.

Additional panels shall be square in shape a have the same background, lettering and symbol colours as the road sign it is supplementary.

Additional panels other than those specified in these regulations may be used after a written approval from the Ministry of Public Works and Housing.

٦,٢. *Description of the additional panels.*

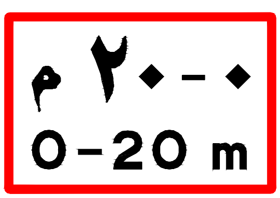
AP . Length of a zone to which a warning applies

The panel, affixed below a warning sign, indicates the length of the hazardous section.

The numerals may be varied in accordance with the circumstances.

The panel should be used whenever a hazardous section exceeds in length what the drivers may expect. Such expectations vary depending on the type of hazard but if the hazardous section exceeds ٥٠٠ metres, the panel should normally be affixed below the warning sign to indicate the length of that section.

AP . Length of a zone to which a regulation applies

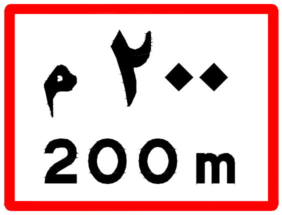


The panel, affixed below a regulatory sign, indicates the extent of the regulation. The panel shall not be used if the regulation ends at the next intersection or if another regulatory sign is indicating the end of the regulation.

The numerals may be varied in accordance with the circumstances.

The distance indicated on the sign should not be too long since it is difficult to estimate distances exceeding ٥٠ metres.

AP̃ . Distance to a dangerous section
or a point a regulation starts to apply



The panel, affixed below a warning sign or a regulatory sign indicates the distance to the hazard or to the regulation.

The numerals may be varied in accordance with the circumstances.

AP̃ . Maximum weight

The panel, affixed below a regulatory sign, indicates that the regulation only applies to vehicles exceeding the indicated weight.

The numerals may be varied in accordance with the circumstances.

The panel may supplementing the following signs:
R١٣, No entry for goods vehicles
R١٤, No entry for any powerdriven vehicle drawing a trailer other than a semi-trailer or a single axle trailer
R١٥, No entry for any powerdriven vehicle drawing a trailer

AP̃ . Time

The panel, affixed below a regulatory sign, indicates that the regulation only applies during the time indicated. The panel may indicate one or more specific days. If the regulation applies on all days within the specified time only the time need to be indicated, not days.

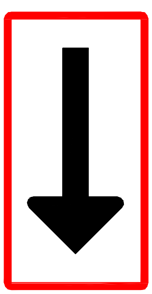
The lettering may be varied in accordance with the circumstances.

AP . Direction of regulation

AP٦,١ Regulation applies in both directions

The panel, affixed below a regulatory sign, indicates that the regulation applies in both directions from the sign.

This panel is normally used to supplement the No parking sign.

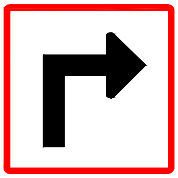


AP٦,٢ End of regulation

The panel, affixed below a regulatory sign, indicates that a regulation ends. The sign the panel is supplementing shall be the same type as the sign that imposed the regulation.

Normally a regulation ends at the next intersection. If the regulation ends at a point ahead of the next intersection this panel should be used unless panel AP٢ is affixed below the sign that impose the regulation.

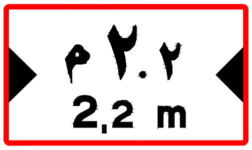
AP Direction of a hazard



The panel, affixed below a warning sign, indicates that the warning applies to a section on an intersecting road.

The panel should be used when the distance from the intersection to the hazard is very short. A typical example when this panel should be used is when a railroad is close to a road and an intersecting road is crossing the railroad.

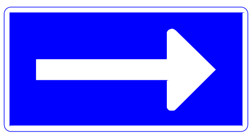
AP Lateral Clearance



The panel may be affixed below sign W٣,١ Carriageway Narrows from Both Sides or sign W٤ Narrow Bridge to indicate the actual lateral space on the narrow section or bridge.

AP Direction of Parking Permission

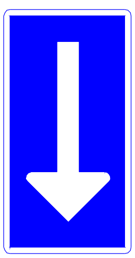
The panel may be affixed below sign P٣ Parking to indicate the direction of parking permission.



AP One Direction

AP Both Directions

AP End of Parking Permission



The panel, affixed below sign P٣ Parking, indicates that the parking permission given by sign P٣ ends. The sign and panel combination may be used if the end of parking permission before the next intersection or is not indicated by some other sign for parking restrictions.

AP Disabled Persons

The panel, affixed below sign P٣ Parking, indicates that the parking space is reserved for disabled persons only.

٧. MAINTENANCE OF TRAFFIC SIGNS

. *General*

Results of neglected maintenance are generally;
- Traffic signs become ineffective and cease to serve their intended purpose
- Danger to road users increases, probability of occurrence of accidents increases
- Discredit to the Ministry

The task of maintenance and repair of traffic signs comprise;
- Inspection of all traffic signs
- Execution of maintenance activities

Maintenance activities could be categorised into;
- Routine maintenance
- Periodic maintenance

Aim

The ultimate aim of proper maintenance of traffic signs is to make them serve the purpose for which they are installed, effectively and at all times. To attain this, traffic signs must always be;
in a good condition
- correctly located
- properly mounted and fixed
- with unobstructed visibility

Maintenance Activities

<u>Routine maintenance</u> activities normally executed are;
- cleaning of signs
- repairing of signs
- repainting sign supports and back of signs
- cutting and pruning vegetation in the vicinity

<u>Periodic maintenance</u> activities normally executed are;
- replacing signs

. *Inspection*

Inventory

In order to facilitate inspection of traffic signs, an inventory is very necessary. An inventory shows,
- different types of traffic signs
- the location of each traffic sign.

It could be the form of line diagram or tabulations. The inventory is very useful when;
- checking whether a sign is missing
- a road sign is to be replaced at its correct location

Field Inspection

Number of inspections for periodic maintenance to be carried out annually;
- at least twice, desirable thrice
- one in the night

Take brief notes of condition and damages observed in traffic signs. Traffic signs should be inspected preferably on foot.

Check list for Inspection

Check that;

- all traffic signs are in their proper location
- all traffic signs are facing in the correct direction and properly oriented
- all sign plates are firmly fixed to the posts.
- all traffic signs are visible and not obscured or hidden by vegetation and other obstructions
- all supporting posts are firmly founded and not loose.

Observe:

- damage of traffic signs due to accident
- flaking or faded paint
- cracked foundations and loose supports of traffic signs
- poor reflecting surfaces or signs in the night
- tilted traffic sign posts and loosely fixed sign plates

. *Defects and Deficiencies*

Defects and deficiencies found in traffic signs should be detected early and remedied promptly by proper maintenance and repair. To properly remedy the defects and deficiencies, it is important to know their causes, which also help in the future to take measures where possible to reduce the maintenance needs. The Table below gives the type of common defects, their causes and the recommended maintenance activities needed.

	Type of defect/ deficiency	Cause	Maintenance activity needed
١.	The traffic sign is dirty/dusty	During dry weather dust raised by vehicles, during wet weather mud and grit splashed by vehicles and may also be due to air pollution	Cleaning
٢.	The traffic sign is partially defaced, obliterated by pasting bills, notices	Vandalism	Cleaning
٣.	Painted surface is faded/flaking off	Weathering, ageing	Repainting
٤.	Components of a traffic sign (bolts, fixtures etc.) removed	Vandalism	Repairing at the site (replacing missing items)
٥.	A sign post is broken/the sign is demolished	Accident	Repairing in the Workshop
٦.	The traffic signs and other traffic control devices are not visible	Vegetation has grown impairing visibility	Cutting and pruning vegetation
٧.	The traffic sign is disfigured and badly obliterated	Vandalism or traffic accident	Replacing sign
٨.	The traffic sign is damaged beyond repair	Traffic accident	Replacing sign

. *Routine Maintenance Activities*

Cleaning

This activity includes cleaning signs, reflectors, and sign posts for retaining their effectiveness, and consideration shall be given to the following:

- frequency of cleaning signs depends on the environment, location and materials used
- wash the surface using a cloth or soft bristle brush, water and detergent solution. Take care not to scratch the surface
- particular care should be taken when cleaning retro reflective sign faces. Use a light detergent and plenty of clean water without grits
- after washing out the dirt remove all traces of detergent with a cloth and by rinsing in clean water
- clean the back of the sign with water and cloth
- use kerosene initially to clean out signs contaminated with bitumen or oil and then wash down with water
- Clean at least twice a year.

Painting

This activity includes painting or repainting back of signs and supporting posts.

(a) General Directions

Surfaces and areas to be painted or repainted should be cleaned free of rust, dirt and other extraneous matter, and consideration shall be given to the following.

- Only clean soft brushes or rollers are to be used
- Painting should be done only during dry weather and only on dry surfaces
- Paints must be thoroughly mixed according to manufacture's instructions. If thinner is to be used, due precautions against fire shall be taken
- Cover components or portions of surface that should not receive a coat of paint applied to the rest of the surface

(b) Directions for repainting Steel Surfaces

- Use a wire brush to remove all loose material from flaky paint surfaces
- Clean area to be repainted using cloth and water and allow the surface to dry thoroughly
- Apply a prime coat evenly to the area after it has been dried thoroughly
- Apply the finishing coat evenly after allowing the prime coat dry.

٧,٥. *Directions for repairing signs at the site*

(a) Correcting badly oriented and or tilted signposts (planted without a concrete base)

- Excavate around the embedded part of the post wide enough to enable a tamper to be used for compaction and deep enough to loosen the post
- move post to upright position and re-orient it. Check with plumb for orientation (the sign face of large informative signs should be at an angle of about ٣٠-٥٠ from the direction of on coming traffic, see chapter ١,٧)
- backfill soil and compact in layers keeping the post firmly held in correct position (if necessary with temporary stakes).

(b) Replacing of bolts and nuts

- The bolts and nuts or screws, along with the necessary washers shall be of GI
- Check and correct (if necessary) alignment of drilled holes before threading bolts or screws
- Prior to tightening of the bolts and screws, a little oil or grease shall be applied.

(c) Setting posts (planted without a concrete base) deeper for extra stability

- Sign posts, without concrete bases, shall be buried to at least ١/٣ the full length of the post, to ensure stability.

٧,٦. *Directions for repairing signs in the Workshop*

- A sign which cannot be repaired at the site shall be dismantled from the post and taken to the workshop for repairs. And then transported back to the site and re-assembled. All nuts and bolts and screws removed in the process shall be replaced (using additional ones for any missing ones)
- a sign that had been posted with a reflective sheeting, needing repairs due to stripping off part of the sheeting, may be repaired by sticking on a piece or pieces of retroreflective sheeting carefully cut and of the correct quality and colour(s), according to the design of the sign.

٧,٧. *Cutting and Pruning of vegetation*

The important points to bear in mind are:
- Vegetation should be cut and pruned to enable the sign to become visible to drivers and be understood at the clear visibility distance. As conditions vary from site to site, this distance of clear visibility will vary accordingly. Also, it is a function of the approaching speed of the vehicles.

٧,٨. *Replacing Signs*

- remove damaged sign, its supporting post and foundation block
- trim sides of foundation pit vertical. The minimum size of foundation for a single standard sign could be taken as ٣٠ cm x ٣٠ cm wide and ٥٠ cm deep. However, this would vary with site conditions
- assemble new sign on new post

٨. PAVEMENT MARKINGS

٨,١. *General*

Pavement Markings are defined as markings or other devices applied to, embedded in, or attached to a pavement surface. Pavement markings must function day and night as well as under adverse weather conditions. While pavement markings are an important and integral element in design of a traffic control scheme, care shall be taken not to overdo pavement markings. Due to their nature and location, pavement markings are subject to continual and rapid deterioration and wear from roadway traffic. To retain conspicuity and function of pavement markings, their maintenance and replacement is, by necessity, an ongoing and continual process with significant cost implications. Thus pavement marking requirements should be considered early in the planning and design of a project.

٨,١,١. Function

Like traffic signs, pavement markings may be classified with respect to the primary function that they serve:

- Regulatory
- Warning
- Guidance

Regulatory pavement markings advise motorists of actions they shall or shall not take. Disregard for a regulatory pavement marking represent an offence. For example, crossing of a solid no passing line is illegal and offending drivers would be subject to citation.

Warning pavement markings advise motorists of the existence of hazardous or potentially hazardous conditions. A dividing line that separates two way traffic warns motorists of the potential hazard of a head on collision with oncoming vehicles if that line is crossed. It is not necessarily a violation to cross that line when turning or passing another vehicle, but the motorist is warned to maintain caution.

Guidance pavement markings help motorists to understand the path that the roadway designer intends for their vehicle to follow. An example of such is a guide line that may be used to mark out the travel path through an intersection for vehicular turns that are unusual and otherwise difficult to understand.

It is important for designers to understand the functional significance of the pavement markings so that their application will be consistent for the use intended. The use of various classes and types of marking in combination with each other and other control devices must be understood and considered by the designer.

٨,١,٢. Marking Types

Pavement markings can further be classified into the following basic types:

- Longitudinal Markings
- Transverse Markings
- Pedestrian Crossing
- Other Markings
- Raised Pavement Markers
- Curb Markings
- Delineators

Longitudinal markings run generally parallel to the longitudinal axis of the line of vehicular traffic on a roadway. In general, the purpose of a longitudinal marking is to convey a continual message to the drivers of a moving vehicle over an extended length of roadway. A lane line is an example of a longitudinal line. It provides a continual message to a driver demarcating the separation of two streams of traffic moving in the same direction. Carelessly crossing a lane line puts one in jeopardy of a sideswipe collision with vehicles in the adjacent traffic stream.

Longitudinal Markings can be characterized as follows:

- Broken lines are permissive in character and may be crossed by a vehicle with due caution.
- Solid lines are restrictive in nature and should not be crossed except in case of an emergency.
- If a road has two or more lanes in each direction and no physical median, a double solid line shall divide the traffic directions.
- Double lines shall consist of two lines equal in width and separated by a gap equal to the width of the line.
- Discontinuities in longitudinal lines, whether solid or broken, shall indicate by their absence locations where turns, merges or diverges are expected to occur.

Transverse Markings are those that are placed at right angles or significantly non-parallel to the longitudinal axis of the roadway. In general, the purpose of a transverse marking is to place an obstruction or barrier across the normal, unimpeded forward movement of a vehicle. A Give Way marking is an example of a transverse pavement marking. The Give Way marking crosses the forward path of a vehicle to advise a driver to stop or be prepared to stop its forward progress in case the situation dictates such action. Because transverse markings must be viewed obliquely from an approaching vehicle, their widths must be substantial to facilitate their detection.

٨,١,٣. Colour

Painted Pavement Markings shall all be either white or yellow. Raised Pavement Markers shall convey to the motorists the same colour message as the painted markings they replace or supplement.

In general, the colour of a marking shall be:

- Yellow for Edge Lines, Warning Markings (humps, parking)
- White for all other markings.
- White, yellow and red may be used on curbs.

All pavement markings shall be reflectorized unless street lightning provides adequate visibility.

٨,١,٤. Marking Maintenance

When pavement markings on existing roadway are in need of upgrade or maintenance, engineering judgement should be exercised in applying the requirements herein. The following guidelines should be considered in such cases:

- When existing pavement markings are either identical to or will be completely covered by new markings as required, the new markings shall be repainted over existing markings.
- When existing markings are in excess of what is required by this Manual (such as additional lane arrows), the existing markings must be retained but not maintained, being allowed to gradually fade from the pavement providing that their presence will not create confusion.
- When existing markings are of a different configuration than those required by this Manual the most appropriate of the following actions should be taken:
  - § If the existing marking is well worn and would be relatively inconspicuous in comparison to the new pavement markings, then the new pavement markings may be applied without regard to the existing markings (care shall be taken in considering the relative conspicuity between the new and old markings at a later point in time when the new markings have been in service and begin to loose some of their original conspicuity).
  - § If practical to do so, existing conflicting markings shall be completely removed from the pavement surface.
  - § If the roadway section is scheduled for future resurfacing, but such is far enough in the future that intermediate pavement marking maintenance is necessary, then the existing markings may be retained and maintained even though they do not fully comply with this Manual.

٨,١,٥. Materials

Pavement Markings may be in cold paint, thermoplastics or ceramic studs.

Generally thermoplastics are used where traffic volume is high or when there is a higher demand on the performance and life-length of the markings.

Thermoplastics are normally used on Primary Roads and paint on Secondary Roads.

٨,٢. *Longitudinal Markings.*

٨,٢,١. Center line and Lane line.

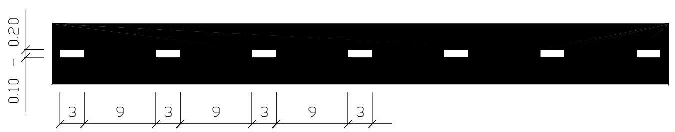
Center lines may be solid or broken or a combination of solid and broken.

– Solid Center lines shall be used when overtaking is prohibited.
– Broken Center lines and Lane lines shall be used when drivers are allowed to cross the line.

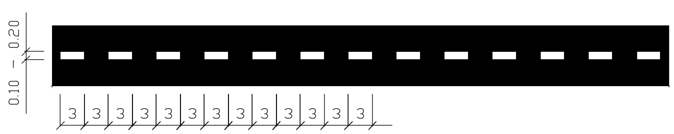
A broken line is formed by segments and gaps in the following ratio:

Area	Segment Length Meter	Gap Length Meter	Width Meter	Colour
Rural	~	đ	⅔ - ⅔ *	White
Urban	~	~	⅔ - ⅔ *	White

*) The line width may be more than ⅔ meters on road sections where better conspicuity is required, for example on high speed roads or demanding road environments.



Broken line in rural areas.



Broken line in urban areas.

Solid line.

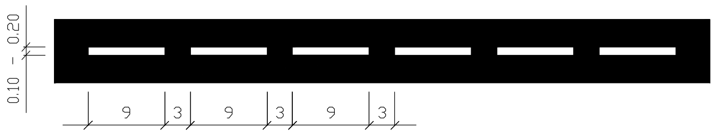
٨,٢,٢. Warning line.

Broken center line may be used as a Warning line on a two-way road where overtaking is hazardous but yet allowed. Such a regulation may be appropriate on a road section where there normally is a large portion of slow moving vehicles, for instance tractors in an agricultural area.

A warning line is formed by segments and gaps in the following ratio:

Area	Segment Length Meter	Gap Length Meter	Width Meter	Colour
Rural	đ	~	⅔ - ⅔ *	White
Urban	đ	~	⅔ - ⅔ *	White

*) The line width may be more than ⅔ meters on road sections where better conspicuity is required, for example on high speed roads or demanding road environments.



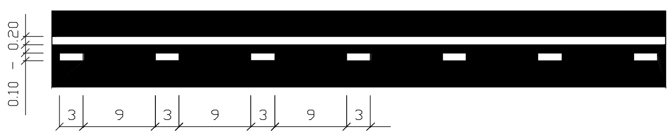
Warning line.

٨,٢,٣. Combination of lines.

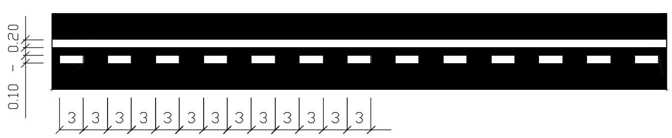
Where center lines are installed, prohibition of overtaking is indicated by a solid line in the direction that overtaking is prohibited. The line combination may be one broken line and one solid line or two solid lines if overtaking should be prohibited in both directions.

Prohibition of overtaking, indicated by a solid center line, shall be established at sharp vertical and horizontal curves and elsewhere on two-lane highways where an engineering study indicates that passing must be prohibited because of inadequate sight distances or other special conditions.

Combination of lines should be used on all paved roads with a width of ٦ meters or more.



Combination of a broken line and a solid line, rural area.



Combination of a broken line and a solid line, urban area.

Combination of two solid lines.

No Passing Lines (Solid lines) is used in place of or on the right side of broken line to impose a mandatory requirement that drivers shall not cross or drive on the left hand side of such marking (with exception of left turns to or from private direct access to property).

٧,٢,٣,١ No Passing Lines at Intersections

No Passing Line shall be used for purpose of traffic control in advance of any controlled intersection (Signal, Stop, Give Way or Pedestrian crossing) on a two-way roadway. It shall replace broken line starting at the distance given in the table below as measured from the STOP marking, GIVE WAY marking or the near edge of intersecting roadway. No Passing Line shall continue towards the intersection until intersecting those markings, other painted or curbed island, or the nearest edge of the intersecting roadway.

USE OF NO PASSING LINES AND CHANNELIZING
LINE AT INTERSECTIONS

Design Speed (km/h)	**Minimum Length (L) (meters)**
٣٠	٨
٤٠	١٦
٥٠	١٦
٦٠	٢٤
٧٠	٢٤
٨٠	٣٢
٩٠	٤٠
١٠٠	٤٨

NOTES:

١. See Figure ٨,٢,٣,١a

٢. Values of L shown are minimum. Greater values may be justified based on engineering judgement. Increases should be made in multiples of ٨ meters.

٣. The values of L given above may also be used as a minimum for Warning Line. Values up to or exceeding ٢L are quite acceptable.

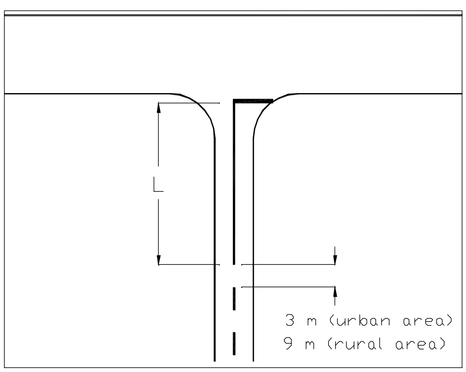


Figure ⅔⅔⅔a The length of No Passing line at Intersection

٨,٢,٣,٢ No Passing Lines at Vertical and Horizontal Bends

The solid No Passing line shall be used when the sight distances are less than the following figures:

Speed Limit (km/h)	**Minimum Passing Sight Distance (meters)**

٥٠	١٥٠
٦٠	١٨٠
٧٠	٢١٠
٨٠	٢٤٠
٩٠	٢٧٠
١٠٠	٣٠٠
١١٠	٣٦٠

The Passing Sight Distance must be measured between two points ١,١ meters above the pavement and from a point ١,٥ meters from the edge of the carriageway as illustrated in the figures below.

Restrictions should be imposed on sections where the range of vision is less than a certain minimum range of vision (S), by means of a solid line laid out in accordance with figures below.

The value to be adopted for S varies with road conditions. Figure ٨,٥,٣a and ٨,٥,٣b show the design of the lines at a hillcrest with a restricted range of vision. Figure ٨,٥,٣c and ٨,٥,٣d show the position of the lines for the same case on a horizontal bend with restricted range of vision.

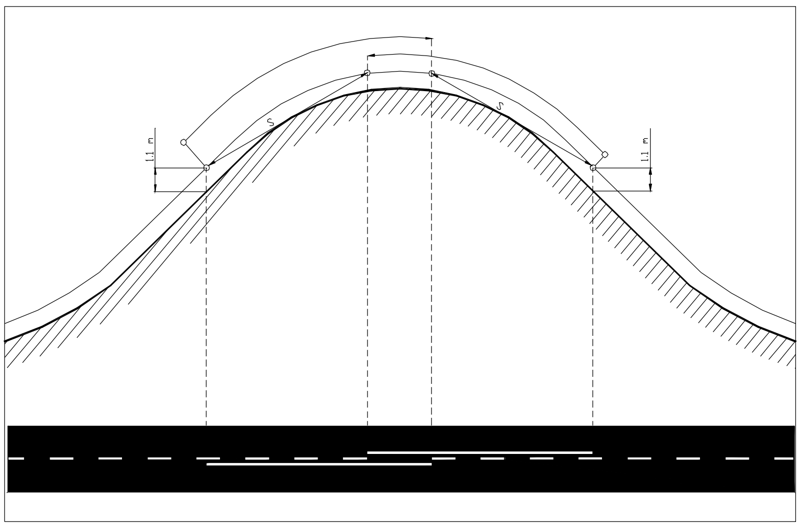


Figure. ⅔ ⅔ a Measuring of the range of vision (S) on a vertical bend.

The range of vision/visibility distance (S) is insufficient from both directions for the motorists to see over the crest. Thus the solid lines overlap.

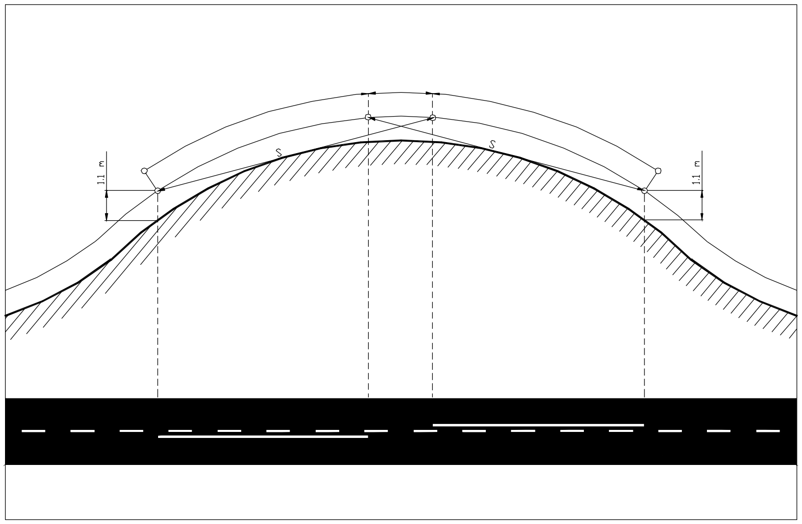


Figure. ⅔ ⅔ b Measuring of the range of vision (S) on a vertical bend.

The range of vision/visibility distance (S) is sufficient from both directions for the motorists to see over the crest. Thus the solid lines do not overlap.

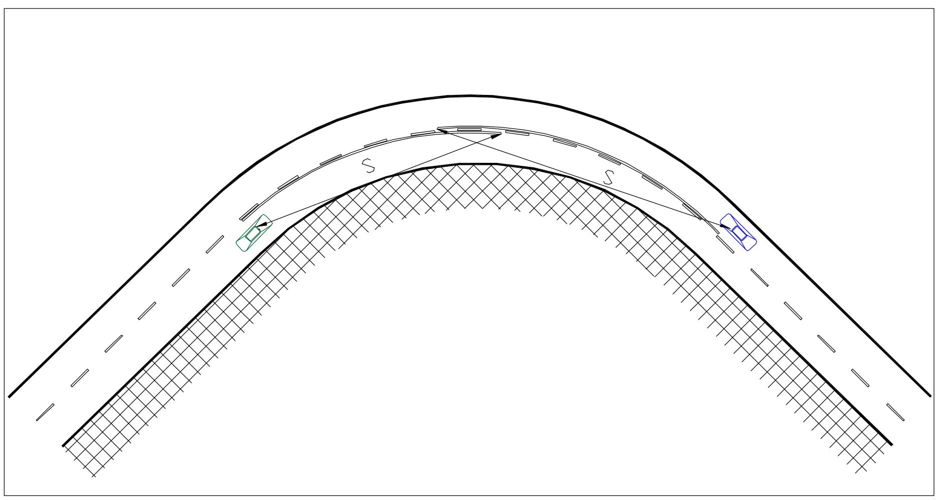


Figure. ⅔ ⅔ c Measuring of the range of vision (S) on a horizontal bend.

The range of vision/visibility distance (S) is insufficient from both directions for the motorists to see beyond the bend. Thus the solid lines overlap.

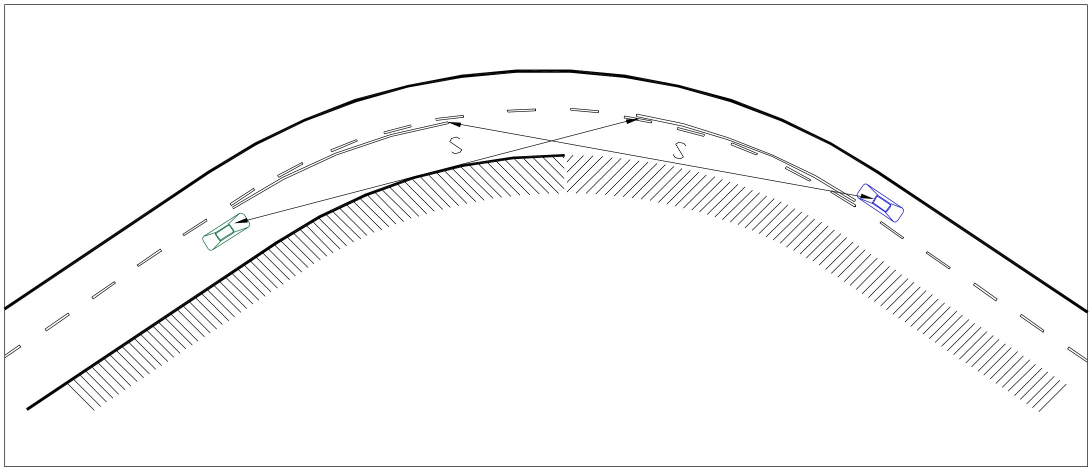


Figure. ⅔⅔d Measuring of the range of vision (S) on a horizontal bend.

The range of vision/visibility distance (S) is sufficient from both directions for the motorists to see beyond the bend. Thus the solid lines do not overlap.

Inside urban areas where the speed limit is ٦٠ km/h or less, solid lines are not needed even if the Passing Sight Distances are less than the prescribed values.

In addition to the described criteria, solid lines may be installed if an engineering study indicates a need of prohibition of overtaking.

٨,٢,٤. Edge Lines.

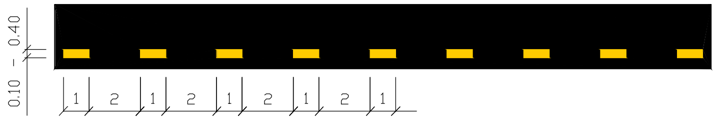
Edge lines shall be used on all primary and secondary roads.

Edge lines shall be yellow in colour.

On freeways and expressways edge line shall be solid. Edge line may be solid on other multi-lane divided roads as well. On other roads edge line shall be broken.

On freeways, expressways and other multi-lane divided roads edge line is normally ٠,٢٠ meters in width. However, if an engineering study indicates the need of enhanced visibility or emphasised guidance from edge lines, the width may be increased to ٠,٣٠ or ٠,٤٠ meters. On other roads edge line shall be ٠,١٠ meters in width but may, if desired, be ٠,٢٠ meters.

Edge lines are very important from a road safety point of view. Therefore it is important to provide new paved roads with edge lines as soon as possible. It is also important to maintain the lines and keep them free from sand, soil, and other things that may reduce the visibility of the lines.



Broken Edge Line

Solid Edge Line

Solid edge lines must not continue through exit and entrance lanes on interchanges and through intersections. At such sections edge line if used shall be broken. See examples below.

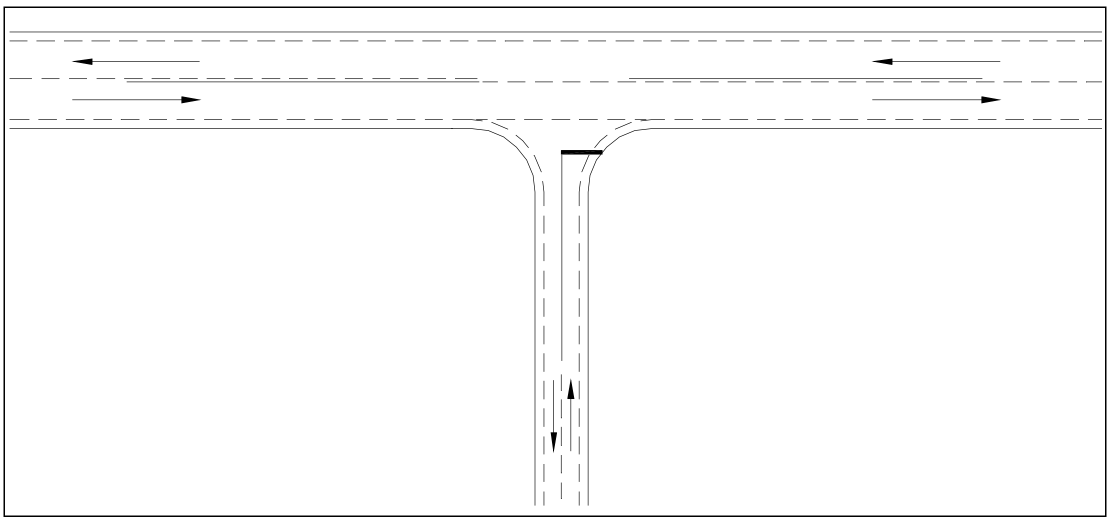


Figure ٨,٤,٤a Marking of broken Edge Line through a T-intersection.

Broken edge line is normally continued through the intersection when the connecting road is a minor road. Note the solid center line that is used to prevent overtaking of approaching vehicles.

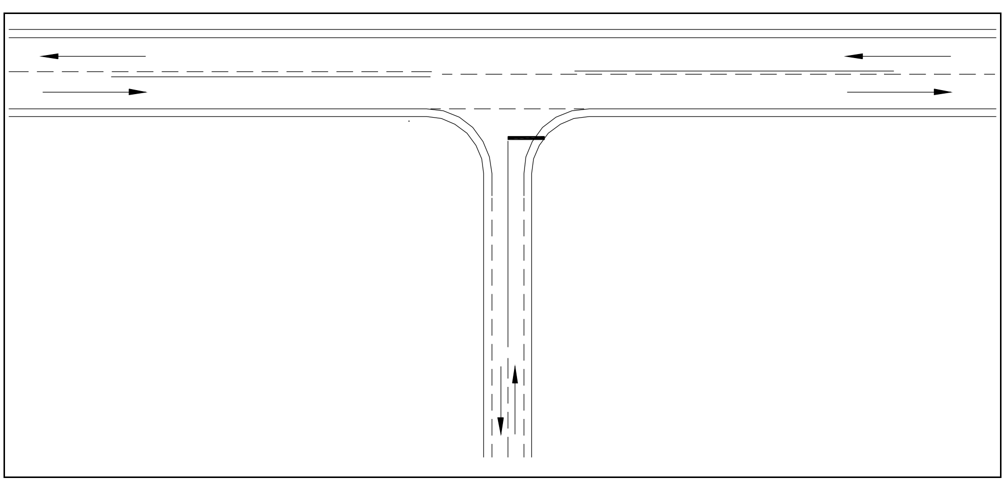


Figure ٨,٤,٤b Marking of solid Edge Lines through a T-intersection.

Solid edge line is normally not used on undivided two-way roads but can, if an engineering study indicates the need, be used on such roads. If solid edge lines are applied, the lines should be broken trough the intersection when the connecting road is a minor road. If the connecting road is a major road the edge line shall be interrupted trough the intersection. Note the solid center line that is used to prevent overtaking of approaching vehicles.

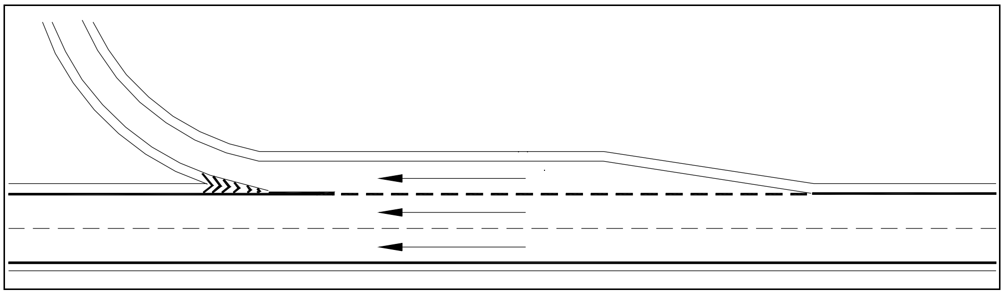


Figure ٨,٤,٤c Typical example on road markings of a deceleration lane.

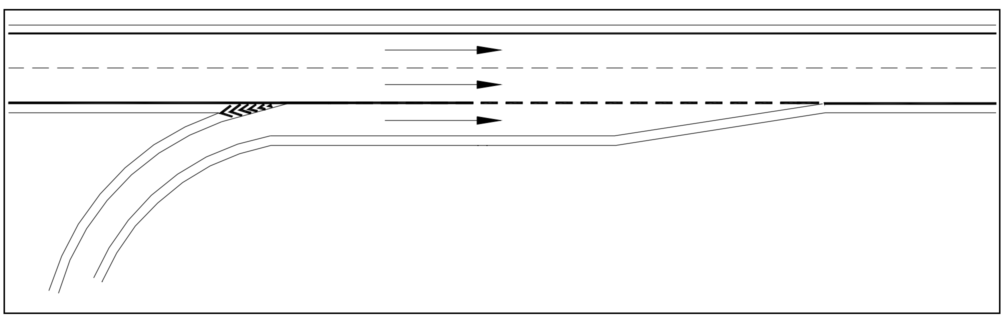


Figure. ⅔ ⅔d Typical example on road markings of an acceleration lane.

٨,٣. *Raised Pavement Markers*

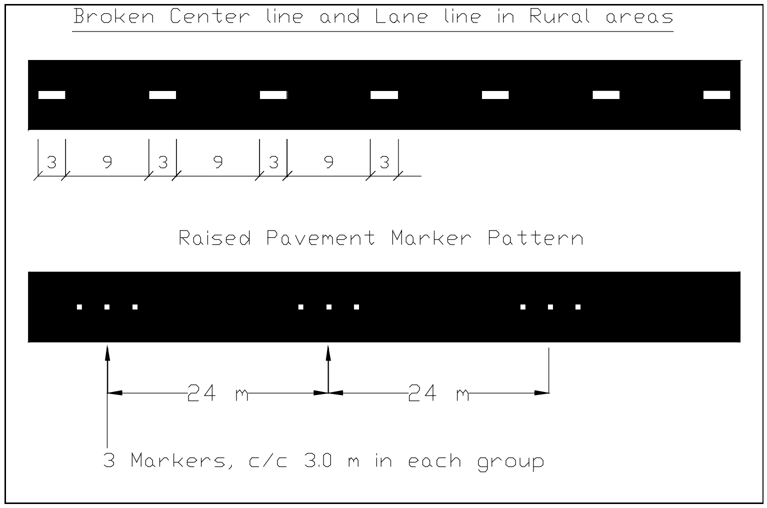
Raised Pavement Markers, RPM, may be used to supplement or substitute longitudinal pavement markings. When used, RPM are arranged in patterns to supplement or substitute solid or broken lines.

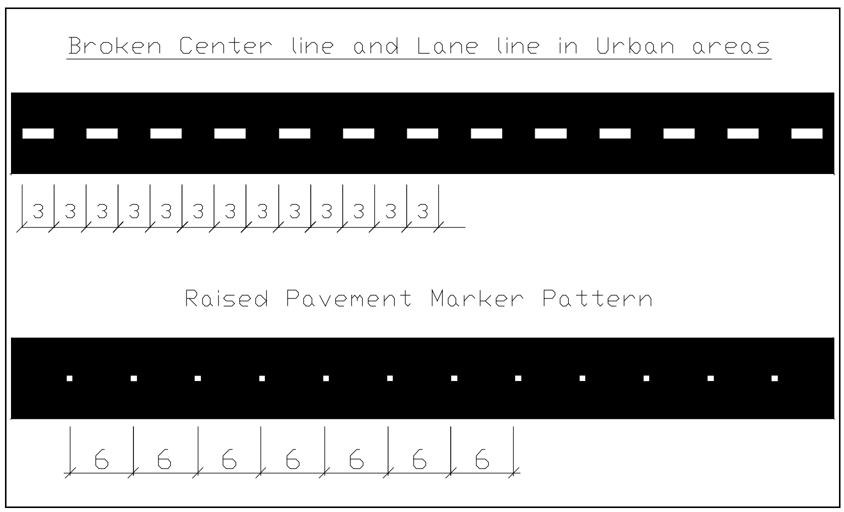
The following characteristics should be applied for RPM:

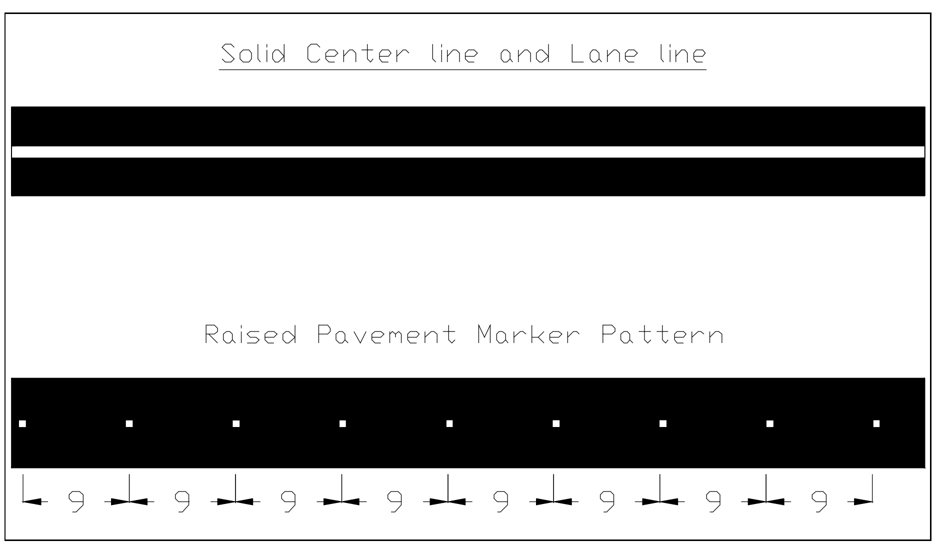
- Generally less than ٢٥ mm in height, round, square or rectangle in shape
- The characteristics of the markers are related to their functional application
- The colour shall conform to the colour of the road markings
- Retroreflective RPM may be used in combination with nonreflective markers
- RPM should be placed close to the painted line

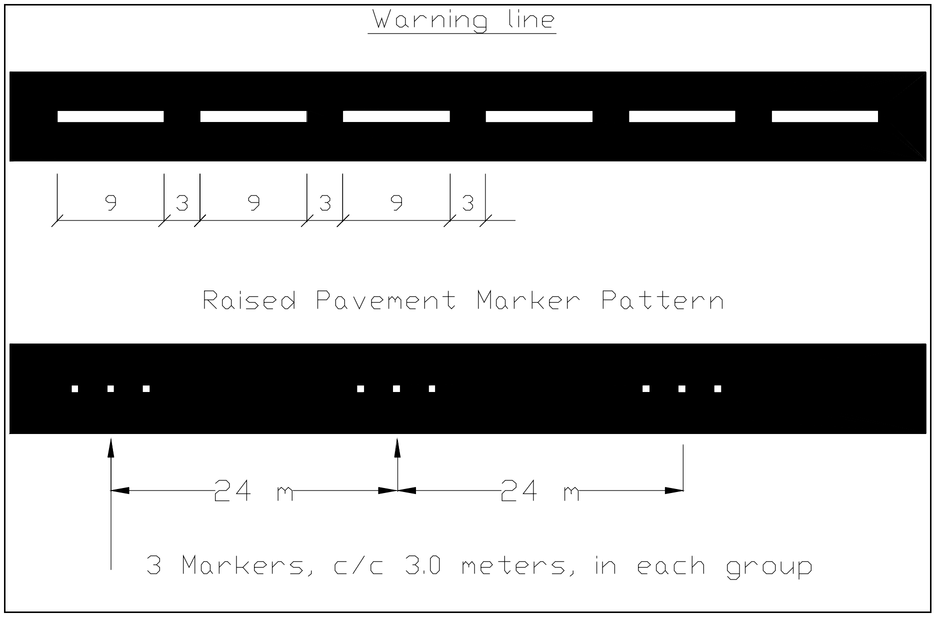
RPM supplementing or substituting edge lines are yellow in colour. RPM supplementing or substituting other markings are white. RPM are normally equipped with reflectors but in urban areas and other places with good street lighting markers without reflectors may be used.

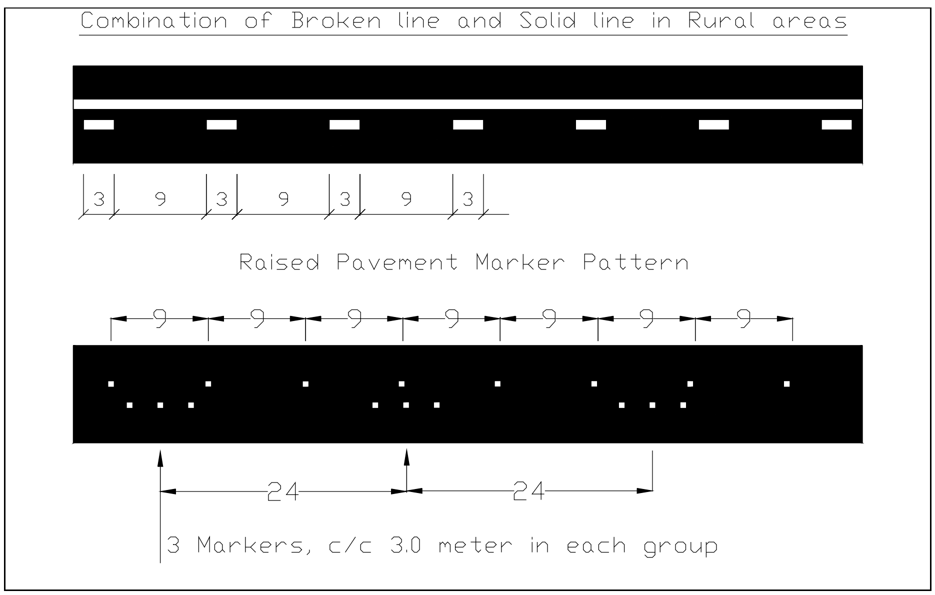
If RPM are supplementing the lines, the spacing should be ٢٤ m in rural areas and ١٢ m in urban areas. If RPM are substituting the lines, the spacing should be according to the figures below.

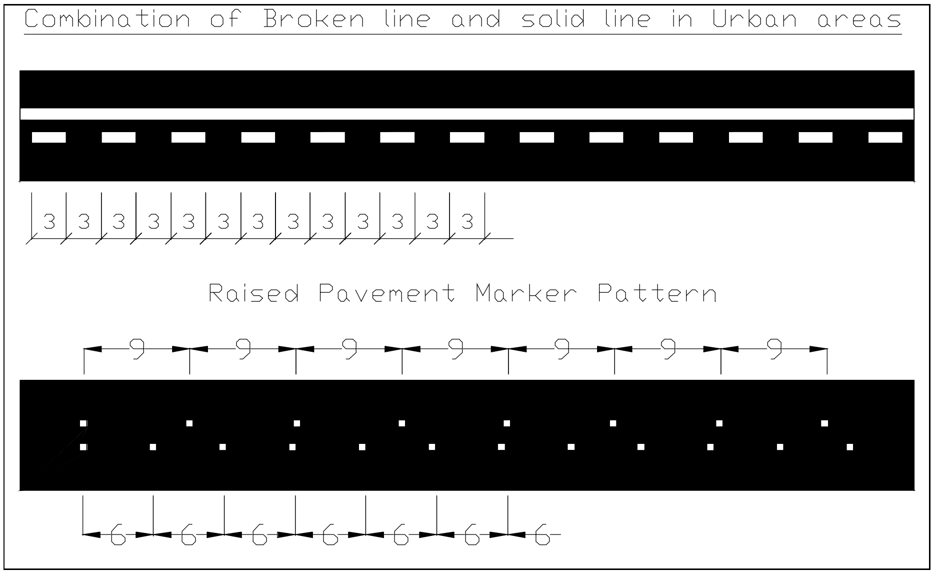

Broken Center line and Lane line in Rural areas
3
9
3
9
3
9
3
Raised Pavement Marker Pattern
24 m
24 m
3 Markers, c/c 3.0 m in each group

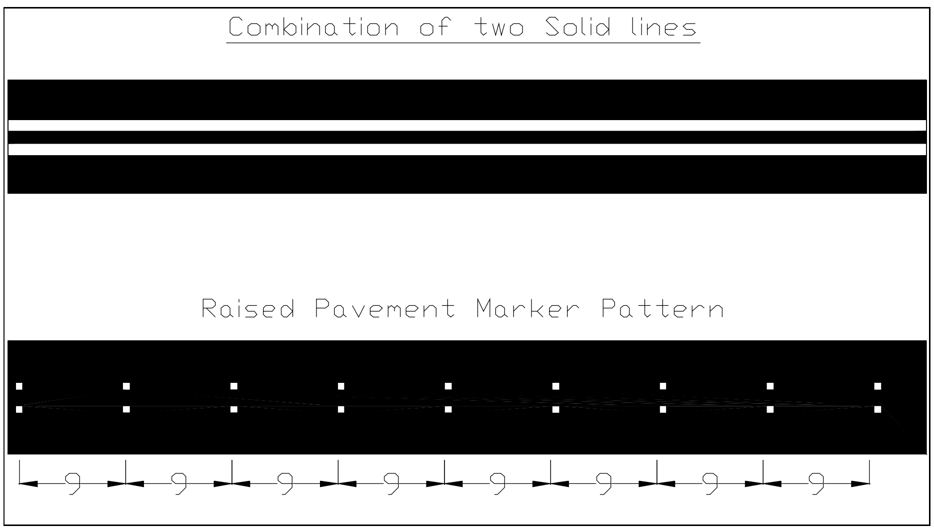

Broken Center line and Lane line in Urban areas
3 3 3 3 3 3 3 3 3 3 3 3 3
Raised Pavement Marker Pattern
6 6 6 6 6 6 6

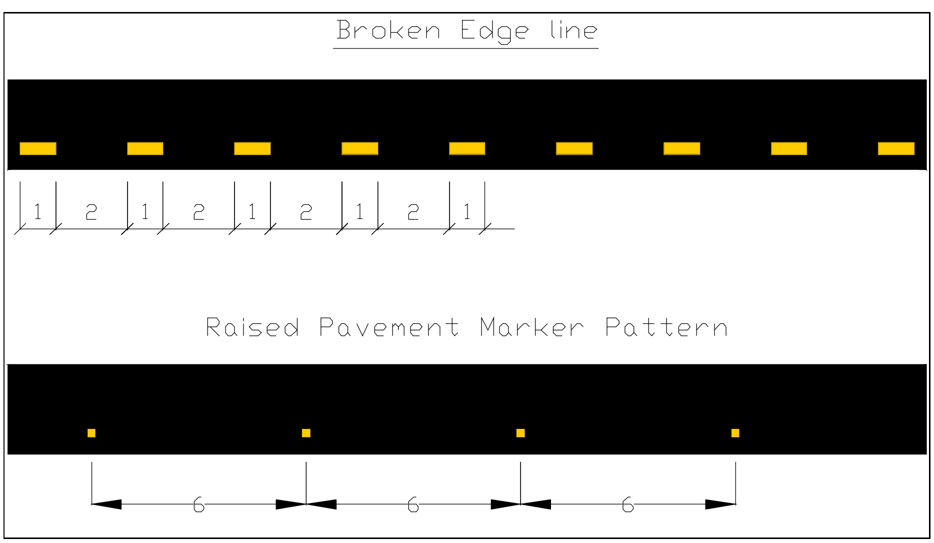

Solid Center line and Lane line
Raised Pavement Marker Pattern
9 9 9 9 9 9 9 9

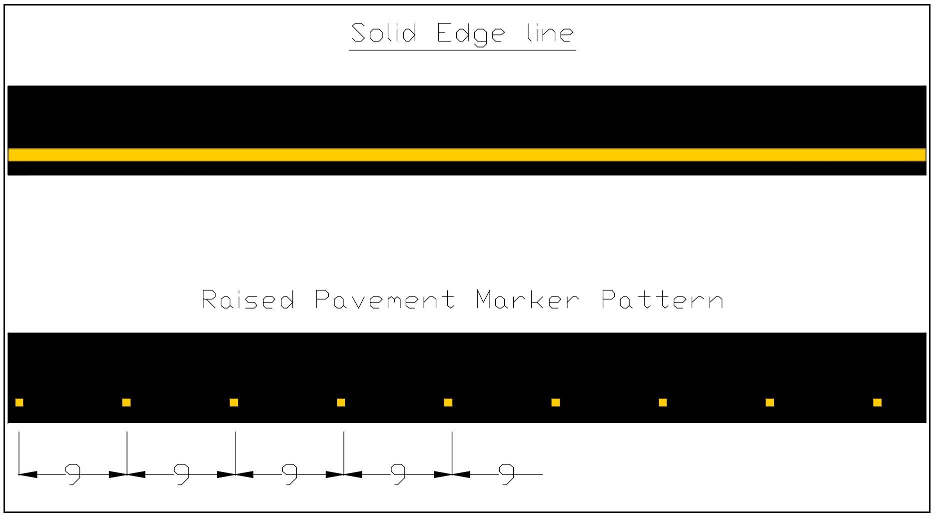

Warning line
9
3
9
3
9
3
Raised Pavement Marker Pattern
24 m
24 m
3 Markers, c/c 3.0 meters, in each group


Combination of Broken line and Solid line in Rural areas
3
9
3
9
3
9
3
Raised Pavement Marker Pattern
9
9
9
9
9
9
9
24
24
3 Markers, c/c 3.0 meter in each group


Combination of Broken line and solid line in Urban areas
3 3 3 3 3 3 3 3 3 3 3 3 3
Raised Pavement Marker Pattern
9 9 9 9 9 9 9
6 6 6 6 6 6 6


Combination of two Solid lines
Raised Pavement Marker Pattern
9 9 9 9 9 9 9 9


Broken Edge line
1 2 1 2 1 2 1 2 1
Raised Pavement Marker Pattern
6 6 6

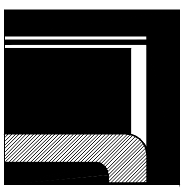


٨,٤. *Transverse Markings*

Transverse Markings consist of road markings across one or more traffic lanes. Transverse Markings are STOP lines, Yield lines, and Pedestrian Crossing markings.

Transverse Markings shall be white.

٨,٤,١. STOP line.



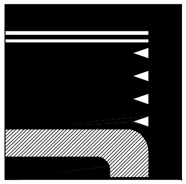
STOP lines shall be used to indicate the point behind which drivers are required to stop in compliance with a STOP sign, a pedestrian crossing or a traffic signal.

Dimensions: STOP line shall be white and its width shall be ٠,٣ meters. Its width may be increased to ٠,٥ meters on high speed rural roads.

Location:

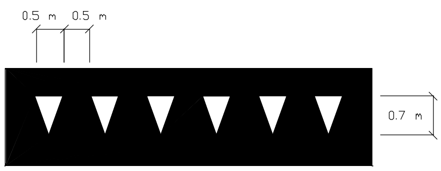
- At intersections with STOP regulation, STOP line shall be located at the point where drivers have the best view over approaching vehicles. Yet the line must be located in such a way that a vehicle stopped behind the line is not intruding on any part of the intersecting road.
- A STOP line at traffic signal shall be located where drivers can se the traffic light. This normally means ١ - ٢ meters ahead of the signal.
- A STOP line before signalised midblock pedestrian crossing shall be at a minimum distance of ٣ meters.

٨,٤,٢. Yield line.



Yield line shall be used to indicate the point behind which drivers are required to stop, when necessary, in compliance with a Yield sign.

Dimensions: Yield line shall consist of triangles with ٠,٥٠ meters base and ٠,٧٠ meters height. Vertices are directed towards the driver who is required to yield.



- At intersections with Yield regulation, Yield line shall be located at a point where drivers have the best view over approaching vehicles. Yet the line must be located in such a way that a vehicle stopped behind the line is not intruding on any part of the intersecting road.
- A Yield line before a signalised mid block pedestrian crossing shall be at a distance of ٣ – ٦ meters.

Below are some examples of application of Yield lines.

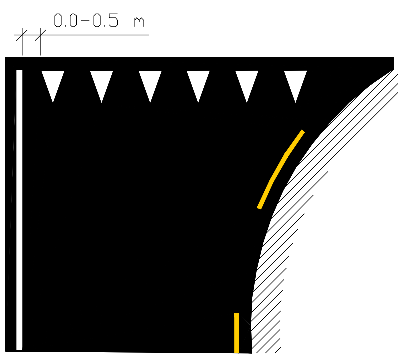


Figure ٨,٤,٢a Yield line, up to ١٠ meters in length. The number of triangles shall be adapted to the width of the approach lane.

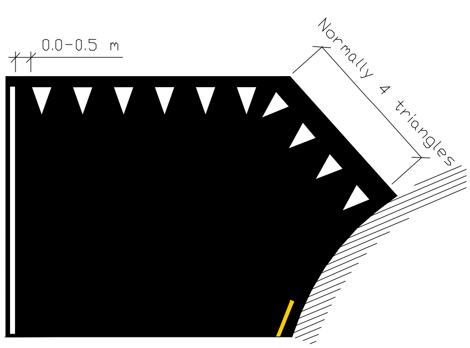
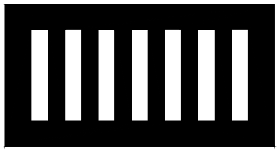


Figure ٨,٤,٢b The Yield line should be bent at ٤٥° angle if its length is more than ١٠ meters. Number of triangles shall be adapted to the width of the approach lane. The portion of the line that is bent normally consists of four triangles.

٨,٤,٣. Pedestrian Crossing.



This marking indicates a point where pedestrians are expected to cross a street or a road.

Dimensions: The marking consists of lines, ٠,٥٠ meters in width and ٠,٥٠ meters spacing. The length of the markings may be adapted to the conditions but should never be less than ٢,٥ meters.

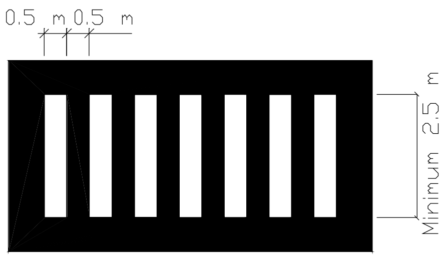


Figure ٨,٤,٣a Dimensioning of a typical Pedestrian Crossing Marking.

Location: Marking of Pedestrian Crossings are done on all paved roads where the sign I١٦ Pedestrian Crossing is erected. Pedestrian Crossings should also be marked at intersections with light signals.

Pedestrian crossings are marked at all intersections where there is substantial conflict between vehicle and pedestrian movements. Marked crosswalks are also provided at other appropriate points of pedestrian concentration or where pedestrians could not otherwise recognise the proper place to cross.

Protection for pedestrians that a Pedestrian Crossing is supposed to provide is very much dependent on drivers' behaviour towards pedestrians entering the crossing. Unfortunately many drivers do not respect the rules to stop when a pedestrian has entered the crossing. Therefore Pedestrian Crossings should not be used indiscriminately. An engineering study should be conducted before they are installed at locations away from traffic signals.

Pedestrian Crossings should not be installed where speed limit exceeds ٦٠ km/h.

Below are some examples of application of pedestrian crossings.

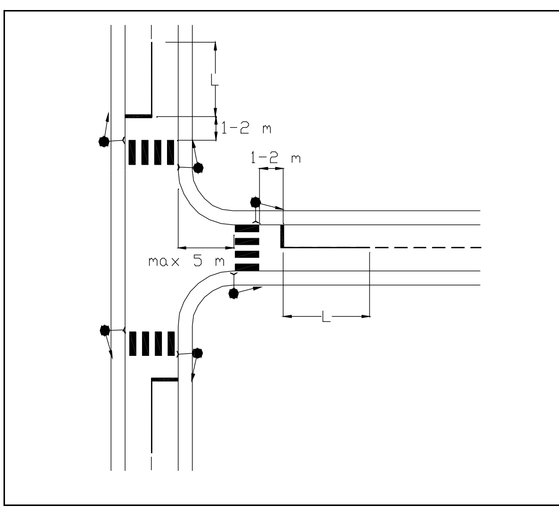


Figure. ⅔⅔a Pedestrian Crossing marking at intersection with traffic signals

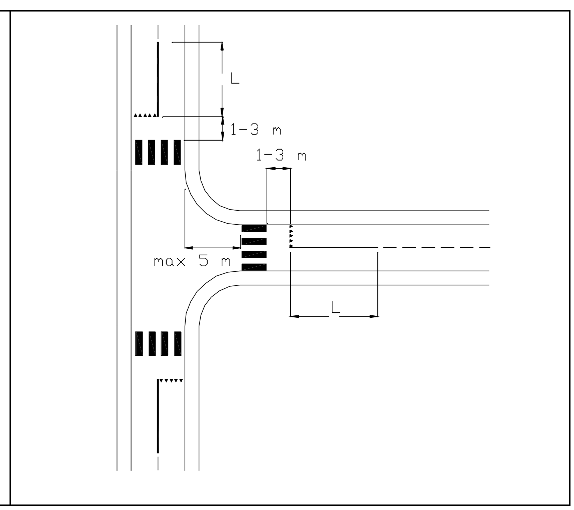


Figure. ⅔⅔b Pedestrian Crossing marking at intersection without traffic signals

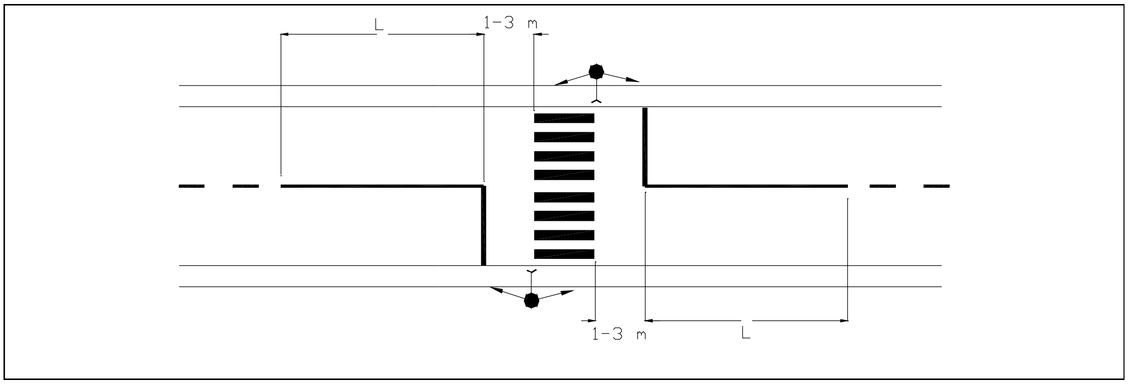


Figure. ⅔ ⅔ c Pedestrian Crossing marking at mid block with traffic signals

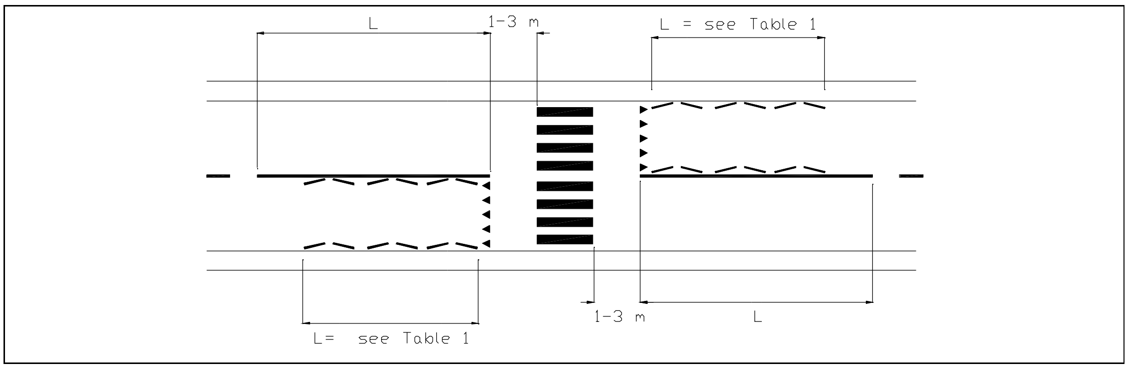


Figure. ⅔ ⅔ d Pedestrian Crossing marking at mid block without traffic signals

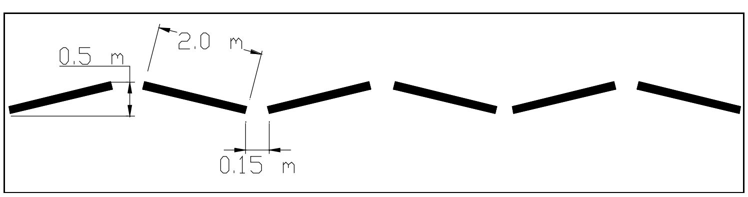
Note:

- The figures above show only the road markings, not the required traffic signs.
- The length (L) of the No Passing line shall be in accordance with chapter ٨,٢,٣,١ No Passing Lines at Intersections.

٨,٤,٤. Zig-zag Lines

Zig-zag lines may be used ahead of midblock pedestrian crossings to emphazise the visibility of the crossing and to indicate the no parking zone ahead of the crossing.

Zig-zag lines are yellow in colour. The length of a zig-zag line is standardized at ٢,٠ m with gaps of ٠,١٥ m. The width of the marking is ١٠٠ mm. These zig-zag lines have to be laid at suitable angles within a width limit of ٥٠٠ mm (see figure below).



Dimensions of zig-zag marking

Zig-zag markings can be laid only if a minimum length of ٤,٠ m is available and can be extended up to a length of ٢٠,٠ m. Details of zig-zag markings are given in Table ١ below.

Table ١
Details of zig-zag markings

Available length for the zig-zag markings (m)	No. of equal length zig-zag markings
٤,٠ = < ٦,٠	٢
٦,٠ = < ٨,٠	٣
٨,٠ = < ١٠,٠	٤
١٠,٠ = < ١٢,٠	٥
١٢,٠ = < ١٤,٠	٦
١٤,٠ = < ١٧,٠	٧
١٧,٠ = < ٢٠,٠	٨

If the carriageway width is less than ٦,٠ m and no center line is marked, ٢ zigzag lines shall be provided on both sides of the road.

Where the carriageway width is more than ٦,٠ m, additional zigzag lines may be provided along the side center of the road.

٨,٥. *Other Markings.*

Other Markings are Lane Selection Arrows, Deflecting Arrows, Obstruction Markings, Parking Space Markings, Words, Curb Markings.

٨,٥,١. Lane Selection Arrows.

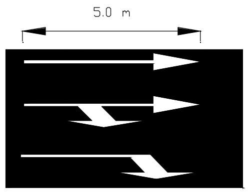
Lane Selection Arrows are used at:

- approaches to intersections containing more than two lanes (figure ٨,٥,١a)
- two lane approaches where only one lane is permitted for the straight on movement (figure ٨,٥,١b)
- two lane approaches where either straight on movement or any of the turning movements are prohibited (figure ٨,٥,١c)
- exit lanes from an expressway or similar (figure ٨,٥,١d)

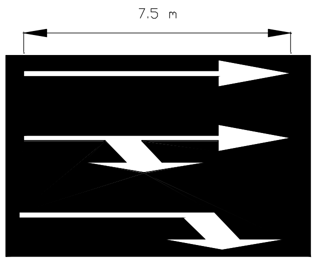
Lane Selection Arrows shall be white.

Lane Selection Arrows shall be located in the center of the lane they concern.

If speed limit exceeds ٦٠ km/h, Lane Selection Arrows must be in large size.



Normal Size



Large Size

Below are some typical examples on the use of Lane Selection Arrows.

Note: Other markings and signs needed are not illustrated since the purpose of the figures is to illustrate the use of Lane Selection Arrows.

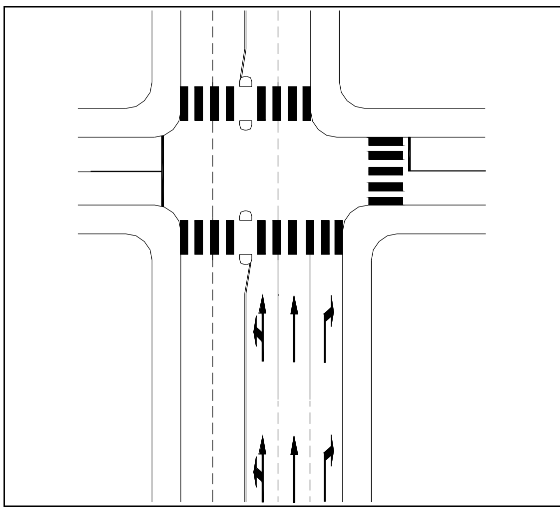


Figure ٨,٥,١a

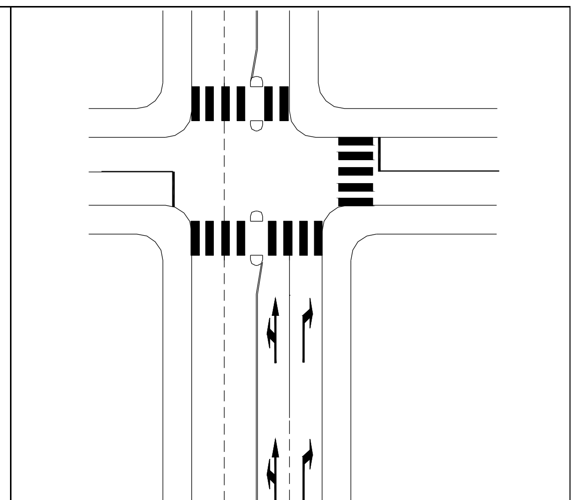
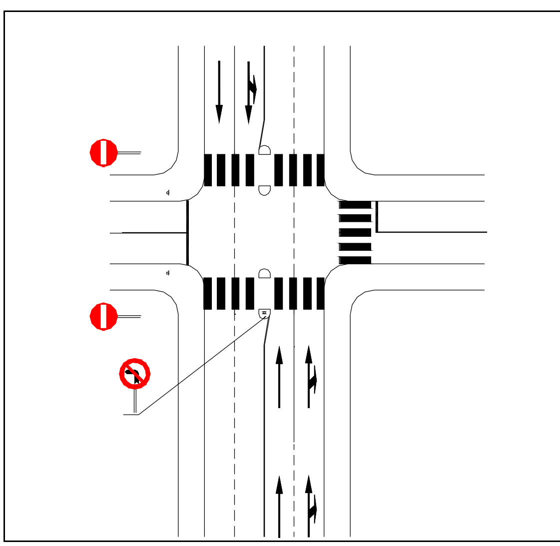


Figure ٨,٥,١b



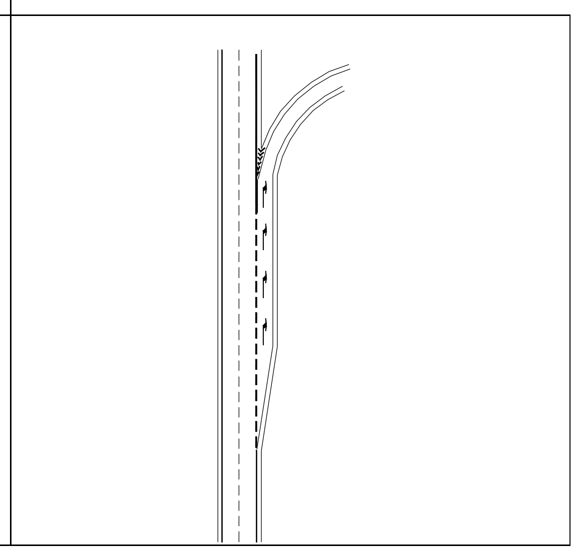


Figure ٨,٥,١c *Figure ٨,٥,١.d*

٨,٥,٢. Deflecting Arrows.

Deflecting Arrows are used to guide traffic at points where the road width changes to a lesser number of through lanes. Deflecting Arrows are located in the center of the lane they concern (see figure ٨,٥,٢b).

Deflecting Arrows may also be used as a warning of a no overtaking section ahead, marked by a solid center line or by a road sign. In such cases Deflecting Arrows are located in the center line, replacing a stroke in the broken line (see figure ٨,٥,٢c).

Deflecting Arrows shall be white.

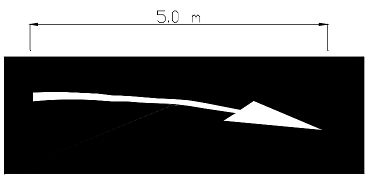


Figure. ⅔ ⅔ a Design and dimension of Deflecting Arrow in normal size and large size.

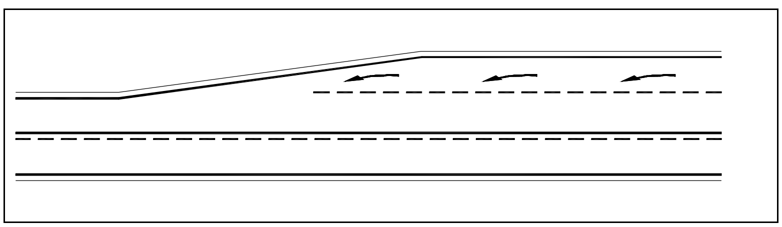


Figure. ⅔ ⅔ b Deflecting Arrows at the end of a climbing lane.

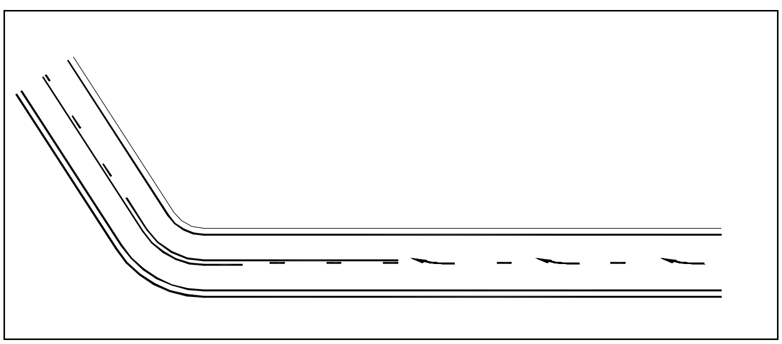


Figure. ⅔ ⅔ c Deflecting Arrows as warning of a solid line ahead.

٨,٥,٣. Obstruction Markings.

Obstruction Markings may be used to prevent drivers from hitting fixed obstacles on the road, such as beginning of medians, traffic islands, curb noses at exit ramps, bridge abutments and piers, etc.

An obstruction may be so located that all traffic must keep right or may be between two lanes of traffic moving in the same direction. The markings in either case must be designed to guide traffic away from the obstruction. The use of solid lines and Obstruction Markings are generally effective.

Obstruction Markings consist of a diagonal line or lines, extending from a center line or a lane line to a point about ٠,٥ m to the right side, or to both sides, of the beginning of the obstruction.

If traffic is required to pass only to the right of the obstruction, the marking shall consist of solid lines and diagonal markings, placed in the triangular area as shown in Figure ٨,٥,٣c.

If traffic may pass either to the right or left of the obstruction, the marking consists of solid lines diverging from the lane line, one to either side of the obstruction. In the area between the solid lines Obstruction Markings shall be placed as shown in Figures ٨,٥,٣a and ٨,٥,٣b.

The lines forming the Obstruction Markings point always in the traffic direction. That means if the marking consists of an open triangle, the vertex shall be pointing towards the oncoming traffic.

Colour of the Obstruction Markings shall be yellow.

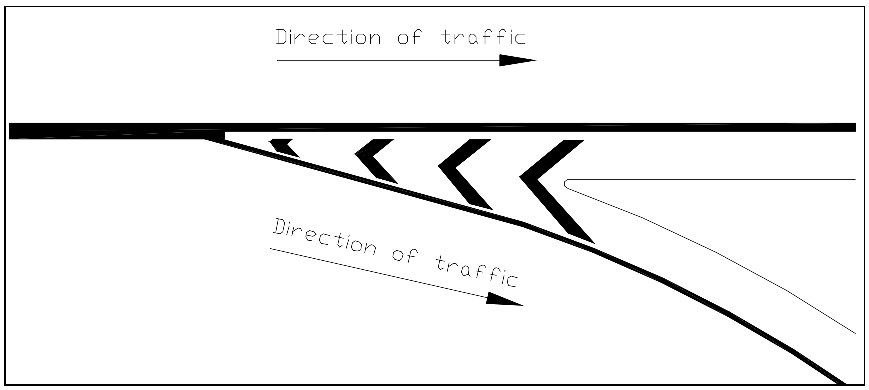


Figure ٨,٥,٣a Obstruction Markings in the gore area of a Exit ramp.

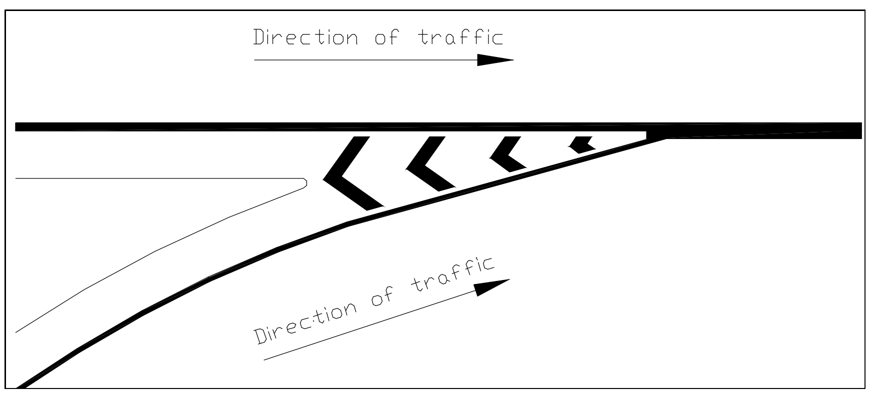


Figure ٨,٥,٣b Obstruction Markings at an Entrance ramp.

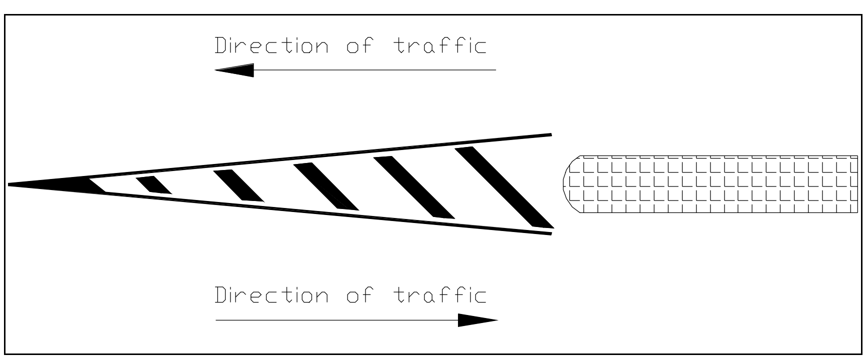


Figure ٨,٥,٣c Obstruction Markings at a Median Island.

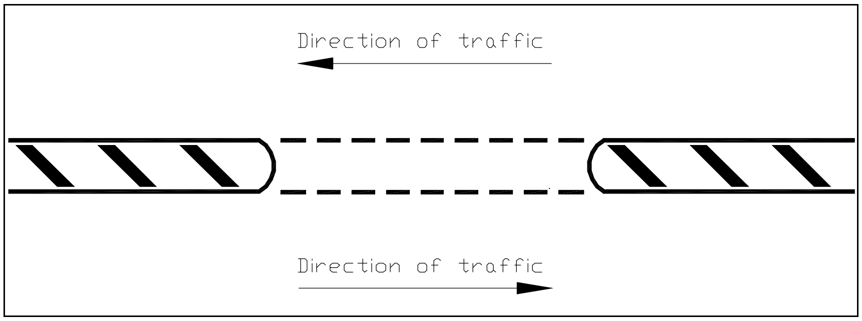


Figure ٨,٥,٣d Opening in an Obstruction Marking.

Widths of the lines forming the Obstruction Markings are according to the figures and the table below.

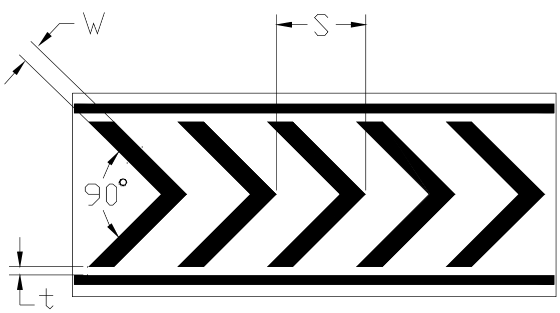


Figure ٨,٥,٣e Dimensions of Chevron Marking

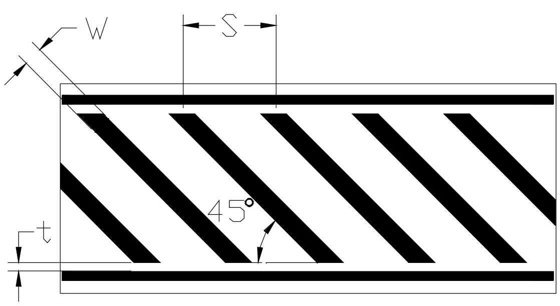


Figure ٨,٥,٣f Dimensions of Diagonal Hatch Marking

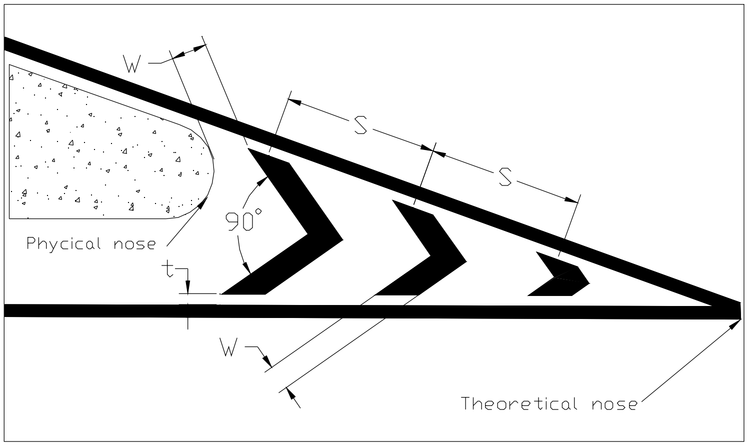


Figure ٨,٥,٣g Dimensions of Chevron Marking at a gore area

OBSTRUCTION MARKING DIMENSIONS

Design Speed Km/h	W mm	S m	t mm
٤٠ - < ٦٠	٢٠٠	١,٠	١٥٠
٦٠ - < ٨٠	٣٠٠	١,٥	١٥٠
٨٠ - < ١٠٠	٥٠٠	٢,٥	٢٠٠
≥ ١٠٠	١٠٠٠	٥,٠	٢٠٠

Solid lines and obstruction markings may be used to form median islands. In some locations it may be more suitable to form a median island by road markings rather than by curbs. Such locations may for example be where the available space is limited so that drivers of large vehicles may be allowed to pass over the island in some situations. In rural areas it must also be more suitable and safer to form a median island by road markings in order to separate left turning traffic from the straight on.

٨,٥,٤. Parking Space Markings.

Parking space markings shall be white.

Marking of parking space limits on urban streets encourages more orderly and efficient use of parking spaces where parking turnover is substantial. It tends to prevent encroachment on approaches to corners, clearance spaces for islands, and other zones where parking is prohibited. Typical parking space markings are shown in Figures ٨,٥,٤a and ٨,٥,٤b.

Parking Space Markings may be solid or open. The width of the lines is ٠,١٠ meters.

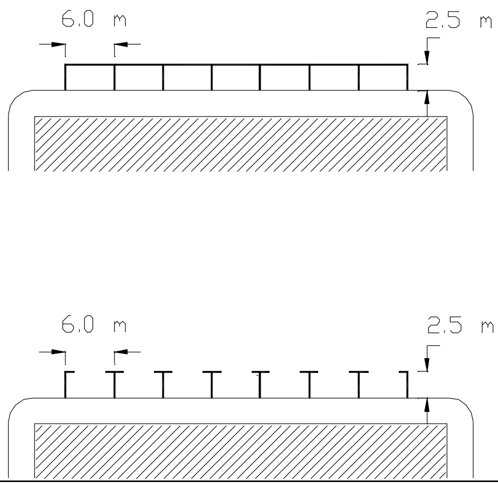


Figure ٨,٥,٤a Typical examples on parallel parking spaces.

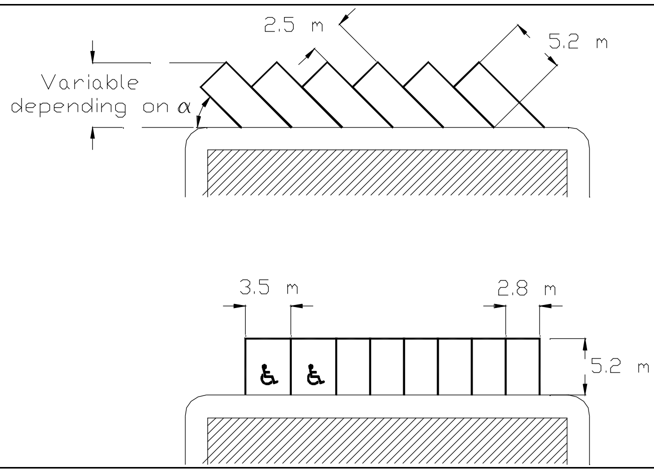


Figure ٨,٥,٤b Typical examples on angle parking spaces.

Note: Parking space dimensions indicated on the figures above are approximate and may be adjusted to fit available space on each spot. However, measures should not be less than indicated.

The disabled symbol marked in a parking space means that only disabled persons are entitled to use that parking space. The width of such parking spaces is ٣,٥ meters to ensure that opening of car doors can be done without problems.

٨,٥,٥. Words and Symbols.

Words and symbols may be marked on the road for guidance, warning or regulation. Normally words and symbols are used only to emphasise information, warning or regulation given by road signs.

Words and symbols shall be white.

The following words and symbols may be used:

Words:

STOP	-	may be marked ahead of a STOP-line.
BUS	-	may be marked at entrances to roads, streets, or lanes reserved for buses only. The word may be repeated along a reserved bus-lane.
TAXI	-	may be marked in parking spaces reserved for taxicabs. If taxi is allowed to use a reserved bus-lane the word TAXI may be marked in

addition to the word BUS.

Symbols:

DISABLED symbol - may be marked in parking spaces to indicate that the space is reserved only for disabled persons.

ROAD NUMBER - may be marked in lanes to guide drivers through complicated intersections.

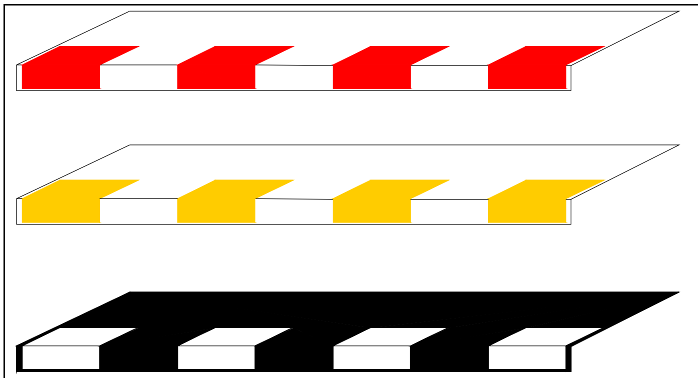
Other words and symbols may be used.

٨,٥,٦. Curb Markings.

Curbs may be painted in white, red or yellow sections.

The paint used for marking of curbs need not be reflective.

The figure below shows the meaning of the different colours of curb markings.



Red = No Parking

Yellow = Parking only for public transport vehicles (bus, taxi)

White = For emphasised visibility only.

٨,٥,٧. Rumble strips

“Rumble Strips” consist of a series of raised strips across the carriageway, arranged into a system of several sets. The desired function of Rumble Strips is to alert drivers when approaching a hazardous location or, even a road work area. By construction, Rumble Strips shall not be speed reducing devices but rather devices that create a noise inside the vehicle, when passing over them. Thus the height of each strip should not be more than a maximum of ١٠ mm. The strips should normally be made of white thermoplastic road marking material.

They should be arranged in sets of ٣ - ٥ strips, and in ٦ - ١٠ such sets, in order to give the desired rumbling effect. The sets should also be arranged with a decreasing distance between each set in order to give the drivers an impression of increasing speed, if they remain at the same speed. The diagrams below indicate typical arrangements of Rumble Strips on a road section with a speed limit of ٦٠ km/h or below and on a road section with a speed limit above ٦٠ km/h.

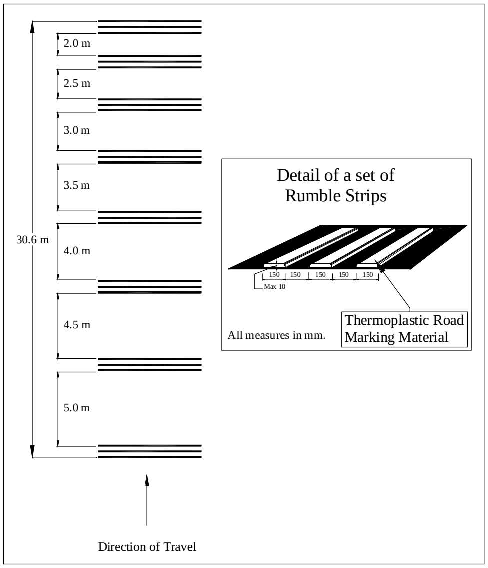


Figure ٨,٥,٧a Typical Layout of a system of Rumble Strips on a road section with a speed limit of ٦٠ km/h or less.

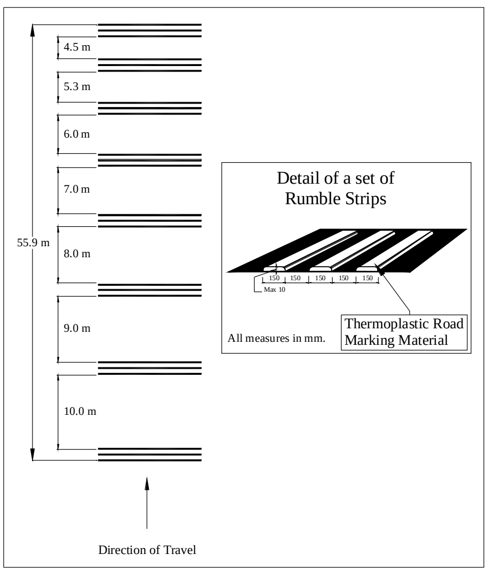


Figure ٨,٥,٧b Typical Layout of a system of Rumble Strips on a road section with a speed limit of more than ٦٠ km/h.

٨,٥,٨. Marking of Humps

Humps installed for traffic calming could be hazardous if drivers do not observe them. White road markings should therefore always be applied on humps. Such marking should be white and applied on the hump at the start of the hump and in both directions. The marking shall consist of white squares, ٠,٥ m arranged in a pattern with two lines of squares with the distance between each square ٠,٥ m as well. The figure below show marking of a road hump.

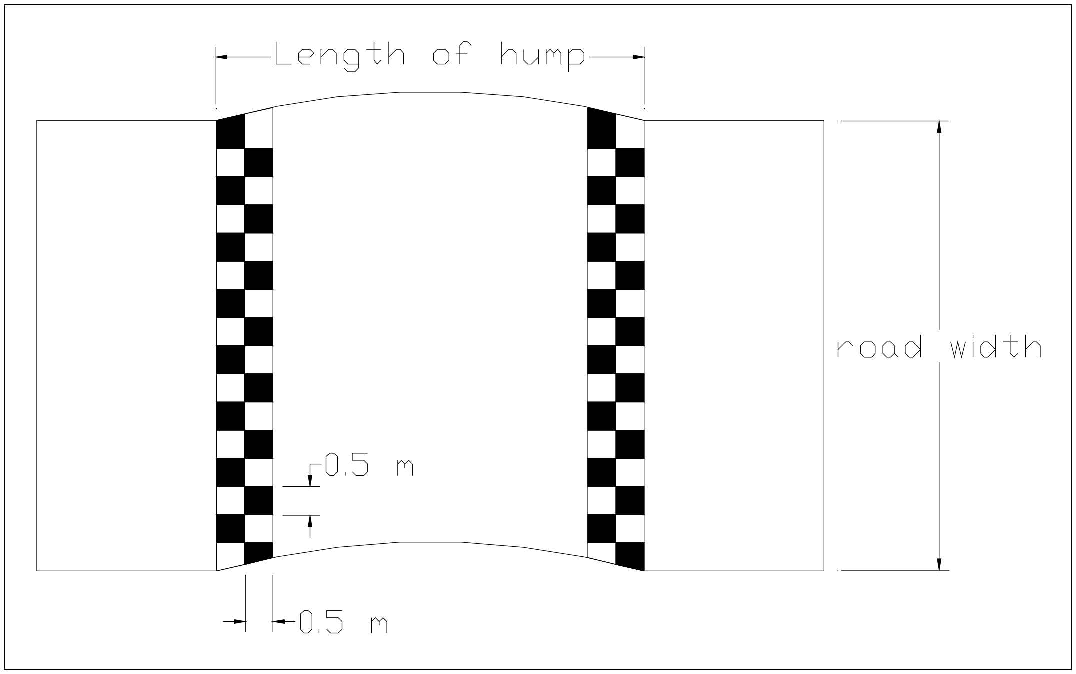


Figure ٨,٥,٨a Marking of road hump

٩. ROADSIDE DELINEATORS

٩,١. *General.*

Road delineators are light retro-reflecting devices mounted in series at the side of the roadway to indicate the roadway alignment. Delineators are effective aids for night driving and considered as guidance devices rather than warning devices. Delineators may be used on long continuous sections of highway or through short stretches where there are changes in horizontal alignment, particularly where the alignment might be confusing or at pavement width transitions.

٩,٢. *Design*

Delineators consists of reflector units capable of clearly reflecting light under normal atmospheric conditions from a distance of at least ٢٠٠ m when lighted by the upper beam of standard automobile lights.

Delineators may be designed to be mounted either on guard-rails or on special delineator posts.

Delineator posts are white and may have a black section at the upper part. Posts may be of plastic, wood or steel.

Reflective elements may be red, white or yellow and consist of either minimum S٢ reflective sheeting or prismatic reflectors.

Reflective elements for delineators mounted on guardrails shall have an area of minimum ٦٠ cm٢. Reflective elements for delineators mounted on posts shall have an area of minimum ١٠٠ cm٢.

Colour of the reflective elements on delineator posts placed on the right hand side of a two-way roadway, and on both sides of a one-way roadway is yellow to conform with the colour of the edge line.

٩,٣. *Placement and Spacing*

Delineator posts, if used, have the top of its reflector unit about ١,٢٠ m above the near roadway edge. Delineators are placed not less than ١,٠ m or more than ٢,٠ m outside the outer edge of the shoulder, or if appropriate, in line with the guardrail. Delineators mounted on guardrails may be placed at a height less than ١,٢ m.

When post mounted delineators are used, they are erected on both sides of the roadway.

Delineators are placed at a constant distance from the edge of the roadway. However, where a guardrail or other obstruction intrudes into the space between the pavement edge and the extension of the line of the delineators, delineators are in line with or inside the innermost edge of the obstruction.

Normally, delineators are spaced ٦٠ m to ١٠٠ m. When normal uniform spacing is interrupted by driveways, crossroads, or similar, interrupted delineators falling within such areas may be moved in either direction, a distance not exceeding one-quarter of the normal spacing. Delineators still falling within such areas are eliminated. On freeways, expressways and similar roads normal delineator spacing is ١٠٠ m.

Spacing is adjusted on approaches and throughout horizontal bends so as several delineators are always visible to the driver. The table below shows suggested maximum spacing for delineators at horizontal bends.

Radius of curve	**Spacing**

١٥	٦
٥٠	١٠
٧٥	١٢
١٠٠	١٥
١٢٥	١٧
١٥٠	٢٠
١٧٥	٢٠
٢٠٠	٢٢
٢٥٠	٢٤
٢٧٥	٢٥
٣٠٠	٢٧

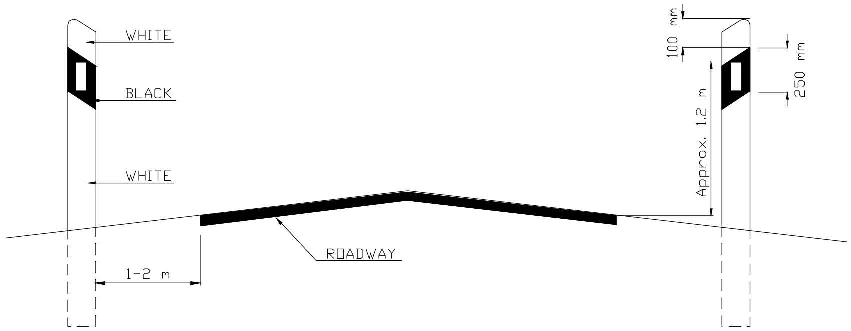


Figure ٩,٣a Typical example of placement and design of Roadside Delineators.

Guardrail mounted Roadside Delineators.

Roadside delineators may be mounted on guardrails or concrete barriers. Such delineators shall be white in colour on both sides of the road.

The size of the delineators is according to the figure below.

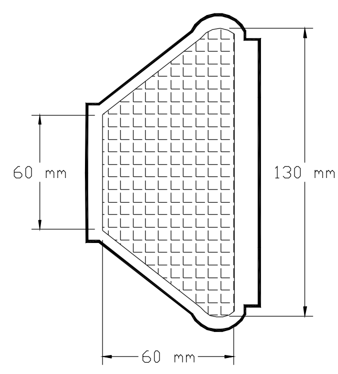


Figure ٩,٣b Typical example of a guardrail delineator.

١٠. TRAFFIC LIGHT SIGNALS

١٠,١. *General*

Traffic Light Signals are devices, that are used for the control and direction of vehicles and pedestrians at:

- Road intersections
- Level Crossings
- Places of road works
- Pedestrian crossings

Traffic Light Signals shall be of the size, colour and type specified below:

١٠,٢. *Light Signals for Vehicular Traffic*

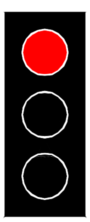
Signals consists of three non-flashing lights, which are red, amber and green respectively arranged vertically and adjacent to each other with the Red lamp being fixed at the uppermost position, followed by the Amber and then Green.

The centres of adjacent lenses shall not be less than ٣٠٥ mm nor more than ٣٦٠ mm.
The diameter of each lens shall not be less than ١٩٣ mm nor more than ٢٢٠ mm.

The centre of the Amber lens shall be fixed at a height not less than ٢,٤ m or more than ٤,٠ m if signals are placed at the centre of the road or at the edge of the road. If signals are placed over the carriageway, height shall not be more than ٩ m or less than ٦ m.

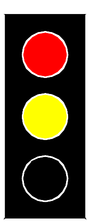
The sequence of traffic light signals is shown below:

Red



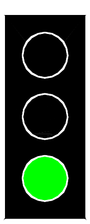
A red light means that traffic shall not proceed; vehicles shall not pass the stop line or, if there is no stop line, shall not pass beyond the signal or, if the signal is placed in the middle or on the opposite side of an intersection, shall not enter the intersection or move on to a pedestrian crossing at that intersection.

Red and Amber



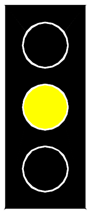
An amber light, which appear at the same time as the red light shall mean that the signal is about to change, but shall not affect the prohibition of passing indicated by the red light.

Green



A green light means that traffic may proceed; however, a green light for controlling traffic at an intersection shall not authorise drivers to proceed if traffic is so congested in the direction in which they are about to proceed and that if they entered the intersection they would probably not have cleared it by the next change of phase.

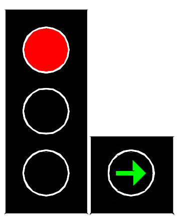
Amber



An amber light, which appears alone shall mean that no vehicle may pass the stop line or beyond the line of the signal unless it is so close to the stop line or signal when the light appears that it cannot safely be stopped before passing the stop line or beyond the line of the signal.

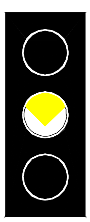
The red, amber and green lights may be replaced by arrows of the same colour on a black background. When lit up, these arrows have the same significance as the lights, but the prohibition or authorisation is restricted to the direction or directions indicated by the arrow or arrows. Arrows signifying that traffic may or may not proceed straight ahead shall point upwards.

Additional Green Arrow



Where a signal includes one or more additional green lights showing one or more arrows, the lighting of such additional arrow or arrows shall, no matter what phase the three-colour system may be in at the time, mean that traffic may proceed in the direction or directions indicated by the arrow or arrows; it shall also mean that, when vehicles are in a lane reserved for traffic in the direction indicated by the arrow or the direction such traffic is required to take, their drivers must proceed in the direction indicated if by stopping they would obstruct the movement of vehicles behind them in the same lane, provided always that vehicles in the traffic stream they are joining must be allowed to pass and that pedestrians must not be endangered.

Flashing Amber



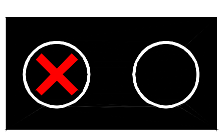
A single amber flashing light shall mean that drivers may proceed but shall do so with particular care.

Amber flashing light may be used in place of a three-colour system at times when traffic is light, particularly at night.

١٠,٣. *Lane Signals.*

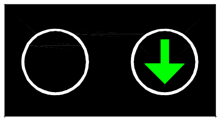
Where green or red lights are placed above traffic lanes shown by longitudinal markings on a carriageway having more than two lanes, red means that traffic may not proceed along the lane over which it is placed and green light means that traffic may proceed. The red light thus placed shall be in the form of two inclined crossed bars and the green light in the form of an arrow pointing downwards.

Red



A red light means that traffic shall not proceed along the lane over which the signal is placed. If the signal over any adjoining lane is showing green light, drivers may change to that lane.

Green



A green light means that traffic may proceed along the lane over which the signal is placed.

١٠,٤. *Light Signals for Pedestrians*

Light signals for pedestrians shall comprise the following:

- A light signal to display a symbol of standing pedestrian as a Red signal
- A light signal to display a symbol of a walking pedestrian as a Green signal

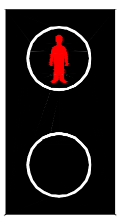
These two signals shall be placed so that they face the pedestrian crossing the road and not the oncoming traffic. These light signals shall be arranged vertically with the Red signal being affixed at the top and the Green being at the bottom.

Minimum vertical clearances from the carriageway shall not be less than ٢,١ m nor more than ٢,٦ m.

Pedestrian signals shall have the following meaning:

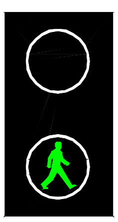
Red

A red light means that pedestrians must not enter the carriageway.



Green

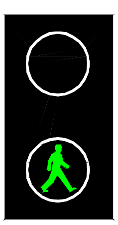
A green light means that pedestrians may cross.



Flashing Green

A flashing green means that the signal is about to turn to Red.

Pedestrians may not cross, but those already on the carriageway may continue to the other side when the flashing green light appears.



A flashing green light means that the period during which pedestrians may cross the carriageway is about to end and the red light is about to appear.

Light signals for pedestrians may be supplemented by audible or tactile signals at pedestrian crossings to facilitate crossings of the carriageway by blind pedestrians.

Pedestrian Actuated signal.

The pedestrian shall obtain the right of way to cross the road, by pressing the button on the push button controller fixed to signal post. Prior to the Green light appears for pedestrians, the signals for vehicular traffic shall turn to Red. When the flashing Green light appears for pedestrians, they should stop crossing the road as the signals for vehicular traffic are about to turn Green and the signals for pedestrians are about to turn to Red.

١١. SIGN SUPPORTS

١١,١. *General.*

The following specifications are the US Specifications for Sign Supports. Some specifications are converted into the Metric System but not all. The result of the calculations for the sign supports may be converted to metric units before ordering or manufacturing Sign Supports and Break Away Devices.

If the sign supports are protected by guard-rails or barriers, there is no need for any Break-away system.

١١,٢. *Large Roadside Signs*

١١,٢,١. Multiple Post Supports

Large roadside signs require substantial supports to insure against failure due to wind loading. The most fundamental treatment for signs that cannot be removed or relocated is to make the supports breakaway.

١١,٢,٢. Breakaway Metal Supports.

The basic concept of the breakaway sign support is to provide a structure that will resist wind loads yet fail, at preselected locations, when struck by a vehicle. The loading conditions for which the support must be designed are as shown in Figure ١١,١. Three critical connection locations and their required characteristics under each loading condition are indicated. When these connections are properly designed, the support will be stable and will possess the breakaway characteristics, when struck by a vehicle, as shown in Figure ١١,٢.

١١,٢,٣. The Breakaway Design Procedure.

Wind Loading.

The design of a breakaway sign support and hinge is based on its capacity to resist imposed wind loads. An example will be presented to illustrate the approach. Assume that a ٤,٨ by ٢,٤ m roadside sign panel is desired as shown in Figure ١١,٣.

To compute the wind load, the formula from the ١٩٧٥ AASHTO specifications will be used. The formula is:

$$p = \text{٠,٠٠٢٥٦}\ (\text{١,٣}V)^{\text{٢}}\ C_d\ C_h$$

where: p = Wind pressure in pounds per square foot

v = Wind speed from map (MPH), ٥٠ year Mean Recurrence Interval (if available)

C_d = Drag coefficient of sign (see Table A)

C_h = Coefficient for height above ground measured to the center of the loaded area (See Table A)

Let the wind velocity be ١٠٠ mph (١٦٠ Km/h). From Table A, the shape factor for the sign is approximately ١,١٩ and height factor is ٠,٨ at ١٢ feet. Using the AASHTO formula for the normal wind pressure on the sign panel, the pressure is computed as:

$p = ٠,٠٠٢٥٦ (١,٣ x ١٠٠)^{٢}\ ١,١٩ x ٠,٨$

$p = ٤١$ psf (١,٩٦ kPa)

The moment diagram resulting from the wind load is presented in Figure ١١,٣. Loading on the vertical supports for the sign panel have been neglected. The required section modulus for A٤٤١ steel is:

$$S = \frac{\text{M (design moment)}}{\text{S (allowable stress)}} \quad \frac{\text{63.0 kip-feet x 12}}{\text{25 kip/in.}^2} = 30.2 \text{ in.}^3 \text{ (495 cm}^3\text{)}$$

Thus, two ٨ by ٢٠ supports having a combined section modulus of ٣٤ in.^{٣} (٥٥٧ cm^{٣}) may be used. After selecting a support size, the next step is to check them for all other AASHTO requirements (see AASHTO Specifications).

١١,٢,٤. Fuse Connections.

The fuse connection must be designed to resist the maximum moment produced in the sign support at the hinge area by the wind loads on the sign. This fuse connection is a friction-type joint, with the resisting moment-producing force being a friction force, which is developed between the fuse plate and the sign support by tension in the attaching bolts. A slotted fuse plate, illustrated in Figure ١١,٤, is used in the connection.

When a vehicle strikes the support, the slip base is broken loose and, as the support begins to deform in the direction of movement, the two bolts holding the fuse plate to the lower part of the support slip out of the slotted bolt holes, and the hinge becomes activated as illustrated in Table B. A recommended procedure for designing the fuse connection is as follows:

- Determine maximum wind moment at the hinge point. This determines the moment capacity of the fuse connection.
- Determine the required bolt force to develop the friction force between the fuse plate and the support. The initial bolt force can be determined by:

١. Required Force = $\frac{m}{d}$

Where: m; is the moment to be resisted
d; is the depth of the post

٢. Individual bolt tension = $\frac{\text{Require force}}{fn}$;

$f = ٠,٢$

Where: f; is the joint friction assumed as ٠,٢
n; is the number of bolts (usually ٢ per post)

٣. Determine bolt size.

- Select the next larger standard bolt size.
- Check the tension that will be produced in the bolts selected. Since the stresses in a bolt will tend to relax over a period of time (creep), the bolt will lose some of the tensile force in it as time passes. In order to produce a friction-type bolted joint, the bolts must have an initial tension force equal to ٧٠ % of the specified minimum tensile strength of the bolts. The required tension for various bolt sizes is shown in Table B.

- The initial bolt tension must be obtained by the calibrated wrench method. Specifications for these methods are outlined in the aforementioned Specification for Structural Joints Using ASTM A ٣٢٥ or A ٤٩٠ Bolts, and should be closely followed at both the design and construction stages.

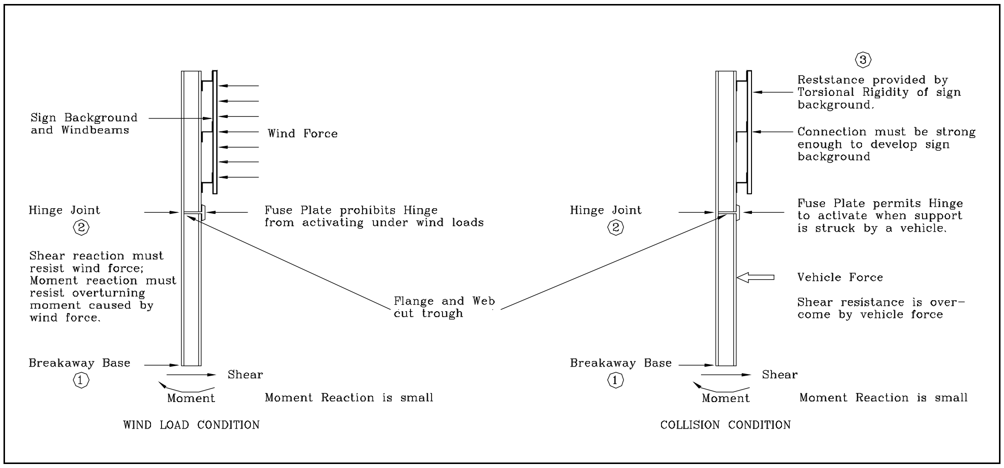


Figure ١١,١. Breakaway Sign Loading Conditions

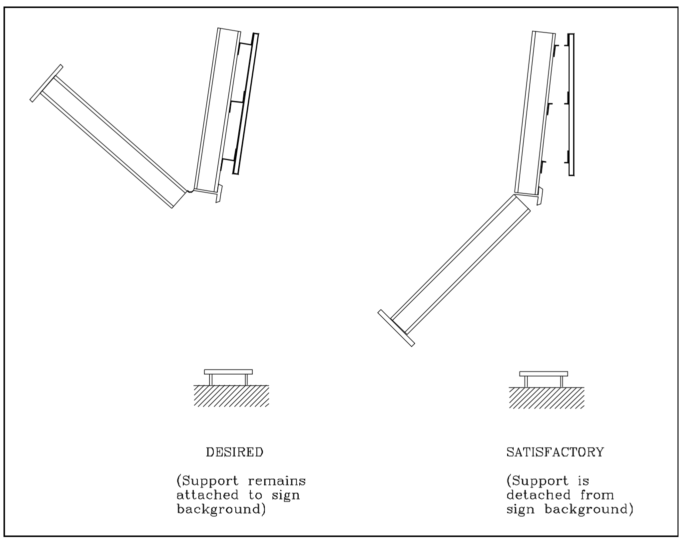


Figure ١١,٢ Collision Behaviour for Breakaway Signs

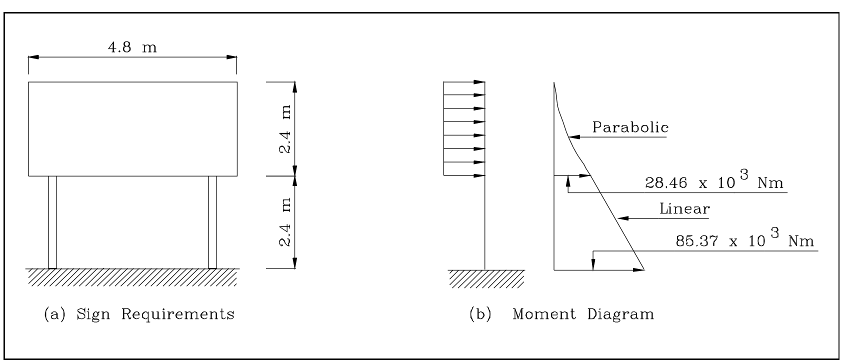


Figure ١١,٣ Wind Loading

Note: The minimum distance from the slip base to the Hinge Joint shall be ٢,٢ meters. Inclined Slip Bases shall be used on signs which have supports within ١,٩ meters of one other.

TABLE A Wind Load Coefficients

Shape of Member	C_d			
	Single Member or Truss			Two Members or Truss – One in front of the Other and Trusses forming Triangular Cross Sections (All Trusses with small solidity ratios)
	$V_d < 32$	$32 < V_d < 64$	$V_d > 64$	
Cylindrical	1.10	$\frac{100}{(V_d)^{1.3}}$	0.45	1.20
Dodecagonal	1.20	$\frac{9.62}{(V_d)^{0.6}}$	0.79	---
Octagonal	1.20	1.20	1.20	---
Flat	1.70	1.70	1.70	2.00
Elliptical (Broadside facing wind)	$1.7\ (D/d_o - 1) + C_{dD}\ (2 - D/d_o)$			
Elliptical (Narrow side facing wind)	$C_{dd}\left[\frac{4 - D/D_o}{3}\right]$			

Where:

V_d = Wind Speed (mph)

D/d_o = Ratio of Major to Minor diameter of Ellipse

C_{dd} = Drag coefficient for a Cylindrical shape of Diameter d_o

C_{dD} = Drag coefficient for a Cylindrical shape of Diameter D

OTHER ELEMENTS C_d

Sign Panel		
	L/W = 1.0	1.12
	= 2.0	1.19
	= 5.0	1.20
	= 10.0	1.23
	= 15.0	1.30
Traffic Signals		1.2
Luminaires with generally rounded surface		0.5
Luminaires with flat sides		1.2

HEIGHT COEFFICIENT (C_h)

Height (feet)	C
0 –15	0.80
15 – 30	1.00
30 – 50	1.10
50 – 100	1.25
100 – 150	1.40
150 – 200	1.50
200 – 300	1.60

TABLE B Fastener Tension

Bolt Size inches (cm)	**Minimum Fastener Tension, Kips (N)**	
	A ٣٢٥ Bolts	**A ٤٩٠ Bolts**

١/٢ (١,٢٧) ٥/٨ (١,٥٩) ٣/٤ (١,٩٠) ٧/٨ (٢,٢٢) ١ (٢,٥٤)	١٢ (٥٣) ١٩ (٨٤) ٢٨ (١٢٤) ٣٩ (١٧٣) ٥١ (٢٢٧)	١٥ (٦٧) ٢٤ (١٠٧) ٣٥ (١٥٦) ٤٩ (٢١٨) ٦٤ (٢٨٥)

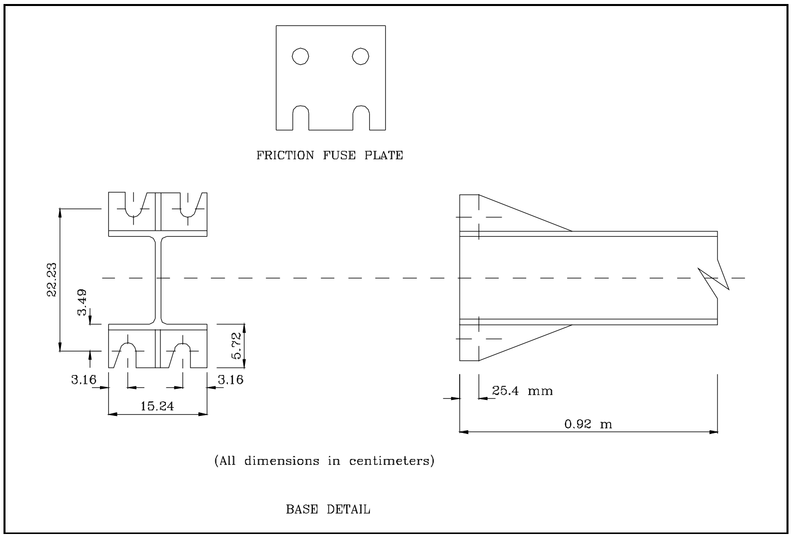


Figure ١١,٤ Breakaway Hardware

Background-to-Post Connection.
The maximum connection force anticipated is ١٠،٠٠٠ lbs. (٤٤،٥٠٠N).

Rotational Stiffness of Sign Background.
A minimum stiffness of ١٠٠ ft-lb/degree (٥،٧٣٠ ft-lb/radian) [١٣٥,٦ Nm/degree (٧،٧٦٥ Nm/radian)] also has been assumed.

Slip Bases.
Single support slip base designs and multiple support slip base designs with an ٨-foot clearance between supports are acceptable without further testing if they are of a type that has been tested and accepted in the past, provided:

- The post does not weigh more than ٤٥ pounds per foot and the total weight below the hinge does not exceed ٦٠٠ pounds (Single supports are not expected to have hinges nor is it expected that their posts plus sign panels would weigh as much as ٦٠٠ pounds).

- The total clamping force at the face does not exceed ٤٥ kips. (One-quarter or less of this amount would be preferred and would be closer to current practice. The clamping force must be controlled by installing bolts with a torque wrench, using torque limiting nuts, or another acceptable method.)

- Washers used with the clamping bolts are of sufficient strength to prevent the washers' cupping into the vee notches.

- The stub height is no more that ٤ inches (١٠ cm).

- Where used, the upper hinge is designed to withstand the bending moment at the hinge. (Little or no factor of safety is expected to be provided in this joint).

Where multiple support slip base design provide less than ٨ feet of clearance between supports, all supports within an ٨-foot part must be considered as acting together. The acceptability of such a design may be determined by crash testing, a layout that will cause the requisite number of supports to be struck, of acceptability may be inferred by multiplying the number of supports by the energy loss that a vehicle would experience striking a single support and using this value to determine the change in momentum. Criteria fore slip base bolt tension are presented in Table C.

١١,٣. *Small Roadside Signs*

١١,٣,١. General

Small signs are defined as those supported on a single support or on two supports less than ٦ feet (١,٨ m) apart. The general expectation is that all supporting elements will be struck by the impacting vehicle. Breakaway devices for this classification of sign may be broadly classified as unidirectional and multidirectional. Both types are discussed below.

TABLE C Bolt Torque for Breakaway Slip Bases

Bolt Size inches (cm)	**Torque inch - lb (Nm)**
٥/٨ (١,٦)	٦٠٠" # (٦٨)
٣/٤ (١,٩)	٩٠٠" # (١٠٢)
٧/٨ (٢,٢)	١،٠٠٠" # (١١٣)
١ (٢,٥)	١،٥٠٠" # (١٦٩)

١١,٣,٢. Unidirectional Breakaway Posts.

The most basic type of unidirectional breakaway sign support is the inclined slip base. The design as depicted in Figures ١١,٥ and ١٠,٦, utilizes a ٤ - bolt slip base inclined at ١٠° or ٢٠° vertically toward the direction of traffic flow. Inclining the slip base results in a vertical component and ensures that the sign rotates and moves upward sufficiently to allow the impacting vehicle to pass freely under the sign without incurring a secondary collision with the top of the automobile. Early designs of this type included a hinge in the support. Subsequent testing demonstrated that the hinge was superfluous when used for small signs, and it has been eliminated from later designs. The design is not suited to lightweight, thin-walled pipe supports which may be liable to crushing at the point of impact, thus delaying base release and causing the sign to strike the top of the automobile. A limitation of this type of support is its directional properties. The support is generally not suited to installations in narrow medians or traffic islands separating opposing traffic flows.

١١,٣,٣. Multidirectional Slip Bases

The impact with a unidirectional slip base from the side can result in a severe impact. In locations where the support is expected to be struck from any one of many angles, the multidirectional slip base is appropriate. Figures ١١,٧ through ١١,٩ illustrate the multidirectional slip base concept. The base is designed to pitch up and over the vehicle for any direction of hit. It is suggested for sign locations in medians, in channelizing islands, and at other locations where the sign can be hit from more than one predominant direction.

Another type of multidirectional breakaway base involves the use of a pipe coupling at or just above the ground line. The pipe threads provide a weakened plane on which failure occurs. The sign post pulls out of the coupling and either passes over the vehicle depending upon the impact speed. Figure ١١,١٠ illustrates this basic design. It is particularly effective as an arterial street breakaway device. This system has successfully used pipes up to ٢ ١/٢ inches in diameter. Test data indicate that such systems will meet the criteria of ١١٠٠ pound-seconds. The use of a stub pipe in the soil with the sign support inserted inside the stub has been used successfully for delineator supports.

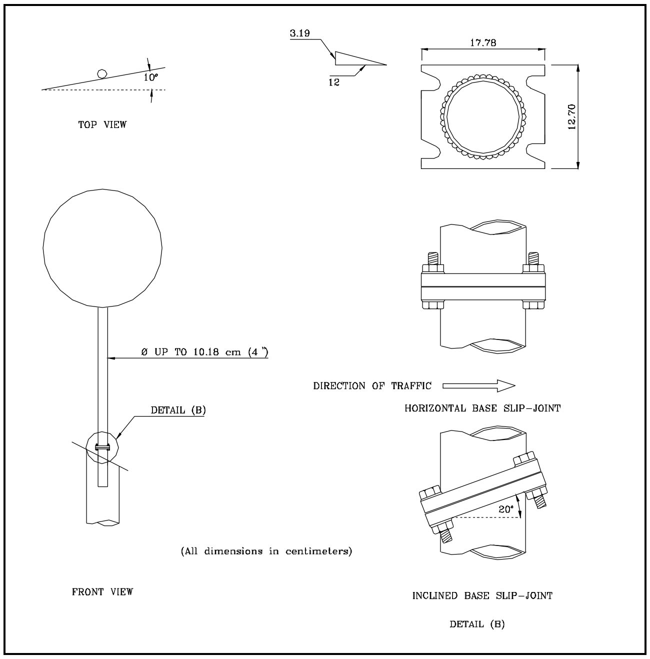


Figure ١١,٥
General Details of Pipe Supports

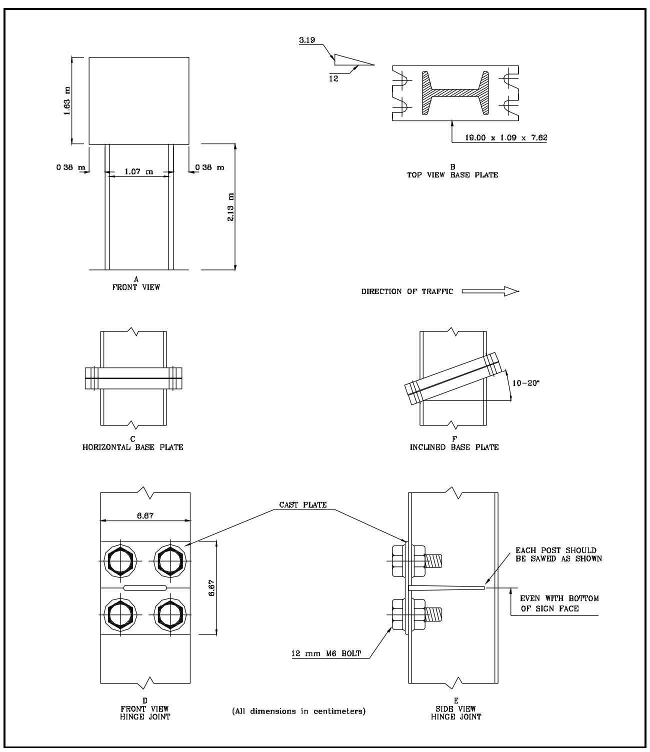


Figure ١١,٦
General Details of Small Signs

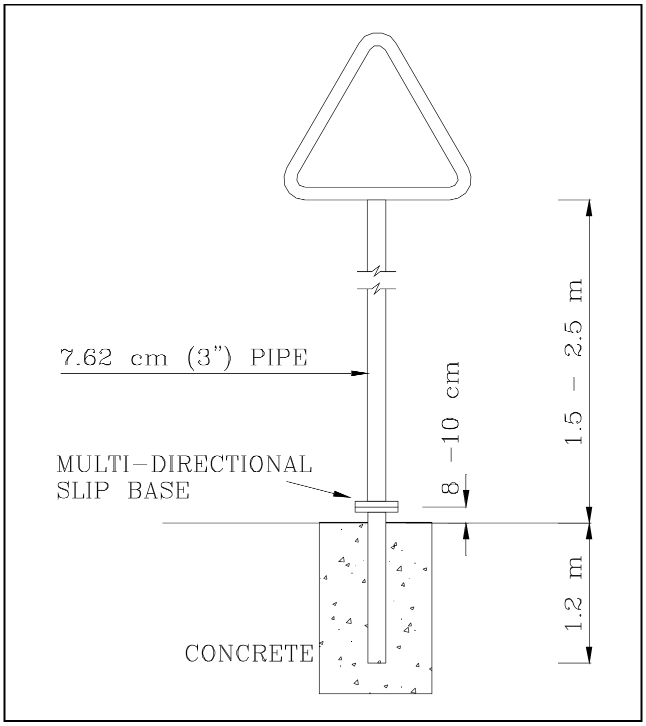


Figure ١١,٧ Small Sign with Multi-Directional Slip Base

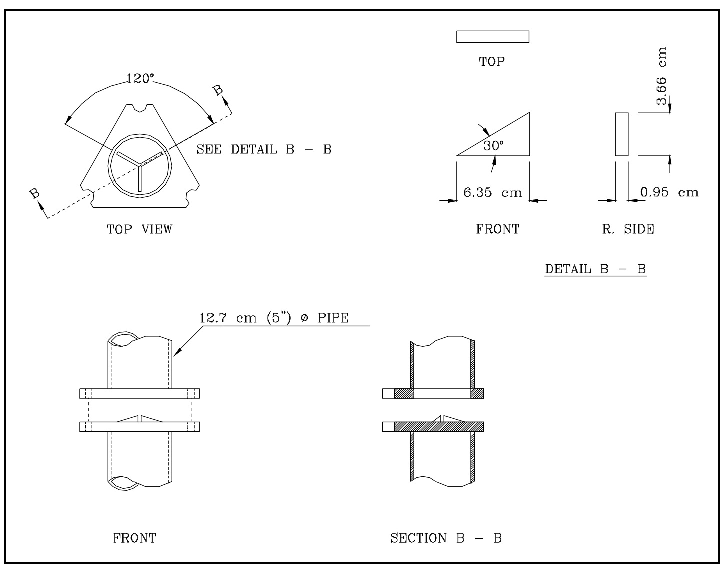
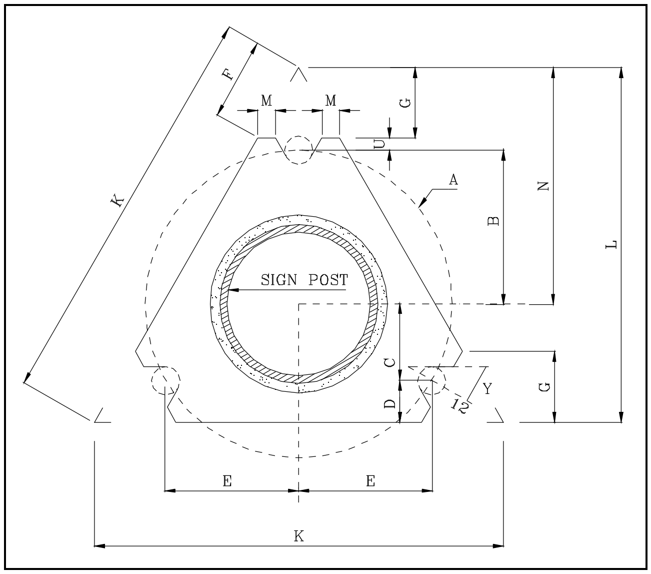


Figure ١١,٨ Recommended Multi-Directional Slip Base with Rise Modification for ٥ inch Pipe Support



Dimensions Nominal Pipe Size	Bolt Size & Torque	Weld Size	t "	Y "	A "	B "	C "	D "	E "	F "	G "	K "	L "	M "	U "	N "
٣" Dia. ٣١/٢ " Dia	٥/٨"O x ٣ ١/٢" T:٤٥٠	٣/٨"	٥/٨	٧	٧	٣ ١/٢	١ ٣/٤	١ ١/٤	٣	٢ ٥/١٦	٢	١٠ ٣/٨	٩	١/٢	١/٢	٦
٤" Dia. ٥ " Dia	٣/٤"O x ٣ ٣/٤" T:٧٥٠	٧/١٦"	٧/٨	٨ ٩/١٦	٩	٤ ١/٢	٢ ١/٤	١ ١/٢	٣ ٧/٨	٢ ٧/٨	٢ ١/٢	١٣	١١ ١/٤	٧/٨	١/٢	٧ ١/٢

Figure ١١,٩ Details of Multi - Directional Slip Base

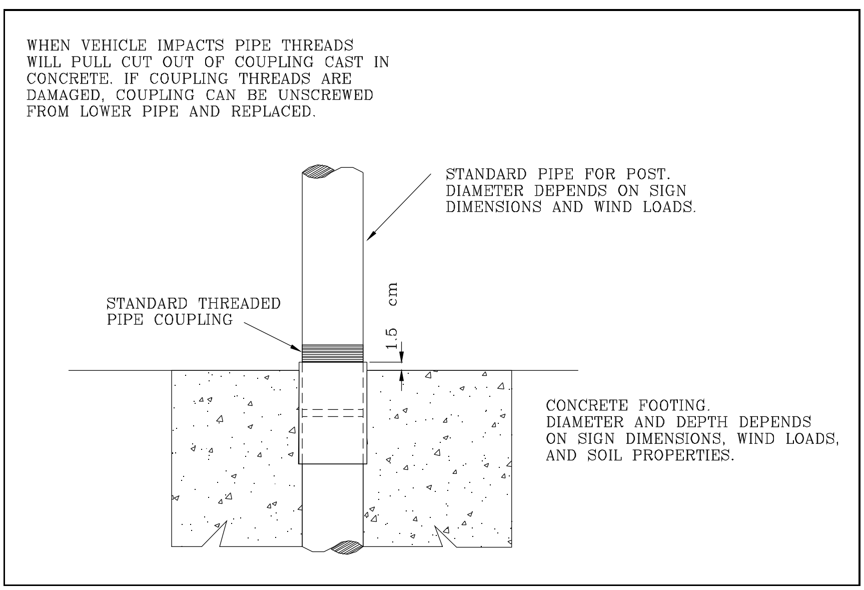


Figure ١١,١٠ Threaded Coupling Breakaway Feature

١١,٣,٤. Special Problems of Breakaway Devices

Breakaway sign supports are subject to maintenance problems which did not exist for rigid supports. Probably most significant is the wind vibration problem which tends to work the bolts in the slip base loose. In some of the early installations, the bolts actually worked completely out of the slot. A keeper plate of thin (٢٠-gage) sheet metal is now commonly used to restrain this action. Combined with routine maintenance of the bolt torque, this practice virtually has eliminated the vibration problem.

Another common problem is overtorquing of the bolts in the slip base. The slip base operates on the weakened shear plane concept. If the bolt are overtorqued, high friction exists between the slip base elements. A large impact force is required to activate the base, and it is not uncommon for the base to "lock up" under these conditions. The bolt tension specified in Table B should be used as a guide. This can be obtained by the use of a torque as indicated in Table C. In the absence of a torque wrench, the one-half turn-of-the-nut method can be used to approximate the proper torque.

Many field tests have revealed that small traffic control devices located less than ٧ ft (٢،١m) above the ground at the bottom of the panel resulted in secondary collisions with the windshield of the impacting vehicle. The secondary collision can take place only if the device is pulled loose from the post during impact. A connection of sufficient strength to insure that the device will not be pulled from the post should be used.

١١،٤. *Base Bending (yielding) Sign Supports*

١١،٤،١. General

Performance of this type of support is much more difficult to predict than other support types. Variations in the depth of embedment, the soil resistance and many other factors influence their dynamic behaviour. For this reason, a reasonably conservative approach to design is currently recommended.

For this type structure, unless acceptability is demonstrated through testing and/or an approved analytical method of which there is none today, posts should be set in soil to a depth no grater than ٣،٥ feet (without concrete foundation collars, soil bearing plates, or anchors) and, within an ٨-foot part, the plastic section module should not exceed:

For single post, ١،٣ in٣ (٢١،٣ cm٣)

For two post, each ٠،٧ in٣ (١١،٥ cm٣)

For three posts, each ٠،٤ in ٣ (٦،٦ cm٣)

Research testing found the following unacceptable (change in momentum >١١٠٠ lb./sec) (٤٤٩ Kb. Sec.):

Steel U-Post:

- ٦ lb./ft. (٨،٩ kg/m) full-length billet steel U-posts (two ٣ lb./ft. (٤،٥ kg/m) posts back-to- back).

Pipe:

- ٢ ١/٢ in. (٦،٣٥ cm) inside diameter full-length standard steel pipe..

Thin-Wall Steel Tubing:

- ٣،٥٠ in. (٨،٨٩ cm) O.D by ٠،٠٨٣ in. (٠،١٧ cm) in a concrete footing.

- ٢،٨٧٥ in. (٧،٣٠ cm) O.D by ٠،١٢٠ in. (٠،٣٠ cm) in a concrete footing.

The plastic section module should be computed on the gross section of all the elements included in the basic post. Some manufacturers suggest that two elements combined together actually behave as two independent units and therefore the plastic modules should be the sum of the two individual module rather than that of the composite section. Recent controlled tests suggest that prior to physical separation of the two elements, the deceleration experienced by a ٢٢٥٠ lb (١٠٢٣ kg) vehicle is well above the acceptable level when the plastic section modules of the combined cross section exceeds the values listed above.

١١,٤,٢. Special Provisions.

If the Sign Support is an I-beam type and the dimension is > ١٤٠ mm, or if there are ٣ or more posts, recalculation according to Table D should be done.

<u>TABLE D</u>

Support	Thickness of Fuse Plate		Bolt Size			
			Multi-Directional		Uni-Directional	
	cm	Inch	cm	Inch	cm	Inch
I - Beam ١٢٠	٢,٢٠	٧/٨	١,٥٩	٥/٨	١,٩٠	٣/٤
I - Beam ١٤٠	٢,٢٠	٧/٨	١,٥٩	٥/٨	١,٩٠	٣/٤

<u>TABLE E Number of Posts related to Sign Area.</u>

Sign Area m٢	Side Slope				
	١ : ٦	١ : ٥	١ : ٤	١ : ٣	١ : ٢
١,٥ - ٥,٠	٢ I-Beam ١٨٠	٢ I-Beam ١٨٠	٢ I-Beam ١٨٠	٢ I-Beam ١٨٠	٢ I-Beam ٢٠٠
٥,٠ - ١٠,٠	٢ I-Beam ٢٤٠	٢ I-Beam ٢٤٠	٢ I-Beam ٢٤٠	٢ I-Beam ٢٤٠	٢ I-Beam ٢٤٠
١٠,٠ - ١٥,٠	٢ I-Beam ٣٠٠	٢ I-Beam ٣٠٠	٢ I-Beam ٣٠٠	٢ I-Beam ٣٠٠	٢ I-Beam ٣٣٠
١٥,٠ - ٢٠,٠	٢ I-Beam ٣٣٠	٢ I-Beam ٣٣٠	٢ I-Beam ٣٣٠	٢ I-Beam ٣٦٠	٢ I-Beam ٣٦٠
٢٠,٠ - ٣٠,٠	٣ I-Beam ٣٣٠	٣ I-Beam ٣٦٠	٣ I-Beam ٣٦٠	٣ I-Beam ٣٦٠	٣ I-Beam ٤٠٠

<u>TABLE F. Foundation Size.</u>

Sign Area m٢	Side Slope	I-Beam Type	Single Foundation Measures (Reinforced Concrete) Depth x Width x Length
١,٥ - ٢,٠	١ : ٦	١٦٠	٠,٨٠ x ١,٠٠ x ١,٥٠
	١ : ٥	١٦٠	٠,٨٠ x ١,٠٠ x ١,٥٠
	١ : ٤	١٦٠	٠,٨٠ x ١,٠٠ x ١,٥٠
	١ : ٣	١٦٠	٠,٨٠ x ١,٠٠ x ١,٥٠
	١ : ٢	١٦٠	٠,٨٠ x ١,٠٠ x ١,٥٠
٢,٠ - ٥,٠	١ : ٦	١٨٠	٠,٨٠ x ١,٠٠ x ٢,٠٠
	١ : ٥	١٨٠	٠,٨٠ x ١,٠٠ x ٢,٠٠
	١ : ٤	١٨٠	٠,٨٠ x ١,٠٠ x ٢,٠٠
	١ : ٣	١٨٠	٠,٨٠ x ١,٠٠ x ٢,٠٠
	١ : ٢	٢٠٠	٠,٨٠ x ١,٠٠ x ٣,٠٠
٥,٠ - ١٠,٠	١ : ٦	٢٤٠	٠,٨٠ x ١,٠٠ x ٣,٠٠
	١ : ٥	٢٤٠	٠,٨٠ x ١,٠٠ x ٣,٠٠
	١ : ٤	٢٤٠	٠,٨٠ x ١,٠٠ x ٣,٠٠
	١ : ٣	٢٤٠	٠,٨٠ x ١,٠٠ x ٣,٠٠
	١ : ٢	٢٧٠	٠,٨٠ x ١,٠٠ x ٣,٥٠
١٠,٠ - ١٥,٠	١ : ٦	٣٠٠	٠,٨٠ x ١,٠٠ x ٣,٥٠
	١ : ٥	٣٠٠	٠,٨٠ x ١,٠٠ x ٤,٠٠
	١ : ٤	٣٠٠	٠,٨٠ x ١,٠٠ x ٤,٠٠
	١ : ٣	٣٠٠	٠,٨٠ x ١,٠٠ x ٤,٠٠
	١ : ٢	٣٣٠	٠,٨٠ x ١,٠٠ x ٤,٠٠

٠,٨٠ x ١,٠٠ x ٤,٥٠ ٠,٨٠ x ١,٠٠ x ٤,٥٠ ٠,٨٠ x ١,٠٠ x ٤,٥٠ ٠,٨٠ x ١,٠٠ x ٤,٥٠ ٠,٨٠ x ١,٠٠ x ٥,٠٠	٣٣٠ ٣٣٠ ٣٣٠ ٣٦٠ ٣٦٠	١ : ٦ ١ : ٥ ١ : ٤ ١ : ٣ ١ : ٢	١٥,٠ - ٢٠,٠
٠,٨٠ x ١,٠٠ x ٤,٥٠ ٠,٨٠ x ١,٠٠ x ٤,٥٠ ٠,٨٠ x ١,٠٠ x ٥,٠٠ ٠,٨٠ x ١,٠٠ x ٥,٠٠ ٠,٨٠ x ١,٠٠ x ٥,٠٠	٣٦٠ ٣٦٠ ٣٦٠ ٣٦٠ ٤٠٠	١ : ٦ ١ : ٥ ١ : ٤ ١ : ٣ ١ : ٢	٢٠,٠ - ٣٠,٠